# Exercise Manual

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# 1 Introduction

A collection of math exercises.

# 2 Numerical Stuff *(N)*

## 2.1 Theory *(T)*

- (NxTx24x12x29x001)
NOTAGS
Among the following numbers, circle the integers:

$$32,\ -20,\ 1/4,\ -2/3,\ 3.5,\ -4,\ 4\frac{2}{3}, -3\frac{1}{3}$$

---

- (NxTx24x12x29x002)
NOTAGS
Among the following numbers, circle the rationals which are **not** integers:

$$32,\ -20,\ 1/4,\ -2/3,\ 3.5,\ -4,\ 4\frac{2}{3}, -3\frac{1}{3}$$

---

- (NxTx24x12x29x003)
NOTAGS
Among the following numbers (when simplified), circle the reals which are **not** rationals:

$$32,\ \sqrt{25},\ \sqrt{2},\ -23,\ 3.5,\ \pi,\ 4\frac{2}{3}, e$$

---

- (NxTx24x12x29x004)
NOTAGS
Draw the number line and place the following numbers on it in order:

$$4,\ -3,\ 2,\ -1,\ 10,\ -5,\ 3$$

What is the largest of these numbers and smallest of these numbers?

---

- (NxTx24x12x29x005)
NOTAGS
Draw the number line and place the following numbers on it in order:

$$3,\ -3,\ 1/2,\ -1/3,\ 3.25,\ -4.25,\ 3\frac{2}{3}, -3\frac{2}{3}$$

What is the largest of these numbers and smallest of these numbers?

---

- (NxTx25x01x20x001)
NOTAGS
Consider the following numbers: 99, $-57$, 19.7, $\sqrt{17}$, $\frac{6}{7}$.

a) State whether or not it is an integer.

b) State whether or not it is a rational number.

---

- (NxTx24x12x29x006)
NOTAGS
Label each statement True or False:

a) $4 < 9$

b) $-9 < -4$

c) $30 = 30$

d) $-10 = 10$

e) $3 \leq 5$

f) $5 \leq 5$

g) $5 < 5$

h) $63 \geq 62.5$

---

- (NxTx24x12x29x007)
NOTAGS

a) Find any positive integer less than 4. How many positive integers less than 4 are there?

b) Find any positive rational (which is not an integer) less than 4. How many positive rationals less than 4 are there?

---

- (NxTx25x01x01x001)
NOTAGS
No calculator!

  a) Which fraction is larger $\frac{3}{8}$ or $\frac{3}{5}$?

  b) Does $\frac{2}{8} = \frac{1}{4}$?

  c) Does $\frac{17}{87} = \frac{3}{87}$?

---

- (NxTx25x01x01x002)
NOTAGS
No calculator!

  a) The rational $\frac{1}{3}$ is between what two consecutive integers?

  b) The rational $\frac{7}{3}$ is between what two consecutive integers?

  c) The rational $-\frac{7}{3}$ is between what two consecutive integers?

---

- (NxTx24x12x31x001)
NOTAGS
Circle the primes: 2, 9, 13, 21, 23, 30, 51

---

- (NxTx24x12x31x002)
NOTAGS
Factor:

  a) 15 =  b) 20 =

---

- (NxTx24x12x31x003)
NOTAGS
Factor:

  a) 13 =  b) 40 =

---

- (NxTx25x01x01x101)
NOTAGS

  a) LCM of 5 and 2 is:  b) LCM of 3 and 2 is:

---

- (NxTx25x01x01x102)
NOTAGS

a) LCM of 3 and 6 is:

b) LCM of 14 and 1 is:

---

- (NxTx25x01x01x003)
NOTAGS

a) LCM of 8 and 6 is:

b) LCM of 14 and 7 is:

---

- (NxTx25x01x01x004)
NOTAGS

a) LCM of 16 and 20 is:

b) LCM of 9 and 15 and 10 is:

---

## 2.2 Negatives and Absolute Value *(NA)*

- (NxNAx24x12x29x001)
NOTAGS
Simplify the following.

a) $--8 =$

b) $---2 =$

c) $-7 =$

---

- (NxNAx24x12x29x002)
NOTAGS
Simplify the following.

a) $|12.37| =$

b) $|-1| =$

c) $|---33| =$

d) $|0| =$

---

## 2.3 Integer Arithmetic *(AI)*

- (NxAIx24x12x29x001)
NOTAGS
Evaluate the following.

  a) $3 + 5 =$

  b) $5 + 3 =$

  c) $3 \cdot 5 =$

  d) $(5)(3) =$

---

- (NxAIx24x12x29x002)
NOTAGS
Evaluate the following.

  a) $-3 \cdot 5 =$

  b) $(-5)(-3) =$

  c) $5 - 3 =$

  d) $3 - 5 =$

---

- (NxAIx24x12x29x003)
NOTAGS
Evaluate and simplify the following.

  a) $20 \div 5 =$

  b) $-20 \div 5 =$

  c) $\dfrac{10}{-2} =$

  d) $\dfrac{-10}{-2} =$

---

- (NxAIx24x12x29x004)
NOTAGS
Evaluate and simplify the following.

  a) $|5 - 3| =$

  b) $|3 - 5| =$

---

- (NxAIx24x12x29x005)
NOTAGS
Label each statement True or False:

  a) $4 + 1 < 2 + 2$

  b) $3 - 2 > 1$

---

- (NxAIx24x12x29x006)
NOTAGS
Label each statement True or False:

  a) $3 - 1 = 10 - 8$

  b) $-10 = (10)(-1)$

---

- (NxAIx24x12x30x001)
NOTAGS
Label each statement True or False:

a) $-3-5 \leq 1-2$ b) $|5+2| \geq |5-15|$ c) $|(-3)(-5)| \leq 2+4$

---

- (NxAIx24x12x30x002)
NOTAGS
Label each statement True or False:

a) $|-7897| \geq 0$ b) $|-513 \cdot 25| \geq (-945)(15)$

---

- (NxAIx24x12x27x003)
NOTAGS
For each expression, calculate it or indicate that it is *undefined*.

a) $0 \div 25 =$ c) $0+23 =$

b) $25 \div 0 =$ d) $34-0 =$

---

- (NxAIx24x12x30x003)
NOTAGS
For each expression, calculate it or indicate that it is *undefined*.

a) $\dfrac{0}{-4} =$ c) $\dfrac{-4}{0} =$

b) $0 \div 4 =$ d) $-4 \div 0 =$

---

- (NxAIx24x12x30x004)
NOTAGS
For each expression, calculate it or indicate that it is *undefined*.

a) $-4 \div 0 =$ c) $0 \div 52 =$

b) $(-52)(0) =$ d) $\dfrac{-4}{0} =$

---

## 2.4 Fraction Arithmetic *(AF)*

- (NxAFx25x01x01x001)
NOTAGS

  a) As a decimal $\frac{1}{2} =$

  b) As a decimal $\frac{-1}{4} =$

---

- (NxAFx25x01x01x002)
NOTAGS

  a) As a fraction $0.75 =$

  b) As a fraction $2.75 =$

---

- (NxAFx25x01x01x003)
NOTAGS

  a) As a fraction $2\frac{1}{2} =$

  b) As a fraction $5\frac{3}{4} =$

---

- (NxAFx25x01x01x004)
NOTAGS

  a) As a mixed number $\frac{3}{2} =$

  b) As a mixed number $\frac{7}{3} =$

---

- (NxAFx24x12x31x001)
NOTAGS

  a) Reduce $\frac{25}{10} =$

  b) Reduce $\frac{-12}{4} =$

---

- (NxAFx24x12x31x002)
NOTAGS
Determine if the given fractions are equivalent.

  a) $\frac{5}{15}$ and $\frac{1}{3}$

  b) $\frac{2}{8}$ and $\frac{2}{4}$

---

- (NxAFx24x12x31x003)
NOTAGS
Determine if the given fractions are equivalent.

a) $\frac{3}{4}$ and $\frac{12}{16}$

b) $\frac{4}{6}$ and $\frac{10}{15}$

---

- (NxAFx24x12x31x004)
NOTAGS

a) Find two fractions that are equivalent to $\frac{2}{7}$:

b) Find two fractions that are equivalent to $\frac{15}{10}$:

---

- (NxAFx24x12x31x005)
NOTAGS
Write each integer as a fraction by putting it over 1.

a) $4 =$

b) $25 =$

c) $0 =$

d) $-3 =$

e) $1 =$

f) $-1 =$

---

- (NxAFx24x12x31x006)
NOTAGS
Multiply and simplify.

a) $\frac{3}{7} \cdot \frac{3}{2} =$

b) $\frac{5}{4} \cdot \frac{7}{3} =$

---

- (NxAFx24x12x31x106)
NOTAGS
Perform the operation.

a) $\frac{1}{4} \cdot \frac{3}{4} =$

b) $3 \cdot \frac{2}{7} =$

---

- (NxAFx24x12x31x206)
NOTAGS
Perform the operation and simplify.

a) $\left(\frac{6}{8}\right)\left(\frac{4}{9}\right) =$

b) $\left(\frac{7}{3}\right)\left(\frac{3}{8}\right) =$

---

- (NxAFx24x12x31x306)
NOTAGS
Evaluate.

a) $\left(\frac{1}{5}\right)\left(\frac{4}{6}\right) =$

b) $\left(\frac{2}{5}\right)(5) =$

---

- (NxAFx25x02x12x002)
NOTAGS
Evaluate $\left(\frac{1}{8}\right)\left(\frac{10}{3}\right) =$

---

- (NxAFx24x12x31x007)
NOTAGS
Divide and simplify.

a) $\frac{3}{7} \div \frac{5}{2} =$

b) $\frac{6}{5} \div \frac{7}{9} =$

---

- (NxAFx24x12x31x008)
NOTAGS
Divide.

a) $\frac{5}{6} \div \frac{2}{9} =$

b) $4 \div \frac{4}{7} =$

---

- (NxAFx24x12x31x009)
NOTAGS
Perform the operation and simplify.

a) $7 \div \frac{6}{4} =$

b) $\frac{5}{6} \div 5 =$

---

- (NxAFx24x12x31x109)
NOTAGS
Perform the operation.

a) $\frac{3}{4} \div 7 =$

b) $\frac{8}{5} \div 4 =$

---

- (NxAFx24x12x31x010)
NOTAGS
Evaluate $\frac{1}{3} \div \frac{1}{7} =$

---

- (NxAFx25x02x12x001)
NOTAGS
Evaluate $\frac{1}{4} \div \frac{5}{6} =$

---

- (NxAFx24x12x31x011)
NOTAGS
Add and simplify.

a) $\frac{3}{7} + \frac{4}{7} =$ b) $\frac{4}{12} + \frac{5}{12} =$

---

- (NxAFx24x12x31x012)
NOTAGS
Perform the operation and simplify.

a) $\frac{6}{5} + \frac{7}{5} =$ b) $\frac{3}{12} + \frac{14}{12} =$

---

- (NxAFx24x12x31x013)
NOTAGS
Evaluate

a) $\frac{4}{9} + \frac{2}{9} =$ b) $\frac{1}{15} + \frac{42}{15} =$

---

- (NxAFx24x12x31x014)
NOTAGS
Subtract and simplify.

a) $\frac{8}{7} - \frac{1}{7} =$ b) $\frac{7}{12} - \frac{3}{12} =$

---

- (NxAFx24x12x31x015)
NOTAGS
Perform the subtraction.

a) $\frac{8}{5} - \frac{7}{5} =$ b) $\frac{43}{12} - \frac{43}{12} =$

---

- (NxAFx24x12x31x016)
NOTAGS
Evaluate

a) $\frac{1}{9} - \frac{2}{9} =$ b) $\frac{20}{15} - \frac{25}{15} =$

---

- (NxAFx24x12x27x001)
NOTAGS
Evaluate

a) $\frac{4}{3} + \frac{2}{6} = \text{—} + \text{—} =$

b) $\frac{2}{5} + \frac{7}{4} = \text{—} + \text{—} =$

---

- (NxAFx25x01x01x102)
NOTAGS
Evaluate

a) $\frac{5}{4} + \frac{1}{6} = \text{—} + \text{—} =$

b) $\frac{5}{12} + \frac{3}{4} = \text{—} + \text{—} =$

---

- (NxAFx25x01x01x103)
NOTAGS
Evaluate

a) $\frac{3}{7} + \frac{3}{4} = \text{—} + \text{—} =$

b) $\frac{5}{8} + \frac{2}{5} = \text{—} + \text{—} =$

---

- (NxAFx25x01x01x104)
NOTAGS
Evaluate

a) $\frac{5}{6} + \frac{1}{4} = \text{—} + \text{—} =$

b) $\frac{7}{3} + \frac{6}{5} = \text{—} + \text{—} =$

---

- (NxAFx25x01x01x005)
NOTAGS
Evaluate

a) $\frac{3}{7} + \frac{1}{2} = \text{—} + \text{—} =$

b) $\frac{2}{6} + \frac{3}{5} = \text{—} + \text{—} =$

---

- (NxAFx25x01x01x006)
NOTAGS
Evaluate

a) $\frac{8}{10} + \frac{7}{4} = \text{—} + \text{—} =$

b) $\frac{1}{2} + \frac{5}{6} =$

---

- (NxAFx25x01x01x007)
NOTAGS
Evaluate

  a) $\frac{1}{1}+\frac{2}{3}=$ b) $1+\frac{3}{4}=$

---

- (NxAFx25x01x01x008)
NOTAGS
Evaluate

  a) $\frac{5}{3}+\frac{7}{4}=$ b) $2+\frac{2}{5}=$

---

- (NxAFx25x01x01x009)
NOTAGS
Evaluate

  a) $4+\frac{3}{2}=$ b) $2\frac{1}{2}+3\frac{1}{2}=$

---

- (NxAFx25x01x01x010)
NOTAGS
Evaluate

  a) $5\frac{2}{7}+3\frac{4}{7}=$ b) $4\frac{1}{8}+2\frac{3}{8}=$

---

- (NxAFx25x01x01x011)
NOTAGS
Evaluate $13\frac{2}{10}+3\frac{2}{5}=$

---

- (NxAFx25x02x12x003)
NOTAGS
Evaluate $\frac{4}{10}+\frac{4}{15}=$

---

- (NxAFx25x01x01x012)
NOTAGS
Evaluate

  a) $\frac{5}{4}-\frac{1}{6}=$ — $-$ — $=$ b) $\frac{5}{12}-\frac{1}{4}=$ — $-$ — $=$

---

- (NxAFx25x01x01x013)
NOTAGS
Evaluate

  a) $\frac{4}{7} - \frac{2}{5} =$ — − — =

  b) $\frac{5}{8} - \frac{2}{5} =$ — − — =

---

- (NxAFx25x01x01x014)
NOTAGS
Evaluate

  a) $\frac{5}{6} - \frac{1}{4} =$ — − — =

  b) $\frac{7}{3} - \frac{6}{5} =$ — − — =

---

- (NxAFx25x01x01x015)
NOTAGS
Evaluate

  a) $\frac{5}{4} - \frac{2}{3} =$

  b) $\frac{5}{3} - \frac{1}{9} =$

---

- (NxAFx25x01x01x016)
NOTAGS
Evaluate

  a) $\frac{1}{1} - \frac{2}{5} =$

  b) $\frac{1}{1} - \frac{1}{6} =$

---

- (NxAFx25x01x01x017)
NOTAGS
Evaluate

  a) $1 - \frac{1}{4} =$

  b) $\frac{8}{7} - \frac{1}{1} =$

---

- (NxAFx25x01x01x018)
NOTAGS
Evaluate

  a) $\frac{5}{2} - 1 =$

  b) $\frac{9}{3} - \frac{4}{7} =$

---

- (NxAFx25x01x01x019)
NOTAGS
Evaluate $\frac{7}{2} - \frac{3}{5} =$

---

- (NxAFx25x02x12x004)
NOTAGS
Evaluate $\frac{16}{6} - 2 =$

---

## 2.5 Order of Operations, and Exponents *(OP)*

### 2.5.1 With Integers *(I)*

- (NxOPxIx25x01x01x020)
NOTAGS
Evaluate

a) $4^2 =$ b) $(-4)^2 =$

---

- (NxOPxIx25x01x01x021)
NOTAGS
Evaluate

a) $(-3)^3 =$ b) $(-3)^4 =$

---

- (NxOPxIx25x01x01x022)
NOTAGS
Evaluate

a) $1^8 =$ b) $0^8 =$

---

- (NxOPxIx25x01x01x023)
NOTAGS
Evaluate

a) $45^1 =$ b) $(-45)^1 =$

---

- (NxOPxIx25x01x01x024)
NOTAGS
Evaluate

a) $-3^2 =$ b) $(-3)^2 =$

---

- (NxOPxIx25x01x01x025)
NOTAGS
Evaluate

a) $-1^5 =$ b) $(-1)^5 =$

---

- (NxOPxIx25x01x01x026)
NOTAGS
Evaluate

a) $-1^6 =$ b) $(-1)^6 =$

---

- (NxOPxIx25x01x01x027)
NOTAGS
Write in exponential form

a) $7 \cdot 7 \cdot 7 =$ b) $(-8) \cdot (-8) =$

---

- (NxOPxIx25x01x01x028)
NOTAGS
Write in exponential form

a) $(2)(2)(2)(2)(2)(2)(2) =$ b) $(-83)(-83)(-83)(-83) =$

---

- (NxOPxIx25x01x01x029)
NOTAGS
Evaluate (reorder first)

a) $7 + 89 + (-7) - 89 =$ b) $(-8) + 34 + 8 + 1 + (-33) =$

---

- (NxOPxIx25x01x01x030)
NOTAGS
Evaluate

a) $3 + 5 \cdot (2) =$ b) $(3 + 5) \cdot (2) =$

---

- (NxOPxIx25x01x01x031)
NOTAGS
Evaluate

a) $3 + 5 \cdot (-2) =$ b) $(3 + 5) \cdot (-2) =$

---

- (NxOPxIx25x01x01x032)
NOTAGS
Evaluate

  a) $1 + 2^3 =$ b) $(1 + 2)^3 =$

---

- (NxOPxIx25x01x01x033)
NOTAGS
Evaluate

  a) $10 - 4 =$ b) $4 - 10 =$

---

- (NxOPxIx25x01x01x034)
NOTAGS
Evaluate

  a) $|4 - 10| =$ b) $|10 - 4| =$

---

- (NxOPxIx25x01x01x035)
NOTAGS
Evaluate

  a) $-(10 - 4) =$ b) $-(4 - 10) =$

---

- (NxOPxIx25x01x01x036)
NOTAGS
Evaluate

  a) $-8 - 3^2 =$ b) $11 - 72 \div 9 =$

---

- (NxOPxIx25x01x01x037)
NOTAGS
Evaluate

  a) $18 - 42 \div 7 =$ b) $13 - 5 - 1 + 9 \div 3 =$

---

- (NxOPxIx25x01x01x038)
NOTAGS
Evaluate

  a) $5 \cdot 6 - (15 - 6) =$ b) $3 \cdot 9 - (35 - 1) =$

---

- (NxOPxIx25x01x01x039)
NOTAGS
Evaluate

  a) $9 + 3 - 12 =$

  b) $17 - 10 - 8 =$

---

- (NxOPxIx25x01x01x040)
NOTAGS
Evaluate

  a) $10 + 6 \div 2 + (3)(-3) =$

  b) $3 \cdot 4 \cdot 2 \div 4 + 3 =$

---

- (NxOPxIx25x01x01x041)
NOTAGS
Evaluate

  a) $2 \cdot 5 \cdot 10 \div 5 + 3 =$

  b) $(2 \cdot 2)^2 =$

---

- (NxOPxIx25x01x01x042)
NOTAGS
Evaluate

  a) $(5 \cdot 2)^2 =$

  b) $13 + 0 \div 7 =$

---

- (NxOPxIx25x01x01x043)
NOTAGS
Evaluate

  a) $9 + 10 \div 5 =$

  b) $[12 \div (4 \div 2)]^2 =$

---

- (NxOPxIx25x01x01x044)
NOTAGS
Evaluate

  a) $[32 \div (8 \div 2)]^2 =$

  b) $12 + 3 \cdot 2 + (3 + 5 \cdot 2) =$

---

- (NxOPxIx25x01x01x045)
NOTAGS
Evaluate

  a) $9 + 4 \cdot 5 + (8 + 4 \cdot 4) =$

  b) $13 + 2(5 - 3) =$

---

- (NxOPxIx25x01x01x046)
NOTAGS
Evaluate

  a) $16 + 5(9 - 4) =$ b) $12 \div 3 \cdot 4 =$

---

- (NxOPxIx25x01x01x047)
NOTAGS
Evaluate

  a) $25 \div 5 \cdot 5 =$ b) $1 + (-9 + 7)^2 - 7 \cdot 2 =$

---

- (NxOPxIx25x01x01x048)
NOTAGS
Evaluate $2 \cdot |4 - 5|^3 - (5 - 4)^2 =$

---

- (NxOPxIx25x01x01x055)
NOTAGS
Evaluate each in TWO ways - 1) just order of operations, 2) by first distributing

  a) $2 \cdot (3 + 2) =$ b) $(3 - 2) \cdot 4 =$

---

- (NxOPxIx25x01x01x056)
NOTAGS
Evaluate each in TWO ways - 1) just order of operations, 2) by first distributing

  a) $(-2) \cdot (1 + 3) =$ b) $(-1 + 1) \cdot 10 =$

---

- (NxOPxIx25x01x01x057)
NOTAGS
Evaluate each in TWO ways - 1) just order of operations, 2) by first distributing

  a) $2 \cdot (4 + 2 - 3) =$ b) $(-3 + 2 - 7) \cdot (-1) =$

---

### 2.5.2 With Fractions *(F)*

- (NxOPxFx25x01x01x148)
NOTAGS
Evaluate $\frac{2}{3} + \frac{5}{6} \cdot \frac{2}{5} =$

---

- (NxOPxFx25x01x01x049)
NOTAGS
Evaluate

a) $\left(\frac{1}{2}+\frac{1}{2}\right)\cdot(5)=$ b) $\frac{6}{5}+\frac{1}{10}\cdot(-2)=$

---

- (NxOPxFx25x01x01x050)
NOTAGS
Evaluate

a) $\frac{1}{2}+\left(\frac{1}{2}\right)^2=$ b) $\left(1+\frac{1}{2}\right)^2=$

---

- (NxOPxFx25x01x01x051)
NOTAGS
Evaluate

a) $\left|\frac{2}{3}-\frac{1}{6}\right|=$ b) $-\left(\frac{2}{5}-\frac{1}{10}\right)=$

---

- (NxOPxFx25x01x01x052)
NOTAGS
Evaluate

a) $-\left(\frac{1}{10}-\frac{2}{5}\right)=$ b) $\frac{1}{3}-\frac{1}{6}\div\frac{1}{2}=$

---

- (NxOPxFx25x01x01x053)
NOTAGS
Evaluate

a) $\frac{1}{3}-4\div\frac{1}{2}=$ b) $\frac{7}{8}\cdot\frac{1}{2}+\frac{1}{4}=$

---

- (NxOPxFx25x01x01x054)
NOTAGS
Evaluate

a) $5\cdot\frac{2}{5}+\left(\frac{1}{2}-6\right)=$ b) $\frac{4}{3}\cdot\frac{1}{8}-\left(\frac{1}{2}-1\right)=$

---

- (NxOPxFx25x02x12x001)
NOTAGS
Evaluate $5 \cdot \frac{3}{10} + \left(\frac{1}{4} - 1\right) =$

---

## 2.6 Numeric Word Problems *(WP)*

### 2.6.1 Involving Integers *(I)*

- (NxWPxIx24x12x29x001)
NOTAGS

a) How many integers are between 12 and 15 (if you include both 12 and 15)?

b) How many integers are between 10 and 300 (if you include both 10 and 300)?

---

- (NxWPxIx24x12x29x002)
NOTAGS
In the year 2014, the highest subway station was the Smith-9 Street station (F and G trains); it is 88 feet *above street level*. The lowest station was the 191st 1 station; it was 180 feet *below street level*. What is the positive difference in heights between these two stations?

---

- (NxWPxIx24x12x30x001)
NOTAGS
Suppose your math class has 20 students, and then 7 students come in late, and then 9 students leave. How many students are in the class now?

---

- (NxWPxIx24x12x30x002)
NOTAGS
Suppose Nike is shipping basketballs. Each truck has 500 basketballs, and there are 10 trucks. How many basketballs do we have all together?

---

- (NxWPxIx24x12x30x003)
NOTAGS
Suppose 32 students get into 4 equal size groups. How big is each group?

---

- (NxWPxIx24x12x30x004)
NOTAGS
Suppose 42 students get into groups of size 6. How many groups are there?

---

- (NxWPxIx24x12x30x005)
NOTAGS
Vic has 10 jars, in each of which he put 5 balls. One day he redistributed the balls equally among 5 jars instead. How many balls are now in each jar?

---

- (NxWPxIx25x01x03x004)
NOTAGS

Vic is driving 30 MPH (miles per hour). If he drives for 3 hours, how far does he drive?

---

- (NxWPxIx25x01x03x005)
NOTAGS

Vic is driving 30 MPH (miles per hour). If he drives 150 miles, how long did he drive for?

---

- (NxWPxIx25x01x03x006)
NOTAGS

What is the average of 1, 10 and 4?

---

- (NxWPxIx25x01x03x007)
NOTAGS

Mary ate 4 apples on both weekend days, and 11 apples each of the 5 weekdays. What is the average number of apples Mary ate each day?

---

### 2.6.2 Word Problems Involving Fractions *(F)*

- (NxWPxFx24x12x31x001)
NOTAGS
Find the area of a square with side length $\frac{5}{3}$.

---

- (NxWPxFx24x12x31x002)
NOTAGS
Find the area of a rectangle with width $\frac{7}{2}$ and height $\frac{8}{3}$.

---

- (NxWPxFx25x02x12x001)
NOTAGS
Find the area and perimeter of a rectangle with width $\frac{3}{2}$ and height 5.

---

- (NxWPxFx25x01x01x001)
NOTAGS
Find the perimeter of a square with side length $\frac{10}{3}$:

---

- (NxWPxFx25x01x01x002)
NOTAGS
Find the perimeter of a rectangle with width $\frac{5}{2}$ and height $\frac{9}{4}$:

---

- (NxWPxFx25x01x01x003)

NOTAGS

---

NOTAGS
- (NxWPxFx25x01x01x004)
NOTAGS

a) A triangle has a base $\frac{7}{2}$ and a height 5. What is its area?

b) A triangle has a base $\frac{10}{3}$ and a height $\frac{3}{20}$. What is its area?

---

- (NxWPxFx25x01x01x005)
NOTAGS

a) Suppose a square is attached to the base of a triangle. The base of the triangle is 7 and its height is 10. What is the area of the whole object?

---

- (NxWPxFx25x01x03x001)
NOTAGS

Mary drank $\frac{2}{5}$ liters of milk and left $2\frac{3}{5}$ liters in the carton. How many liters of milk were in the carton before Mary drank?

---

- (NxWPxFx25x01x03x002)
NOTAGS

a) Mary divided 10kg of clay equally between 5 people. How much clay did Mary give each person?

b) Vic divided 3kg of clay equally between 5 people. How much clay did Vic give each person?

---

## 2.7 Exponents and Scientific Notation *(ES)*

- (NxESx25x01x15x013)
NOTAGS
Multiply.

a) $445 \times 10^3 = 445000$ (example)

b) $42965 \times 10^4 =$

---

- (NxESx25x01x15x014)
NOTAGS
Multiply.

a) $153.5 \times 10^3 =$

b) $179.65 \times 10^{-4} =$

---

- (NxESx25x01x15x015)
NOTAGS
Multiply.

a) $215.34 \times 10^{-1} =$

b) $423653 \times 10^{-7} =$

---

- (NxESx25x01x15x016)
NOTAGS
Multiply.

a) $0.003 \times 10^{-1} =$

---

- (NxESx25x01x15x017)
NOTAGS
The following numbers are written in scientific notation. Write them in the usual way (by multiplying).

a) $2.3 \times 10^3 = 2300$ (example)

b) $2.55 \times 10^5 =$

---

- (NxESx25x01x15x018)
NOTAGS
The following numbers are written in scientific notation. Write them in the usual way (by multiplying).

  a) $4.534 \times 10^{-4} =$ b) $2.3 \times 10^{-3} =$

---

- (NxESx25x01x15x019)
NOTAGS
The following numbers are written in scientific notation. Write them in the usual way (by multiplying).

  a) $1.23 \times 10^{4} =$ b) $8.004 \times 10^{-4} =$

---

- (NxESx25x01x15x020)
NOTAGS
The following numbers are written in scientific notation. Write them in the usual way (by multiplying).

  a) $1.76 \times 10^{8} =$ b) $7.25 \times 10^{-6} =$

---

- (NxESx25x01x15x021)
NOTAGS
The following numbers are written in scientific notation. Write them in the usual way (by multiplying).

  a) $2.418 \times 10^{-1} =$

---

- (NxESx25x01x15x022)
NOTAGS
Write the following numbers in scientific notation.

  a) $23.3 = 2.33 \times 10^{1}$ (example) b) $20355 =$

---

- (NxESx25x01x15x023)
NOTAGS
Write the following numbers in scientific notation.

  a) $534.44 =$ b) $0.00003 =$

---

- (NxESx25x01x15x024)
NOTAGS
Write the following numbers in scientific notation.

a) $1.45355 =$ b) $5449994.34 =$

---

- (NxESx25x01x15x025)
NOTAGS
Write the following numbers in scientific notation.

a) $0.001001 =$ b) $255443.234 =$

---

- (NxESx25x01x15x026)
NOTAGS
Write the following numbers in scientific notation.

a) $50000000 =$ b) $24.54 \times 10^4 =$

---

- (NxESx25x01x15x027)
NOTAGS
Write the following numbers in scientific notation.

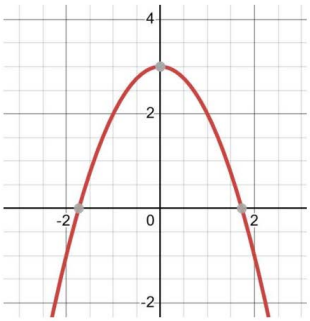
a) $322 \cdot 10^{-3} =$ b) $534.44 \times 10^{34} =$

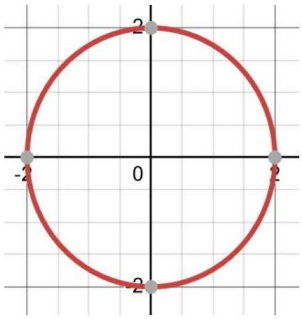
---

# 3 Functions *(F)*

## 3.1 Definition Issues *(DEF)*

- (FxDEFx25x01x20x003)
NOTAGS
Consider the two graphs below. For each one, state whether or not it is the graph of a function.




---

- (FxDEFx25x01x22x001)
NOTAGS
For each operation, state whether or not it is a function (yes/no) and give the reason. If it is a function, write it in function notation.

  a) The operation assigning each country its population.

  b) The operation taking a number to half its value.

  c) The operation assigning to each number its third roots.

  d) The operation assigning to each number its fourth roots.

---

## 3.2 Viewed as Relations *(REL)*

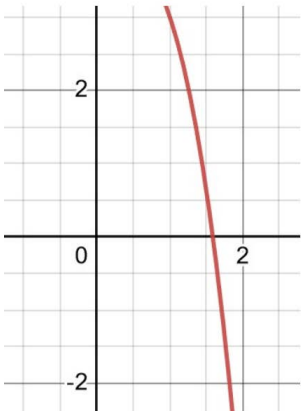
- (FxRELx25x01x20x001)
NOTAGS
Consider the following two relations. For each one, is it a function or not?

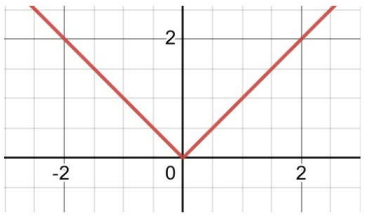
  a) $\{(2,4),(4,2),(3,4),(1,1)\}$ b) $\{(2,4),(3,5),(2,6),(7,9)\}$

---

## 3.3 One to One Functions *(ONE)*

- (FxONEx25x01x20x004)
NOTAGS
Consider functions graphed below. For each one, state whether or not it is one-to-one.




---

- (FxONEx25x01x23x001)
NOTAGS
Consider the following two relations. For each one, is it a function or not? If it is a function, is it injective?

a) $\{(-2,4),(4,2),(2,4),(1,1),(5,6)\}$ b) $\{(0,0),(3,-5),(-2,-6),(6,2)\}$

---

- (CREATIVExxxFxONEx25x04x16x001)
NOTAGS
For each function decide whether or not you think we should view it as one-to-one, or perhaps not one-to-one, but close enough! Explain yourself, and give reasons for each of your answers (this question is open-ended, and may have a range of answers).

a) The function whose input is a person's age, and whose output is that person's height at that age.

b) Suppose $g(t)$ = the temperature of a cup of coffee as it cools, where $t$ is the number of minutes the coffee has been cooling.

c) Suppose $d(t)$ = the temperature outside today, $t$ hours past 3am today. Consider this function up till 11pm today.

d) Imagine you could measure the quality of your mood with a single number between 1 and 10, let $M(d)$ = your mood $d$ days past January 1 of this year.

---

- (TESTxFxONEx25x02x12x001)
type: TEST
Consider the following 3 relations. For each one, is it a function or not? If it is a function, is it injective?

  a) $\{(0,4),(4,0),(-4,2)\}$

  b) $\{(1,3),(2,4),(1,7)\}$

  c) $\{(0,0),(4,2),(2,0)\}$

---

- (FxONEx25x09x15x001)
type: TEST
Consider the following 3 relations. For each one, is it a function or not? If it is a function, is it injective?

  a) $\{(0,5),(5,0),(3,1),(-5,1)\}$

  b) $\{(7,2),(2,3),(4,7),(5,0)\}$

  c) $\{(1,3),(2,4),(1,7)\}$

---

- (GPTx26x02x20x001)
type: TEST
Consider the following 3 relations. For each one, determine whether it is a function. If it is a function, determine whether it is injective.

  a) $\{(1,4),(2,5),(3,6),(4,7)\}$

  b) $\{(0,-1),(2,3),(5,3),(7,8)\}$

  c) $\{(-2,5),(-2,7),(3,1)\}$

---

- (GPTx26x02x20x004)
type: TEST
Evaluate $f(-3)$, $f(9)$, and $f(11)$ where
$f(x) = \begin{cases} 2x & \text{if } x \leq 9 \\ x+7 & \text{if } x > 9 \end{cases}$
Also determine whether $f$ is one-to-one.

---

## 3.4 Function Properties (increase, decrease, max, min) *(PROP)*

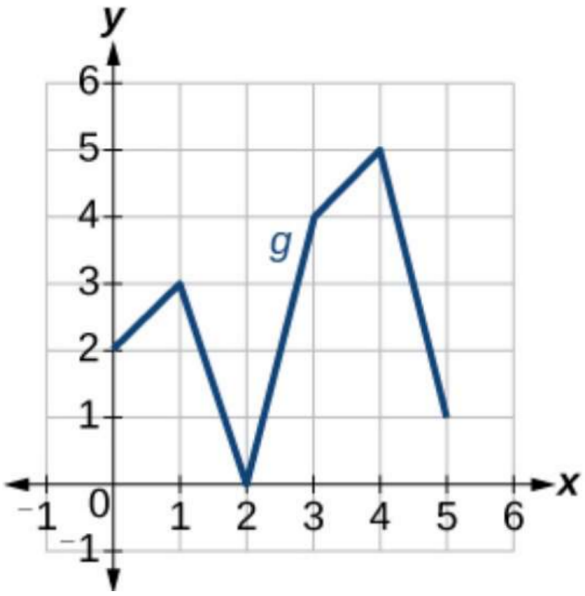
- (FxPROPx25x01x20x006)
NOTAGS
Consider the function graphed below.

  a) Find the intervals where it increases.

  b) Find the intervals where it decreases.

  c) Find all local maxima (both $x$ and $y$ coordinates).

  d) Find all local minima (both $x$ and $y$ coordinates).




---

- (FxPROPx25x09x15x006)
type: TEST
Consider the function graphed below. For all answers, choose the nearest integer value (example: if an $x$ coordinate looks like about 9.1, just take it to be 9).

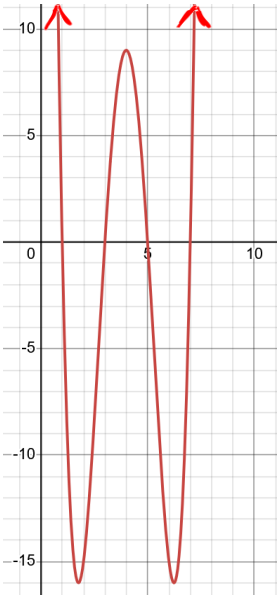
  a) Find the intervals where it increases.

  b) Find the intervals where it decreases.

  c) Find all local maxima (both $x$ and $y$ coordinates).

  d) Find all local minima (both $x$ and $y$ coordinates).

  e) Find all the absolute extrema (both $x$ and $y$ coordinates).




---

- (FxPROPx25x01x24x001)
NOTAGS
If a function $g$ is increasing on the interval $(1, 7)$ and decreasing on the interval $(7, 10)$ then what can be said about the local extremum of $g$ on $(1, 10)$?

---

- (FxPROPx25x01x24x002)
NOTAGS
Consider the line $f(x) = 46x - 87$. What are its local extrema? What are its absolute extrema?

---

- (FxPROPx25x01x24x003)
NOTAGS
Suppose $f(t) = \dfrac{7}{t}$ and $g(t) = \dfrac{7}{t^2}$. Where is each function increasing and where is each function decreasing?

---

- (FxPROPx25x01x24x004)
NOTAGS
Draw a graph with a local maximum, but no absolute maximum.

---

- (FxPROPx25x01x24x005)
NOTAGS
Draw a graph with an absolute maximum, but no absolute minimum, which does have a relative minimum.

---

- (FxPROPx25x01x24x006)
NOTAGS
Draw a graph of a function $f(x)$ with no absolute maximum, but such that $|f(x)| < 10$.

---

- (FxPROPx25x01x24x007)
NOTAGS
For $h(x) = 5\sqrt{x}$ what are its absolute extrema?

---

- (FxPROPx25x01x24x008)
NOTAGS
What are the absolute extrema of the graph of $y = x^2 + 1$?

---

- (FxPROPx25x01x24x009)
NOTAGS
What are the absolute extrema of the graph of $y = -3x^2 + 1$?

---

- (TESTxFxPROPx25x02x14x001)
type:TEST
What are the absolute extrema of the graph of $y = (x-2)^2 + 1$? (give the $x$ and $y$ coordinates)

---

- (FxPROPx25x09x15x001)
type:TEST
What are the absolute extrema of the graph of $y = (x+3)^2 - 1$? (give the $x$ and $y$ coordinates)

---

- (TESTxFxPROPx25x02x14x002)
type:TEST
Consider the function $f(x) = |x-3|$.

  a) On what intervals is $f$ increasing? On what intervals is $f$ decreasing?

  b) What are the local max/min of $f$? (give the $x$ and $y$ coordinate, and indicate if it is a max or min).

---

- (GPTx26x02x20x005)
type: TEST
Consider the function $f$ given by the following table.

| $x$ | -4 | -2 | 0 | 3 | 6 |
|---|---|---|---|---|---|
| $f(x)$ | 10 | 8 | 5 | 1 | -2 |

  a) Evaluate $f(0)$.

  b) Evaluate $f(-4)$.

  c) For what $x$ does $f(x) = 1$?

  d) Is the function increasing or decreasing?

---

## 3.5 Basic Function Evaluation *(E)*

- (FxEx25x01x15x001)
NOTAGS
Let $f(x) = 5x - 2$. Evaluate $f(2)$ and $f(0)$.

---

- (FxEx25x01x15x002)
NOTAGS
Let $f(t) = 10 - 5t$. Evaluate $f(2)$ and $f(0)$.

---

- (FxEx25x01x15x003)
NOTAGS
Let $g(x) = 4 + 3x^2$. Evaluate $g(2)$ and $g(-2)$.

---

- (FxEx25x01x15x004)
NOTAGS
Let $h(x) = 5 + 2x^3$. Evaluate $h(2)$ and $h(-2)$.

---

- (FxEx25x02x12x001)
NOTAGS
Let $f(x) = 5x + 2x^2$. Evaluate $f(\frac{3}{2})$ and $f(-\frac{1}{2})$.

---

- (FxEx25x01x15x005)
NOTAGS
Let $h(u) = \dfrac{5+u}{u-5}$. Evaluate $h(0)$ and $h(-1)$.

---

- (FxEx25x01x20x002)
NOTAGS
Let $h(x) = 13 + 5 \cdot (x-2)^2$.

  a) Evaluate $h(0)$.

  b) Evaluate $h(2)$.

---

- (FxEx25x01x15x009)
NOTAGS

  Let $f(x) = 5x + 100$. Put the following values in order without evaluating: $f(20), f(10), f(100)$. Explain.

---

- (FxEx25x01x15x010)
NOTAGS

  Let $f(x) = -5x$. Put the following values in order without evaluating: $f(20), f(10), f(100)$. Explain.

---

- (FxEx25x01x15x011)
NOTAGS

  Let $f(x) = 5x - 10$. For what $x$ is the function value positive?

---

- (FxEx25x01x15x012)
NOTAGS

  Let $f(t) = 10 - 5t$. For what $t$ is the function value negative?

---

- (FxEx25x01x15x013)
NOTAGS

  Let $g(x) = 4 + 3x^2$. Are $g(20242024)$ and $g(-20242024)$ equal? Why?

---

- (FxEx25x01x15x014)
NOTAGS

  Let $h(x) = 5 + 2x^3$. Are $h(20242024)$ and $h(-20242024)$ equal? Why?

---

- (FxEx25x02x05x001)
NOTAGS

Consider the piecewise defined function given by $f(x) = \begin{cases} x & if\ x < -2 \\ 2x - 1 & if\ -2 \leq x \leq 3 \\ -4x & if\ x > 3 \end{cases}$

Find the following values.

a) $f(-5) =$

b) $f(-2) =$

c) $f(0) =$

d) $f(3) =$

e) $f(4) =$

---

- (TESTxFxEx25x02x12x101)
NOTAGS

Let $h(x) = x^4 - 2x^3$. Evaluate $h(2)$ and $h(-2)$.

---

- (FxEx25x09x15x002)
type: TEST

Let $g(x) = 3x^3 - x^4$. Evaluate $g(2)$ and $g(-2)$.

---

- (GPTx26x02x20x003)
type: TEST

Let $g(x) = 2x^3 - x^4$. Evaluate $g(1)$ and $g(-1)$.

---

- (TESTxFxEx25x02x12x002)
NOTAGS

Evaluate $f(0)$, $f(4)$, and $f(5)$ where $f(x) = \begin{cases} x^2 & if\ x < 5 \\ 5 - x & if\ x \geq 5 \end{cases}$

Also, is $f$ a one-to-one function?

---

- (FxEx25x09x15x004)
type: TEST

Evaluate $f(0)$, $f(3)$, and $f(4)$ where $f(x) = \begin{cases} x^2 & if\ x \leq 3 \\ 10 + x & if\ x > 3 \end{cases}$

Also, is $f$ a one-to-one function?

---

- (GROUPxFxEx25x02x07x001)
NOTAGS

The horizon is the line where the earth and sky meet. Your distance from the horizon depends on your height above sea level. The function $f(h) = \sqrt{2h}$ gives the distance to the horizon (in miles), where $h$ is your height (in feet) above sea level.

Suppose your group went to the beach and you are standing right at sea level with your feet in the water. The distance of the horizon from your gaze will depend on your height. For each person in your group, determine how far away the horizon appears to be (note: find your height in feet; for example a kid that is 3 feet 6 inches tall is 3.5 feet tall).

---

## 3.6 Function Evaluation From Table *(ETAB)*

- (FxETABx25x01x23x001)
NOTAGS
Consider the function $f$ given by the following table.

| $x$ | 2 | 4 | 6 | 17 | 15 | 10 |
|---|---|---|---|---|---|---|
| $f(x)$ | $-3$ | 4 | $-13$ | 2 | 0 | 14 |

a) Evaluate $f(17)$

b) Evaluate $f(2)$

c) For what $x$ does $f(x) = -13$

d) Solve $f(x) = 6$.

---

- (FxETABx25x01x23x002)
NOTAGS
Consider the function $f$ given by the following table.

| $x$ | $-4$ | 4 | 0 | 17 | 5 | 10 |
|---|---|---|---|---|---|---|
| $f(x)$ | $-3$ | 4 | $-13$ | 2 | 0 | 0 |

a) Evaluate $f(0)$

b) Solve $f(x) = 0$

c) For what $x$ is $f(x)$ negative?

d) Which $x$ value is a "fixed point" of $f$?

---

- (TESTxFxETABx25x02x12x001)
NOTAGS
Consider the function $f$ given by the following table.

| $x$ | 2 | $-3$ | 6 | 1 | 0 | $-9$ |
|---|---|---|---|---|---|---|
| $f(x)$ | 0 | 1 | $-1$ | 0 | $-3$ | 9 |

a) Evaluate $f(0)$

b) Evaluate $f(-3)$

c) For what $x$ does $f(x) = 0$

---

- (FxETABx25x09x15x001)
type: TEST
Consider the function $f$ given by the following table.

| $x$ | $-3$ | $-1$ | 0 | 1 | 7 | 9 |
|---|---|---|---|---|---|---|
| $f(x)$ | 9 | 7 | 6 | 0 | $-3$ | $-5$ |

a) Evaluate $f(0)$

b) Evaluate $f(-3)$

c) For what $x$ does $f(x) = -1$

d) Is the function increasing or decreasing?

---

## 3.7 Function Evaluation At Variable Input *(EVAR)*

- (FxEVARx25x01x23x002)
NOTAGS
Let $g(x) = 2x - 1$. Evaluate $g(y + 3)$.

---

- (FxEVARx25x01x23x001)
NOTAGS
Let $f(x) = 5 + 2x^3$. Evaluate $f(t)$ and $f(-u)$.

---

- (FxEVARx25x01x23x003)
NOTAGS
Let $h(t) = t^2 - t$. Evaluate $h(-x)$ and $h(x - 1)$.

---

- (FxEVARx25x02x03x003)
NOTAGS
For the function $f(x) = \dfrac{3x^2 - 1}{x^2}$, find (and simplify when possible)

  a) $f(2) =$

  b) $f(-1) =$

  c) $f(r) =$

  d) $f(x - 1) =$

  e) $f(-x) =$

  f) $f(x^3) =$

---

- (FxEVARx25x02x05x001)
NOTAGS
Let $f(x) = x^2 + x - 2$ and $g(x) = \dfrac{x}{x^2 + 3}$. Find (and simplify) the following.

  a) $f(x + 4)$

  b) $f(x^2 - 1)$

  c) $g(x^2)$

  d) $g(u)$

---

- (TESTxFxEVARx25x02x12x001)
NOTAGS
Let $g(t) = 3t^2$. Evaluate $g(-x)$ and $g(x - 2)$.

---

- (FxEVARx25x09x15x001)
type: TEST
Let $h(t) = -4t$. Evaluate $h(-x)$ and $h(x - 2)$.

---

- (GPTx26x02x20x006)
type: TEST
Let $h(t) = 5t$. Evaluate $h(-x)$ and $h(x + 3)$.

---

## 3.8 One Variable as Function of Other Variable *(VAR)*

- (FxVARx25x01x23x001)
NOTAGS
Consider the equation $30x = 10y$. Write $y$ as a function of $x$.

---

- (FxVARx25x01x23x002)
NOTAGS
Consider the equation $6x = 8y$. Write $x$ as a function of $y$.

---

- (FxVARx25x01x23x003)
NOTAGS
Consider the equation $6t = 8y + 4$. Write $y$ as a function of $t$.

---

- (FxVARx25x01x23x004)
NOTAGS
Consider the equation $u = 3u + 8x^3 - 4$. Write $u$ as a function of $x$.

---

## 3.9 Evaluation and Simplifying for Log/Exp *(ELE)*

- (FxELEx25x01x20x010)
NOTAGS
Evaluate the following expressions:

  a) $\log_2(16)$

  b) $\log_2\left(\frac{1}{16}\right)$

---

- (FxELEx25x01x20x011)
NOTAGS
Evaluate the following expressions:

  a) $\log_{2018}(1)$

  b) $\log_{2018}(0)$

---

- (FxELEx25x01x20x012)
NOTAGS
Evaluate the following expressions:

  a) $\log_{17}(17^{899})$

  b) $13^{\log_{13}(255)}$

---

- (FxELEx25x01x20x013)
NOTAGS
Evaluate the following expression (be careful!):
$\log_2\left(4^2\right)$

---

- (FxELEx25x01x20x008)
NOTAGS
Evaluate/simplify or state if undefined (and why):
$\log_{\frac{1}{2}}\left(\frac{1}{2}\right)$

---

- (FxELEx25x01x20x009)
NOTAGS
Simplify.

  a) $\log_{\frac{1}{2}}(8)$

  b) $\log_{38}(1)$

---

- (FxELEx25x01x20x110)
NOTAGS
Simplify.

  a) $\log_{38}(0)$

  b) $\log_x(x^{89})$

---

- (FxELEx25x01x20x111)
NOTAGS

  Simplify. $15^{\log_{15}(3y+7)}$

---

- (FxELEx25x01x20x112)
NOTAGS
Simplify $\log_{\frac{1}{2}}\left(\frac{1}{16}\right)$

---

- (FxELEx25x01x22x006)
NOTAGS
Simplify each expression:

  a) $\ln(e^{3x})$

  b) $\log_2\left(\frac{1}{4}\right)$

  a) $4^{\log_4(x-9)}$

---

- (FxELEx25x02x04x001)
NOTAGS
Simplify each expression:

  a) $\log_3 \frac{1}{3}$

  b) $\log_6 \sqrt{6}$

---

- (FxELEx25x01x20x027)
NOTAGS
Let $f(x) = x^{\frac{5}{3}}$ and $g(x) = x^{\frac{5}{2}}$.
Which is larger, $f(2018)$ or $g(2018)$? Justify without a calculator.

---

- (TESTxxxFxELEx25x04x28x001)
NOTAGS
Evaluate.

a) $\log_{17}(1)$ b) $\log_4(16)$ c) $\log_3(\frac{1}{27})$

---

- (FxELEx25x11x17x001)
type: TEST
Evaluate each expression or state why it is undefined.

a) $\log_{15}(1)$ b) $\log_{15}(-1)$ c) $\log_{19}(\frac{1}{19})$

---

## 3.10 Log Properties *(LP)*

- (FxLPx25x02x04x001)
NOTAGS
Evaluate

a) $\log_{102} 102^4$ b) $\log_3(\log_8 2)$

---

- (FxLPx25x02x04x003)
NOTAGS
Simplify.

a) $\log_a a^{\frac{1}{5}}$ b) $\log_a \sqrt[3]{a}$

---

- (FxLPx25x02x04x004)
NOTAGS
Simplify.

a) $2^{\log_2 7}$ b) $a^{\log_a \frac{1}{5}}$

---

- (FxLPx25x02x04x005)
NOTAGS
Simplify.

a) $10^{\log \sqrt{4}}$ b) $e^{\ln 3x^2}$

---

- (FxLPx25x02x04x006)
NOTAGS
Use the properties of logarithms to expand the following expressions.

a) $\log_9(5y) =$

b) $\log_8 x^7 =$

---

- (FxLPx25x02x04x007)
NOTAGS
Use the properties of logarithms to expand the following expressions.

a) $\log_b(3x^2y^3)$

b) $\log_8 \dfrac{x^{\frac{1}{2}}}{y^3}$

---

- (FxLPx25x02x04x008)
NOTAGS
Use the properties of logarithms to expand the following expressions.

a) $\log_5 \sqrt[5]{\dfrac{x^2}{y}}$

b) $\ln\left[\dfrac{x^4\sqrt{x^2+3}}{(x+3)^5}\right]$

---

- (FxLPx25x02x04x009)
NOTAGS
Use the properties of logarithms to condense the following expressions.

a) $\log x + \log 5 =$

b) $\log_8 x + 3\log_8 y =$

---

- (FxLPx25x02x04x011)
NOTAGS
Use the properties of logarithms to condense the following expressions.

a) $\frac{1}{2}(\log x + \log y) =$

b) $\frac{1}{3}(\log_7 x + 4\log_7 y) - 3\log_7(x+y) =$

---

- (FxLPx25x02x04x012)
NOTAGS
Use the change of base formula to write the following logarithms as logarithms in the indicated base.

a) $\log_7 12$; write it in base 10.

b) $\log_9 127$; write it in base $e$.

---

- (FxLPx25x02x04x013)
NOTAGS
Use the change of base formula to write the following logarithms as logarithms in the indicated base.

  a) $\log_{25} 9$; write it in base 5 and simplify.

  b) $\log_{\frac{1}{2}} 8$; write it in base 2 and simplify.

---

- (TESTxFxLPx25x02x04x002)
NOTAGS
Simplify.

  a) $\log_a a^4$

  b) $\log_a \frac{1}{a^7}$

---

- (TESTxFxLPx25x01x22x010)
NOTAGS

  a) Condense $3\log_{10}(A) + 4\log_{10}(B)$ into a single log.

  b) Use log properties to expand $\log_7\left(\frac{x^5}{y^2}\right)$.

---

- (FxLPx25x11x17x001)
type: TEST

  a) Condense $3\log(a) + \log(c) - \log(a^2)$ into a single log and simplify.

  b) Use log properties to expand $\log_4\left(\frac{x}{yw}\right)$.

---

## 3.11 Function Evaluation for Piece-wise Defined *(PC)*

- (FxPCx25x01x20x014)
NOTAGS
Let

$$f(x) = \begin{cases} x - 2, & x \leq -1, \\ 2x + 1, & x > -1. \end{cases}$$

Evaluate the following.

  a) $f(-2)$

  b) $f(-1)$

  c) $f(0)$

---

- (FxPCx25x01x24x001)
NOTAGS
Let

$$f(x) = \begin{cases} x^2, & x < 5, \\ -x, & x \geq 5. \end{cases}$$

Evaluate the following.

a) $f(65)$

b) $f(-5)$

c) $f(5)$

---

## 3.12 Evaluation (Mixed) *(EM)*

- (FxEMx25x01x22x009)
NOTAGS
Evaluate the following:

a) $5\,i\,(2\,i - 3)$

b) $25^{-\frac{1}{2}}$

a) $\log_5(25)$

---

## 3.13 Function Categorization *(C)*

- (FxCx25x01x20x002)
NOTAGS
For each of the following functions, determine:

  - Whether it is a polynomial function.
  - Whether it is a rational function.

a) $f(u) = \dfrac{4u}{5u^7 - 87}$

b) $h(y) = \sqrt{y^3 + 5y + 1}$

---

- (FxCx25x01x20x003)
NOTAGS
For each of the following functions, determine:
  - Whether it is a polynomial function.
  - Whether it is a rational function.

  a) $g(x) = \frac{x}{7}$

  b) $g(x) = \frac{7}{x}$

---

- (FxCx25x01x22x007)
NOTAGS

  a) Is $f(x) = 13^x$ an exponential function? If so, find its base.

  b) Is $f(x) = 1^x$ an exponential function? If so, find its base.

  c) Is $f(x) = (-\frac{1}{2})^x$ an exponential function? If so, find its base.

---

## 3.14 Basic Function Graphing *(G)*

- (FxGx24x12x28x001)
NOTAGS
Draw a graph which is NOT a function. Why is it not a function?

---

- (FxGx24x12x28x002)
NOTAGS
Draw a graph which IS a function. Why is it a function?

---

- (FxGx24x12x28x102)
NOTAGS
Draw a graph which is a one-to-one function. Why is it a one-to-one function?

---

- (FxGx24x12x28x003)
NOTAGS
Draw a graph which is an injective function. Why is it an injective function?

---

- (FxGx24x12x28x004)
NOTAGS
Draw a graph which is a function but not one-to-one. Why is it a non one-to-one function?

---

- (FxGx24x12x28x005)
NOTAGS
Draw a graph which is a function but not injective. Why is it a non injective function?

---

- (FxGx24x12x28x006)
NOTAGS
Draw the graph of a function whose domain is all reals, and which is decreasing left of 4 and increasing right of 4. Is it possible for such a function to be one-to-one? Explain.

---

## 3.15 Finding Intercepts *(INT)*

- (FxINTx25x01x21x002)
NOTAGS
Consider the graph of $y = x^2+x-6$. Find all its intercepts (both $x$- and $y$-intercepts).

---

## 3.16 Parabola Basics *(PB)*

- (FxGPx25x02x18x001)
NOTAGS
Consider the parabola $f(x) = 2x^2 + 8x + 3$. Find its axis of symmetry.

---

- (FxGPx25x02x18x002)
NOTAGS
Consider the parabola $f(x) = 3x^2 - 6x + 30$. Find its axis of symmetry.

---

- (FxGPx25x02x18x005)
NOTAGS
Write the parabola $h(x) = (x - 2)^2 + 1$ in in general form.

---

- (FxGPx25x02x18x006)
NOTAGS
Write the parabola $f(x) = (x + 4)^2 - 2$ in in general form.

---

- (HWxFxGPx25x02x18x103)
type: HW
Write the parabola $g(x) = x^2 - 12x + 32$ in vertex form.

---

- (FxGPx25x02x18x004)
NOTAGS
Write the parabola $g(x) = x^2 - x$ in vertex form.

---

- (TESTxFxGPx25x02x18x001)
NOTAGS
Write the parabola $f(x) = x^2 + 2x - 3$ in vertex form.

---

- (TESTxFxGPx25x02x18x002)
NOTAGS
Write the parabola $f(x) = (x - 1)^2$ in in general form.

---

## 3.17 Graphing Parabolas *(GP)*

- (HWxFxGPx25x02x18x102)
type: HW
What is the domain and range of the quadratic function $f(x) = (x - 7)^2 + 3$.

---

- (FxGPx25x02x18x003)
NOTAGS
What is the domain and range of the quadratic function $f(x) = (x + 8)^2 - 9$.

---

- (TESTxFxGPx25x02x18x007)
NOTAGS
What is the domain and range of the quadratic function $f(x) = 2x^2 - 6x$.

---

- (TESTxFxGPx25x02x18x006)
NOTAGS
What is the domain and range of the quadratic function $g(x) = 3x^2 - 6x - 9$.

---

- (FxGPx25x01x21x011)
NOTAGS
Consider $f(x) = 2x^2 + 8x + 3$.

a) Does $f$ open up or down?

b) Find its axis of symmetry.

c) Find the vertex.

d) What are the max/min values (if any)?

e) Sketch a rough graph (label the vertex).

---

- (FxGPx25x01x22x001)
NOTAGS
Consider the parabola $y - 9x^2 = x$. Do *not* graph it, but do the following:

  a) Write the equation in standard form.

  b) Find the axis of symmetry.

  c) Find its $x$- and $y$-intercepts.

---

- (FxGPx25x01x22x002)
NOTAGS
Consider the parabola $y + 3x = -x^2$.

  a) Find its intercepts.

  b) Find the vertex.

  c) Find a pair of points symmetric about the axis.

  d) Graph the parabola.

---

- (FxGPx25x02x03x001)
NOTAGS
For the quadratic function $f(x) = (x - 4)^2 - 1$

  a) Find its vertex

  b) Find its $x$- and $y$-intercepts.

  c) Give the equation of the axes of symmetry.

  d) Draw the graph.

  e) Determine its domain and range.

---

- (FxGPx25x02x03x002)
NOTAGS
For the quadratic function $g(x) = 4 - (x-1)^2$

  a) Find its vertex

  b) Find its $x$- and $y$-intercepts.

  c) Give the equation of the axes of symmetry.

  d) Draw the graph.

  e) Determine its domain and range.

---

- (FxGPx25x02x03x003)
NOTAGS
For the quadratic function $h(x) = 3x^2 - 2x - 4$

  a) Find its vertex

  b) Find its $x$- and $y$-intercepts.

  c) Give the equation of the axes of symmetry.

  d) Draw the graph.

  e) Determine its domain and range.

---

- (FxGPx25x01x22x004)
NOTAGS
The number of people ($P$) infected $t$ days after a flu outbreak is modeled by $P(t) = -t^2 + 100t + 20$.

  a) How many days after outbreak is the maximum number sick?

  b) What is that maximum number of people infected at once?

---

- (FxGPx25x01x22x005)
NOTAGS
A patient's body temperature $T$ (in Fahrenheit) $x$ hours after acetaminophen is $T(x) = 0.4x^2 - 2.6x + 103$.

  a) When will the temperature be minimum?

  b) What is that minimum temperature?

---

- (HWxFxGPx25x01x22x003)
type: HW
Consider $y = x^2 + 5x + 6$.

  a) Find all intercepts.

  b) Find the point on the graph with maximum $y$ (or explain if none).

  c) Find the point on the graph with minimum $y$ (or explain if none).

  d) Graph the equation.

---

- (HWxFxGPx25x02x03x004)
type: HW
For the quadratic function $f(x) = 2x - x^2 - 2$

  a) Find its vertex

  b) Find its $x$- and $y$-intercepts.

  c) Give the equation of the axes of symmetry.

  d) Draw the graph.

  e) Determine its domain and range.

---

- (TESTxFxGPx25x02x18x004)
NOTAGS
What is the domain and range of the quadratic function $f(x) = (x - 3)^2 + 1$.

---

- (TESTxFxGPx25x02x18x005)
NOTAGS
What is the domain and range of the quadratic function $g(x) = x^2 + 6x + 4$.

---

- (TESTxxxFxGPx25x03x27x001)
NOTAGS
The amount of money in dollars ($M$) in the school's bank account $t$ days after it opens is modeled by $M(t) = 2t^2 - 20t + 85$.

  a) How many days after the school opens is the bank account's minimum reached?

  b) What is that minimum amount of money in the bank account?

---

## 3.18 Detailed Function Graphing *(GG)*

- (FxGGx24x12x28x001)
NOTAGS
Graph the function, find its asymptotes, and find its domain and range:

$$g(x) = \frac{1}{(x-3)^2}$$

Also, describe its end behavior, and use arrow notation to describe the behavior at the undefined value.

---

- (FxGGx24x12x28x002)
NOTAGS
Graph the function, find its asymptotes, and find its domain and range:

$$g(x) = 2 + \frac{1}{(x-3)^5}$$

Also, describe its end behavior, and use arrow notation to describe the behavior at the undefined value.

---

- (FxGGx25x01x20x014)
NOTAGS
Sketch the graph of the function $g(x) = 1 - x^2$, and indicate the $x$- and $y$-intercepts.

---

- (FxGGx25x01x20x021)
NOTAGS
Sketch the two functions functions on the same coordinate plane.

a) $f(x) = x^2 + 1$ b) $g(x) = -2x^2 - 1$

---

- (FxGGx25x01x20x022)
NOTAGS
Graph the following function: $h(x) = x^2 - x - 56$

---

- (FxGGx25x01x22x015)
NOTAGS
Sketch $f(x) = 3^x$ and $g(x) = \log_3(x)$ on the same axes. Label any intercepts and coordinates.

---

- (FxGGx25x01x20x015)
NOTAGS
Sketch $f(x) = 10^x$ and $g(x) = \log_{10}(x)$ on the same axes (you may scale the $y$-axis).
What are the asymptotes of each graph?
Find any two points that are symmetric across the line $y = x$.

---

- (FxGGx25x01x20x016)
NOTAGS
Sketch $f(x) = 10^x$ and $g(x) = 10^{-x}$ on the same axes.
What are the asymptotes of each graph?
Find any two points that are symmetric across the $y$-axis.

---

- (FxGGx25x02x03x001)
NOTAGS
For the function $f(x) = 2(x-2)^2(x+1)$ , do the following:

a) Determine the end behaviour of the graph.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity. Determine the local behaviour (cross or touch?) at the $x$-intercepts.

d) Make a sketch of the graph.

---

- (FxGGx25x02x03x002)
NOTAGS
For the function $f(x) = -2x^2(x-2)(x+2)^2$ , do the following:

a) Determine the end behaviour of the graph.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity. Determine the local behaviour (cross or touch?) at the $x$-intercepts.

d) Make a sketch of the graph.

---

- (FxGGx25x02x03x003)
NOTAGS
For the function $f(x) = 3x(x+1)^2(x-1)^3$ , do the following:

a) Determine the end behaviour of the graph.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity. Determine the local behaviour (cross or touch?) at the $x$-intercepts.

d) Make a sketch of the graph.

---

- (FxGGx25x02x03x006)
NOTAGS
For the function $f(x) = -3x^3 + 3x^2 - x + 1$ , do the following:

a) Determine the end behaviour of the graph.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity. Determine the local behaviour (cross or touch?) at the $x$-intercepts.

Hint: Use division in factoring.

d) Make a sketch of the graph.

---

- (FxGGx25x02x04x001)
NOTAGS
For the function $f(x) = \dfrac{x+1}{x-1}$, do the following:

a) Determine the end behaviour and find horizontal asymptotes.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity.

d) Find the vertical asymptotes.

e) Make a sketch of the graph.

---

- (FxGGx25x02x04x002)
NOTAGS
For the function $f(x) = \dfrac{3x^2}{x^2-9}$, do the following:

a) Determine the end behaviour and find horizontal asymptotes.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity.

d) Find the vertical asymptotes.

e) Make a sketch of the graph.

---

- (FxGGx25x02x04x003)
NOTAGS
For the function $h(x) = \dfrac{x-4}{x^2-x-6}$, do the following:

a) Determine the end behaviour and find horizontal asymptotes.

b) Find the $y$-intercept.

c) Find the $x$-intercepts with their multiplicity.

d) Find the vertical asymptotes.

e) Make a sketch of the graph.

---

- (FxGGx25x02x04x004)
NOTAGS
For the function $g(x) = \dfrac{2x+5}{x^3-13x+12}$, do the following:
  a) Determine the end behaviour and find horizontal asymptotes.
  b) Find the $y$-intercept.
  c) Find the $x$-intercepts with their multiplicity.
  d) Find the vertical asymptotes.
  e) Make a sketch of the graph.

---

- (FxGGx25x02x04x005)
NOTAGS
Graph the function $f(x) = 2^x$ and $g(x) = \log_2(x)$ on the same axes.

---

- (FxGGx25x02x04x006)
NOTAGS
Graph the function $f(x) = 2^{x+3}$ and $g(x) = \log_2(x) - 3$ on the same axes.

---

- (TESTxxxFxGGx25x04x28x001)
NOTAGS
Graph the following function, drawing its asymptotes as dashed lines (also, give the equation of each asymptote).

$$g(x) = -1 + \frac{1}{(x+2)^3}$$

---

## 3.19 Find Domain *(D)*

- (FxDx24x12x28x001)
NOTAGS
What is the domain of the function $f(x) = 17x^8 - x^6 + 23$?

---

- (FxDx25x01x22x003)
NOTAGS
For $g(u) = \dfrac{u+3}{(u-9)(u+21)}$, for which inputs is $g$ undefined?

---

- (FxDx24x12x28x002)
NOTAGS
What is the domain of the following function?
$f(x) = \dfrac{x^2+25}{4(x-3)(2x+5)}$

---

- (FxDx24x12x28x003)
NOTAGS
When is the following function undefined?
$f(x) = \dfrac{x^2 + 25}{x^2 - x - 20}$

---

- (FxDx25x01x20x004)
NOTAGS
Let $f(x) = \dfrac{(-9 + x)(2 + x)}{2(x - 1)(x + 17)}$.

  a) Evaluate $f(0)$.

  b) When is the function undefined?

---

- (FxDx25x01x20x006)
NOTAGS
Consider the following function: $f(u) = 10 + \sqrt{u}$.

  a) Evaluate $f(4)$.

  b) What is the domain of $f$?

---

- (FxDx25x01x22x004)
NOTAGS
Define $g(u) = \dfrac{-2}{\sqrt{u}}$. Where is $g$ undefined? What is its domain?

---

- (FxDx25x01x20x001)
NOTAGS
Let $f(x) = \dfrac{x(9 + 6x)(x - 5)}{4(x - 113)(3x + 66)}$.

  a) Evaluate $f(0)$.

  b) When is the function undefined?

  c) What is the domain of the function?

  d) Which $x$ values make $f(x) = 0$?

---

- (FxDx25x01x20x009)
NOTAGS

Let $f(x) = \dfrac{(2x^2 + x)(x - 3)}{(x + 5)(x^2 + x - 2)}$

a) Evaluate $f(3)$, $f(0)$, and $f(-1)$.

b) Determine the values of $x$ for which the function is undefined.

c) Determine the values of $x$ for which the function is defined.

d) What is the domain of the function?

e) For which inputs to the function is the output 0?

f) Find the values of $x$ such that $f(x) = 0$.

---

- (FxDx25x01x24x002)
NOTAGS
Find the domain of $f(x) = 43\sqrt{x + 25}$.

---

- (FxDx25x01x24x003)
NOTAGS
Find the interval where the function is undefined: $f(x) = 6\sqrt{5x}$.

---

- (FxDx25x01x20x011)
NOTAGS
Find the domain of $f(x) = 7\sqrt{5x + 10}$.

---

- (FxDx25x01x24x001)
NOTAGS
Find the domain of $f(x) = 7\sqrt[3]{5x + 10}$.

---

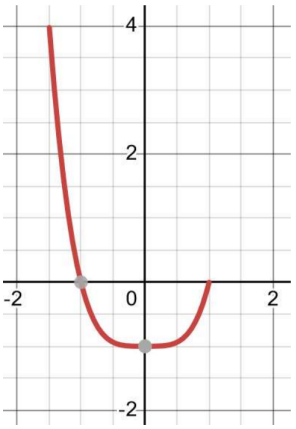
- (FxDx25x01x24x004)
NOTAGS
Find the interval where the function is undefined: $f(x) = \sqrt{4x - 6}$.

---

- (FxDx25x01x20x012)
NOTAGS
Find the domain of $g(x) = \dfrac{x^2 + 7}{2x - 4}$.

---

- (FxDx25x01x20x013)
NOTAGS
Consider the function $F$ graphed below.




a) Evaluate $F(-1)$, $F(0)$, $F(1)$.

b) What is the domain and range of $F$?

---

- (FxDx25x02x05x001)
NOTAGS
Find the domain of $f(x) = \dfrac{1}{x^2 + x - 12}$

---

- (FxDx25x02x05x002)
NOTAGS
Find the domain of $g(x) = \dfrac{3x + 2}{x^2 + 2x - 2}$

---

- (FxDx25x02x05x003)
NOTAGS
Find the domain of $h(x) = \sqrt{3 - x}$

---

- (FxDx25x02x05x004)
NOTAGS
Find the domain of $G(x) = \log_5(3x - 6)$

---

- (FxDx25x01x24x101)
NOTAGS
Let

$$h(x) = \begin{cases} x - 4, & x < 3, \\ 4 - x, & x > 3. \end{cases}$$

What is the domain of $h$?

---

- (FxDx25x01x24x102)
NOTAGS
Let
$$g(x) = \begin{cases} 4, & x \leq 2, \\ \dfrac{3}{x-1}, & x > 2. \end{cases}$$
What is the domain of $g$?

---

- (FxDx25x01x24x104)
NOTAGS
Let
$$f(x) = \begin{cases} \dfrac{3}{x-1}, & x \leq 2, \\ 4, & x > 2. \end{cases}$$
What is the domain of $f$?

---

- (FxDx25x01x22x010)
NOTAGS
Find an example of a rational function whose domain is all real numbers.

---

- (CREATIVExxxFxDx25x04x16x001)
NOTAGS
Suppose a scientific study defines a function $f(a)$, where the input ( $a$ ) is the age of a person, and the output is the average height of a person that age.
Decide what the domain should be, giving reasons for your answer (there is not one clear correct answer ...).

---

## 3.20 Find Zeroes *(Z)*

- (FxZx24x12x28x001)
NOTAGS
What are the zeroes of the following function?
$f(x) = \dfrac{x^2 + 25}{4(x-3)(2x+5)}$

---

- (FxZx25x01x20x005)
NOTAGS
Let $f(x) = \dfrac{(-9+x)(2+x)}{2(x-1)(x+17)}$.
For which inputs is the output 0?

---

- (FxZx25x01x22x102)
NOTAGS
Consider $g(u) = \dfrac{u+3}{(u-9)(u+21)}$.

a) Evaluate $g(0)$.

b) For which inputs is $g(u) = 0$?

---

- (FxZx25x01x22x011)
NOTAGS
Find an example of a rational function whose output is never zero.

---

- (FxZx25x02x18x001)
NOTAGS
For each function, find the zeroes and their multiplicities.

a) $f(x) = (x-3)^2(x+1)^4$

b) $g(x) = (x+4)(x-12)^3(x-1)$

---

- (FxZx25x02x18x002)
NOTAGS
For each function, find the zeroes and their multiplicities.

a) $f(x) = (2x-1)^3(x+7)$

b) $h(x) = (4x+5)^2(7x)^5$

---

- (FxZx25x02x19x001)
NOTAGS
Find the roots and their multiplicities for the function $f(x) = 5x^5 - 45x^3$

---

- (TESTxFxZx25x02x18x001)
NOTAGS
For each function, find the zeroes and their multiplicities.

a) $f(x) = (x-7)^6(x+5)^9$

b) $h(x) = (4x+12)^2(x^7)$

---

## 3.21 Mixed: Domain and Zeroes *(DZ)*

- (TESTxFxDZx25x02x13x001)
NOTAGS
Let $f(x) = \dfrac{x^2 - 3x - 4}{-2(x+10)(5x-10)}$

  a) When is the function undefined?

  b) What is the domain of the function?

  c) Evaluate $f(0)$.

  d) Which $x$ values make $f(x) = 0$?

---

- (FxDZx25x09x15x001)
type: TEST
Let $f(x) = \dfrac{x^2 - 25}{-2(x+3)(4x-12)}$

  a) When is the function undefined?

  b) What is the domain of the function?

  c) Evaluate $f(0)$.

  d) Which $x$ values make $f(x) = 0$?

---

- (GPTx26x02x20x007)
type: TEST
Let $f(x) = \dfrac{x^2 - 16}{(x-4)(x+2)}$.

  a) When is the function undefined?

  b) What is the domain?

  c) Evaluate $f(0)$.

  d) For which $x$ does $f(x) = 0$?

---

- (TESTxxxFxDZx25x04x28x001)
NOTAGS
Let $f(x) = \dfrac{x^2+7}{3x^2-2x-1}$.

  a) What is the domain?

  b) What are the vertical asymptotes?

  c) What is the horizontal asymptote?

---

- (TESTxxxFxDZx26x05x01x004)
NOTAGS
Let $f(x) = \dfrac{4x^2+7}{x^2-x-12}$.

  a) What is the domain?

  b) What are the vertical asymptotes?

  c) What is the horizontal asymptote?

---

## 3.22 End Behavior and Basic Limits *(LIM)*

- (FxLIMx25x02x18x001)
NOTAGS
Determine the end behavior of the functions.

  a) $g(x) = 3x^5$ b) $f(x) = -7x^5$

---

- (FxLIMx25x02x18x002)
NOTAGS
Determine the end behavior of the functions.

  a) $g(x) = 9x^4$ b) $f(x) = -5x^6$

---

- (FxLIMx25x02x18x004)
NOTAGS

Determine $\lim_{x\to\infty} f(x)$ for each function.

  a) $f(x) = 9x^4$ b) $f(x) = 15x^3$

---

- (FxLIMx25x02x18x005)
NOTAGS

Determine $\lim_{x\to-\infty} g(x)$ for each function.

a) $g(x) = 3x^5$ b) $g(x) = 4x^6$

---

- (FxLIMx25x01x22x001)
NOTAGS
Determine the end behavior of the function $g(x) = 3x^5 + 7x^2 - x + 12$.

---

- (FxLIMx25x02x19x002)
NOTAGS
Determine the end behavior of the function $f(x) = 3x^3 + 7x^4 - 17x$.

---

- (FxLIMx25x02x19x004)
NOTAGS
Determine the end behavior of the function $h(x) = 3x^7 + 7x^3 - 10x^7$.

---

- (HWxFxLIMx25x02x18x003)
type: HW
Determine the end behavior of the functions.

a) $g(x) = -90x^2$ b) $f(x) = 5x^3$

---

- (HWxFxLIMx25x02x19x003)
type: HW
Determine the end behavior of the function $f(x) = 3x^3 + 7x^4 - 17x$.

---

- (TESTxFxLIMx25x02x19x007)
type: TEST
Determine the end behavior of the function $f(x) = 3x^5 + 7x^2 - 17x^3$.

---

- (TESTxFxLIMx25x02x18x001)
type: TEST
Determine the end behavior of the functions.

a) $h(x) = 10x^3$ b) $f(x) = -5x^4$

---

- (TESTxFxLIMx26x03x22x001)
type: TEST
Determine the end behavior of the functions.

a) $g(x) = -3x^6$ b) $h(x) = 12x^7$

---

- (TESTxFxLIMx26x05x01x001)
type: TEST
Determine the end behavior of the functions.

  a) $g(x) = -3x^8$

  b) $h(x) = 4x^3 + 10x^7$

---

- (TESTxFxLIMx25x02x18x002)
NOTAGS

  Determine $\lim_{x\to\infty} f(x)$ for each function.

  a) $f(x) = 9x^3$

  b) $f(x) = -x^5$

---

## 3.23 Arithmetic On Functions *(AR)*

- (FxARx25x01x20x010)
NOTAGS
Let $f(x) = x^3$ and $g(x) = x - 1$.

  a) Find $(f + g)(-3)$

  b) Find $(f - g)(-3)$

  c) Find $(f * g)(-3)$

  d) Find $\left(\frac{f}{g}\right)(-3)$

---

- (FxARx25x01x24x001)
NOTAGS
Let $f(x) = x^3$ and $g(x) = x - 1$. For each of the following, find the simplified and expanded form.

  a) $(f + g)$

  b) $(f - g)$

  c) $(f * g)$

---

- (FxARx25x01x24x002)
NOTAGS
Let $f(x) = \dfrac{1}{x^3}$ and $g(x) = \sqrt{x+1}$. Find the domain of each of the following.

  a) $(f + g)$

  b) $(f - g)$

  c) $(f * g)$

  d) $\left(\frac{f}{g}\right)$

---

- (FxARx25x01x24x003)
NOTAGS
Find an example of function $f$ and $g$ so that $\left(\frac{f}{g}\right)$ is defined everywhere.

---

- (TESTxFxARx25x02x14x001)
NOTAGS
Let $f(x) = \frac{2}{x-5}$ and $g(x) = \sqrt{2x-4}$. Find the domain of each of the following.

  a) $(f+g)$

  b) $(f-g)$

  c) $(f*g)$

---

- (FxARx25x09x15x001)
type: TEST
Let $f(x) = \frac{2}{3x-5}$ and $g(x) = \sqrt{5x-20}$. Find the domain of each of the following.

  a) $(f+g)$ b) $(f-g)$ c) $(f*g)$

---

- (TESTxFxARx25x02x14x002)
NOTAGS
Let $f(x) = x^2 - 1$ and $g(x) = \frac{2}{x}$. Find the domain of each of the following.

  a) $\frac{f}{g}$ b) $\frac{g}{f}$

---

- (GPTx26x02x20x009)
type: TEST
Let $f(x) = x^2 - 9$ and $g(x) = \frac{2}{x-2}$. Find the domain of:

  a) $\frac{f}{g}$ b) $\frac{g}{f}$

---

- (FxARx25x09x15x002)
type: TEST
Let $f(x) = x^2 - 4$ and $g(x) = \frac{3}{x+1}$. Find the domain of each of the following.

  a) $\frac{f}{g}$ b) $\frac{g}{f}$

---

- (GPTx26x02x20x008)
type: TEST
Let $f(x) = \dfrac{3}{x-1}$ and $g(x) = \sqrt{2x-6}$. Find the domain of:

a) $(f+g)$ b) $(f-g)$ c) $(f \cdot g)$

---

## 3.24 Composition Of Functions *(CP)*

- (FxCPx25x01x24x001)
NOTAGS
If the order is reversed when composing two functions, can the result ever be the same as the answer in the original order of the composition? If yes, give an example. If no, explain why not.

---

- (FxCPx25x01x20x012)
NOTAGS
Let $f(x) = 2x$ and $g(x) = x + 10$.

a) Find $(f \circ g)(5)$

b) Find $g(f(5))$

c) Find $f(f(5))$

d) Find $(g \circ g)(5)$

---

- (FxCPx25x02x05x001)
NOTAGS
Given $f(x) = 2x + 1$ and $g(x) = x^2 + 3$, find and simplify the following.

a) $(f \circ g)(3)$

b) $(f \circ g)(x)$

c) $(g \circ f)(x)$

d) $(f \circ g \circ f)(x)$

---

- (GPTx26x02x20x010)
type: TEST
Let $f(x) = 3x + 2$ and $g(x) = 4x - 5$.

a) Find $(f \circ g)(2)$.

b) Simplify $f(g(x))$.

c) Simplify $(g \circ f)(x)$.

---

- (FxCPx25x10x27x001)
type: TEST
Let $f(x) = -9 - 3x$ and $g(x) = 2 + x^3$.

  a) Evaluate $(g \circ f)(-2)$

  b) Simplify $(f \circ g)(x)$

---

- (FxCPx25x11x05x004)
type: TEST
Let $f(x) = 5 - 2x$ and $g(x) = 2 - x^2$.

  a) Evaluate $(f \circ g)(-5)$

  b) Simplify $(g \circ f)(x)$

---

- (FxCPx25x02x05x002)
NOTAGS
Given $f(x) = \dfrac{1}{x}$ and $g(x) = \dfrac{2}{x-1}$, find and simplify the following.

  a) $(f \circ g)(3)$

  b) $(f \circ g)(x)$

  c) $(g \circ f)(x)$

  d) $(f \circ g \circ f)(x)$

---

- (FxCPx25x01x24x002)
NOTAGS
Let $f(x) = \dfrac{4}{x}$ and $h(x) = \dfrac{1}{x+1}$. Find $f(h(x))$ (simplify and expand).

---

- (FxCPx25x01x24x003)
NOTAGS
Let $g(u) = \sqrt{u}$ and $h(x) = x^4$. Find $h(g(u))$ (simplify and expand).

---

- (FxCPx25x01x24x004)
NOTAGS
Let $f(x) = \dfrac{1}{x-4}$ and $g(x) = \dfrac{2}{x} + 4$. Find $f(g(x))$ and $g(f(x))$.

---

- (FxCPx25x01x20x016)
NOTAGS
Let $f(x) = 5x^2 + 1$ and $g(x) = \sqrt{x+1}$.

  a) Find $f(g(x))$ (simplify).

  b) Find the domain of $f \circ g$.

---

- (FxCPx25x01x24x006)
NOTAGS
For $f(x) = \sqrt{x}$ and $g(x) = x^2$, write the domain of $f \circ g$ in interval notation. Also, find the domain of $g \circ f$.

---

- (FxCPx25x01x24x005)
NOTAGS
For $f(x) = \dfrac{1}{x}$ and $g(x) = \sqrt{x-1}$, write the domain of $f \circ g$ in interval notation.

---

- (FxCPx25x01x20x017)
NOTAGS
Let $H(x) = (2x+1)^5$. Find functions $f$ and $g$ so that $H = f \circ g$.

---

- (FxCPx25x01x24x019)
NOTAGS
Let $H(x) = \dfrac{1}{3x^2+1}$. Find functions $f$ and $g$ so that $H = f \circ g$.

---

- (FxCPx25x01x24x018)
NOTAGS
Let $H(x) = \dfrac{1}{(2x+1)^2}$. Find functions $f$ and $g$ so that $H = f \circ g$.

---

- (FxCPx25x02x05x101)
NOTAGS
Let $f(x) = x^2 + x - 2$ and $g(x) = \dfrac{x}{x^2+3}$. Find (and simplify) the following.

a) $f(g(x))$ b) $g(f(x))$

---

- (TESTxFxCPx25x02x14x001)
NOTAGS
Let $f(x) = 3x$ and $g(x) = x^2 - 2$.

a) Find $(f \circ g)(3)$ b) Find $(g \circ f)(3)$

---

- (TESTxFxCPx25x02x14x002)
NOTAGS
Let $f(x) = x^2$ and $g(x) = x + 3$. Simplify the following.

a) $f(g(x))$ b) $g(f(x))$

---

- (FxCPx25x09x15x001)
type: TEST
Let $f(x) = 2x - 1$ and $g(x) = 5x + 3$.

a) Find $(f \circ g)(-1)$ b) Simplify $f(g(x))$ c) Simplify $(g \circ f)(x)$

---

- (TESTxFxCPx25x02x14x003)
NOTAGS
Let $f(x) = x - 1$ and $g(x) = \dfrac{1}{x+4}$. Answer the following using interval notation.

  a) Find the domain of $f \circ g$.

  b) Find the domain of $g \circ f$.

---

- (FxCPx25x09x15x003)
type: TEST
Let $f(x) = \sqrt{x}$ and $g(x) = \dfrac{4}{x^2}$. Answer the following using interval notation.

  a) Find the domain of $f \circ g$.

  b) Find the domain of $g \circ f$.

---

## 3.25 Inverses *(IN)*

- (FxINx26x01x30x001)
NOTAGS
For each function, is it invertible or not? (graph and use the horizontal line test)

  a) $f(x) = 31x + 1$

  b) $g(x) = x^4 - 8$

  c) $h(t) = \dfrac{1}{(t-3)^2}$

---

- (FxINx24x12x28x001)
NOTAGS
Consider the function $f(x) = 5x - 1$. And consider its inverse function $f^{-1}$.

  a) Before finding the full inverse function, just calculate $f^{-1}(-1)$ and $f^{-1}(0)$.

  b) Check algebraically if $g(x) = (1/5)x + 1$ is the inverse function.

  c) If $g$ is NOT the inverse, then find the inverse algebraically.

---

- (FxINx25x01x21x001)
NOTAGS
Consider the function $f$ given by the following table.

| $x$ | 2 | 4 | 6 | 10 | 15 | 17 |
|---|---|---|---|---|---|---|
| $f(x)$ | 3 | 4 | 13 | 2 | 6 | 14 |

Use the table to evaluate each.

a) $f(2)$ b) $f(10)$

a) $f^{-1}(14)$ b) $f^{-1}(4)$

a) $f^{-1}(2)$

---

- (FxINx25x01x21x003)
NOTAGS
Suppose $f(x) = 4x - 3$. Find the inverse $f^{-1}(x)$ using algebra.

---

- (FxINx25x01x21x004)
NOTAGS
For each pair of functions, check if they are inverses by verifying $f(g(x)) = x$ and $g(f(x)) = x$.

a) $f(x) = 2x,\ g(x) = x + 2$ b) $f(x) = 7 - x,\ g(x) = 7 - x$

---

- (FxINx25x02x05x001)
NOTAGS
Show that $f(x) = 4x - 7$ and $g(x) = \dfrac{x+7}{4}$ are inverses.

---

- (FxINx25x02x05x002)
NOTAGS
Show that $f(x) = \dfrac{2}{x-5}$ and $g(x) = \dfrac{2}{x} + 5$ are inverses.

---

- (FxINx25x02x05x003)
NOTAGS
Show that $f(x) = -3x + 1$ and $g(x) = \dfrac{x-1}{-3}$ are inverses.

---

- (FxINx25x02x05x004)
NOTAGS
CORRECT??
Show that $f(x) = \dfrac{x-2}{2x+1}$ and $g(x) = \dfrac{-x-2}{2x-1}$ are inverses.

---

- (FxINx25x02x05x005)
NOTAGS
Find the inverse of the function $f(x) = 2x - 1$.

---

- (FxINx25x02x05x006)
NOTAGS
Find the inverse of the function $g(x) = \dfrac{1}{x} + 1$.

---

- (FxINx25x02x05x007)
NOTAGS
Find the inverse of the function $h(x) = x^2 - 4$, with domain $(-\infty, 0]$ (so $x \leq 0$).

---

- (FxINx25x02x05x008)
NOTAGS
Find the inverse of the function $f(x) = \dfrac{x-1}{x+1}$.

---

- (TESTxxxFxINx25x03x28x001)
NOTAGS
For each pair of functions, check if they are inverses by checking if both $f(g(x)) = x$ and $g(f(x)) = x$.

a) $f(x) = \dfrac{x}{2} + 1,\ g(x) = 2(x-1)$ b) $f(x) = 9x + 1,\ g(x) = -9x - 1$

---

- (TESTxxxFxINx26x03x22x002)
type: TEST
For each pair of functions, check if they are inverses by checking if both $f(g(x)) = x$ and $g(f(x)) = x$.

a) $f(x) = x + 1,\ g(x) = -x - 1$ b) $f(x) = \dfrac{x}{4} - 1,\ g(x) = 4(x+1)$

---

- (TESTxxxFxINx25x02x26x001)
NOTAGS
Show that $f(x) = \dfrac{-3}{x+1}$ and $g(x) = \dfrac{-3}{x} - 1$ are inverses.

---

- (TESTxxxFxINx25x03x26x002)
NOTAGS
Find the inverse of the function $f(x) = \dfrac{2x-5}{3}$.

---

- (TESTxxxFxINx26x03x22x003)
type: TEST
Find the inverse of the function $g(x) = \dfrac{4-2x}{8}$

---

## 3.26 Graph Transformations *(TR)*

- (FxTRx25x02x05x001)
NOTAGS
How does $f(x+21)-8$ transform the graph of $f(x)$?

---

- (FxTRx25x02x05x002)
NOTAGS
How does $g(x-1)-42$ transform the graph of $g(x)$?

---

- (FxTRx25x02x05x003)
NOTAGS
How does $g(25x)$ transform the graph of $g(x)$?

---

- (FxTRx25x02x05x004)
NOTAGS
How does $g(\frac{x}{3})$ transform the graph of $g(x)$?

---

- (FxTRx25x02x05x005)
NOTAGS
Write a formula for the function obtained when the graph of $f(x)=x^3$ is shifted up 1 unit and to the left 2 units.

---

- (FxTRx25x02x05x006)
NOTAGS
Write a formula for the function obtained when the graph of $f(x)=|x|$ is shifted up 3 units and to the right 7 units.

---

- (FxTRx25x02x05x007)
NOTAGS
Write a formula for the function obtained when the graph of $g(x)=\dfrac{1}{x}$ is shifted down 4 units and to the right 5 units.
Write the resulting function as a single rational expression!

---

- (FxTRx25x01x21x010)
NOTAGS
Graph $f(x)=x^2$. Then:

  a) Let $g(x)=(x+3)^2-1$. Describe in words the transformation from graph $f$ to graph $g$.

  b) Let $h(x)=-x^2$. Describe in words the transformation from graph $f$ to graph $h$.

---

- (FxTRx25x02x05x008)
NOTAGS
Determine where the function $f(x) = (x-4)^2$ is increasing and where it is decreasing. Use graph transformations.

---

- (GPTx26x02x20x002)
type: TEST
What are the absolute extrema of the graph of $y = (x-4)^2 + 3$? (Give the $x$ and $y$ coordinates.)

---

- (FxTRx25x02x05x009)
NOTAGS
Determine where the function $g(x) = -3 + (x+42)^3$ is increasing and where it is decreasing. Use graph transformations (this problems is easy!).

---

- (FxTRx25x02x05x010)
NOTAGS
Determine where the function $g(x) = \dfrac{1}{x-20} - 30$ is increasing and where it is decreasing. Use graph transformations.

---

- (FxTRx25x02x05x011)
NOTAGS
Consider $f(x) = x^2 + 5$. On one axis graph the basic function it is transformed from. Then on another axis, graph $f(x)$ using graph transformations.

---

- (FxTRx25x02x05x012)
NOTAGS
Consider $g(x) = |x+3| - 5$. On one axis graph the basic function it is transformed from. Then on another axis, graph $g(x)$ using graph transformations.

---

- (FxTRx25x02x05x013)
NOTAGS
Consider $g(x) = 2 + \dfrac{1}{3x}$. On one axis graph the basic function it is transformed from. Then on another axis, graph $g(x)$ using graph transformations.

---

- (FxTRx25x02x05x014)
NOTAGS
Consider $h(x) = 4 \cdot \sqrt{x-2}$. On one axis graph the basic function it is transformed from. Then on another axis, graph $h(x)$ using graph transformations.

---

- (FxTRx25x02x05x015)
NOTAGS
Consider $h(x) = 2^{x-2} - 2$. On one axis graph the basic function it is transformed from. Then on another axis, graph $h(x)$ using graph transformations.

---

- (FxTRx25x02x05x016)
NOTAGS
Consider $f(x) = -3x^2 - 4$. On one axis graph the basic function it is transformed from. Then on another axis, graph $f(x)$ using graph transformations.

---

- (FxTRx25x02x05x017)
NOTAGS
Consider $f(x) = 4^{-x}$. On one axis graph the basic function it is transformed from. Then on another axis, graph $f(x)$ using graph transformations.

---

- (TESTxFxTRx25x02x14x001)
NOTAGS
How does $f(x-4) - 8$ transform the graph of $f(x)$?

---

- (TESTxFxTRx25x02x14x002)
NOTAGS
Write a formula for the function obtained when the graph of $f(x) = x^4$ is shifted down 3 units and to the left 5 units.

---

- (FxTRx25x09x15x002)
type: TEST
Write a formula for the function obtained when the graph of $f(x) = \sqrt{x}$ is shifted up 2 units and to the right 3 units.

---

- (TESTxFxTRx25x02x14x003)
NOTAGS
Suppose $g(x)$ is function defined everywhere. Suppose $h(x) = 4\ g(x)$.

  a) How is the graph of $h$ transformed from the graph of $g(x)$?

  b) If $g(-2) = 3$, then what is $h(-2)$?

---

- (TESTxFxTRx25x02x14x004)
NOTAGS
Let $h(x) = (x-3)^2 + 1$.

  a) On one axis, graph the basic function that $h(x)$ is transformed from (right below here).

  b) Then on another axis, graph $h(x)$ using graph transformations (right below here).

---

- (FxTRx25x02x14x004)
type: TEST
In your graphs, clearly label the axes so that the transformation from $f$ to $h$ is clear.

  a) Right below here graph the function $f(x) = x^2$

  b) Right below here graph the function $h(x) = (x-1)^2 + 2$

---

- (GPTx26x02x20x011)
type: TEST
Let $h(x) = (x+2)^2 - 4$.

  a) Graph the basic function from which $h(x)$ is transformed.

  b) Graph $h(x)$ using transformations.

---

- (GPTx26x02x20x013)
type: TEST
Write a formula for the function obtained when the graph of $f(x) = \sqrt{x}$ is shifted down 1 unit and left 4 units.

---

- (GPTx26x02x20x014)
type: TEST
Consider the function $f(x) = -(x-2)^2$.

  a) Where is it increasing?

  b) Where is it decreasing?

  c) Find the absolute extrema (if it does not have a max or min, state this).

  d) Find the local extrema (if it does not have a max or min, state this).

---

- (GROUPxFxTRx25x02x10x001)
NOTAGS
Graph $f(x) = \sin(x)$ in Desmos. Use Desmos (and your brain!) to do the following:

  a) Add an additional plot of $g(x) = 5\sin(x)$. Descibe how this transforms $f(x)$ and make a rough sketch of $f$ and $g$ on the same axes.

  b) Add an additional plot of $h(x) = \frac{1}{5}\sin(x)$. Descibe how this transforms $f(x)$ and make a rough sketch of $f$ and $h$ on the same axes.

  c) Add an additional plot of $A(x) = \sin(5x)$. Descibe how this transforms $f(x)$ and make a rough sketch of $f$ and $A$ on the same axes.

  d) Add an additional plot of $B(x) = \sin(\frac{1}{5}x)$. Descibe how this transforms $f(x)$ and make a rough sketch of $f$ and $B$ on the same axes.

---

## 3.27 Function Applications *(APP)*

- (FxAPPx25x01x23x001)
NOTAGS
Let $D(t)$ be your distance (in miles) from BCC $t$ hours after noon on January 1, 2025. Explain the meaning of each statement:

  a) $D(3) = 10$

  b) $D(1) = 0$

  c) $D(0) = 2$

---

## 3.28 Functions *(MIXED)*

- (FxMIXEDx25x01x21x008)
NOTAGS
Consider the function $f(x) = 2 + \sqrt{x}$. Do the following.

  a) Graph it.

  b) Find domain and range.

  c) Is it one-to-one?

  d) Does it have an inverse?

  e) If it has an inverse, find the domain and range of that inverse.

---

- (FxMIXEDx25x01x21x009)
NOTAGS
Consider the function $g(x) = |x + 1|$ Do the following.

  a) Graph it.

  b) Find domain and range.

  c) Is it one-to-one?

  d) Does it have an inverse?

  e) If it has an inverse, find the domain and range of that inverse.

---

- (FxMIXEDx25x01x22x005)
NOTAGS
Find the domain and zeroes of $g(x) = \frac{4x}{x^2-100}$.

---

- (FxMIXEDx25x01x22x008)
NOTAGS
For $f(x) = 3+2^{x+1}$, give the horizontal asymptote, domain, range, end behavior, and sketch its graph.

---

- (FxMIXEDx25x01x22x011)
NOTAGS
Consider $f(x) = \log_3(x-2)$. Find its domain, range, and vertical asymptote. Sketch the graph.

---

- (TESTxxxFxMIXEDx25x04x28x001)
NOTAGS
For each function, find its domain and range.

a) $h(x) = 3 + \sqrt{7x-1}$ b) $f(x) = -6 + \ln(5x+2)$

---

## 3.29 Functions with Domain/coDomain

- (Fx25x10x12x001)
type: HW , topics: FUNCT
Let $A = \{0, 2, 4, 6, 8\}$ and $B = \{0, 1, 2, 3, 4, 5, 6\}$.
Let the function $f : A \to B$ be defined by $f(n) = n/2$.

a) What are the domain and codomain of $f$?

b) Is $f$ injective?

c) What is the range of $f$?

d) Why is $f$ not surjective? Is there any surjective function from $A$ to $B$?

---

- (Fx25x10x12x002)
type: HW , topics: FUNCT
Let $A = \{0, 2, 4, 6, 8\}$ and $B = \{3, 4, 5, 6\}$.

a) Let the function $h : A \to B$ be defined by:

$$h(0) = 6, \quad h(2) = 5, \quad h(4) = 4, \quad h(6) = 3, \quad h(8) = 6.$$

Is $h$ surjective?

b) Why is $h$ not injective? Is there any injective function from $A$ to $B$?

---

- (Fx25x10x12x003)
type: HW , topics: FUNCT
Let $X = \{u \in \mathcal{Z} \mid 1 \leq u \leq 100\}$ and $Y = \{u \in \mathcal{Z} \mid 10 \leq u \leq 109\}$. Let the function $h : X \to Y$ be defined by $h(n) = n + 9$. Prove that $h$ is bijective. Why does $h$ have an inverse? What is $h^{-1}(109)$?

---

- (Fx25x11x05x002)
type: TEST , topics: FUNCT
Let $X = \{u \in \mathcal{Z} \mid 1 \leq u \leq 100\}$ and $Y = \{u \in \mathcal{Z} \mid -100 \leq u \leq -1\}$. Let the function $h : X \to Y$ be defined by $h(n) = -101 + n$. Prove that $h$ is bijective. Why does $h$ have an inverse? What is $h^{-1}(-50)$?

---

- (Fx25x10x12x004)
type: HW , topics: FUNCT
Let $X = \{u \in \mathcal{Z} \mid 1 \leq u \leq 10\}$. Let the function $h : X \to X$ be defined by $h(n) = \begin{cases} n+1 \text{ if } n \text{ is odd} \\ n/2 \text{ if } n \text{ is even} \end{cases}$

  a) Describe the range of $h$. Is $h$ onto?

  b) Use the same definition for $h$, but now let $X = \{u \in Z \mid 1 \leq u \leq 100\}$. Describe the new range.

---

- (Fx25x10x27x004)
type: TEST , topics: FUNCT
Let $X = \{u \in \mathcal{Z} \mid 1 \leq u \leq 100\}$. Let the function $f : X \to X$ be defined by

$$f(n) = \begin{cases} n \text{ if } n \text{ is odd} \\ n-1 \text{ if } n \text{ is even} \end{cases}$$

  a) Is $f$ injective?

  b) Describe the range of $f$. Is $f$ onto?

  c) Does $f$ have an inverse function?

---

- (Fx25x12x11x001)
type: TEST , topics: FUNCT
Let $X = \{u \in \mathcal{Z} \mid 1 \leq u \leq 10\}$ and let $Y = \{v \in \mathcal{Z} \mid 0 \leq v \leq 20\}$.
Define $f : X \to Y$ by $f(x) = 20 - 2x$.

  a) Is $f$ injective?

  b) Describe the range of $f$. Is $f$ onto?

  c) Is $f$ bijective?

  d) Does $f$ have an inverse function?

---

- (Fx25x10x12x005)
type: HW , topics: FUNCT
For each function $Z \times Z \to Z$, determine if it is i) onto?, ii) one-to-one?, iii) bijective?

a) $f(a, b) = a - b$ b) $g(x, y) = x^2 + x^4$

---

- (Fx25x11x05x006)
type: TEST , topics: FUNCT
For each function $Z \times Z \to Z$, determine if it is i) onto?, ii) one-to-one?, iii) bijective?

a) $f(a, b) = a^6 + b^4 - 2$ b) $g(x, y) = x^3 + y$

---

- (Fx25x10x27x001)
type: TEST
Consider the function $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ given by the following table.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | 2 | 4 | 3 | 5 | 1 |

Use the table to evaluate each (for evaluations, just give the answer).

a) $f(4)$ c) $f^{-1}(4)$

b) $f(1)$ d) $f^{-1}(3)$

Is $f$ injective? Is $f$ surjective? Is $f$ bijective?

---

- (Fx25x11x05x003)
type: TEST
Consider the function $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ given by the following table.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | 2 | 3 | 4 | 5 | 1 |

Use the table to evaluate each (for evaluations, just give the answer).

a) $f(4)$ c) $f^{-1}(4)$

b) $f(1)$ d) $f^{-1}(3)$

Is $f$ injective? Is $f$ surjective? Is $f$ bijective?

---

# 4 Algebra *(A)*

## 4.1 Translation *(T)*

- (AxTx25x01x02x001)
NOTAGS
Express the following WITHOUT calculating.

  a) 2 more than 8

  b) 7 less then 100

---

- (AxTx25x01x02x002)
NOTAGS
Express the following WITHOUT calculating.

  a) The sum of 71 and 87

  b) 20 more than $x$

---

- (AxTx25x01x02x003)
NOTAGS
Express the following WITHOUT calculating.

  a) The product of 34 and 16

  b) The product of 32 and an unknown

---

- (AxTx25x01x02x004)
NOTAGS
Express the following WITHOUT calculating.

  a) The product of two unknowns.

  b) The quotient of 292 and 13

---

- (AxTx25x01x02x005)
NOTAGS
Express the following WITHOUT calculating.

  a) The difference of 34 and 892

  b) The difference between 45 and $x$

---

- (AxTx25x01x02x006)
NOTAGS
Express the following WITHOUT calculating.

  a) The sum of an unknown and 100

  b) The sum of $a$ and $b$

---

- (AxTx25x01x02x007)
NOTAGS
Express the following WITHOUT calculating.

  a) The sum of two unknowns  b) 15 less than 200

---

- (AxTx25x01x02x008)
NOTAGS
Express the following WITHOUT calculating.

  a) 1 less than $y$  b) 13 less than an unknown quantity

---

- (AxTx25x01x02x009)
NOTAGS
Express the following WITHOUT calculating.

  a) 16 less than an unknown quantity  b) Double 17.

---

- (AxTx25x01x02x010)
NOTAGS
Express the following WITHOUT calculating.

  a) Double an unknown  b) 8 times an unknown

---

- (AxTx25x01x02x011)
NOTAGS
Express the following WITHOUT calculating.

  a) A third of 200  b) A third of $x$

---

- (AxTx25x01x02x012)
NOTAGS
Express the following WITHOUT calculating.

  a) The difference between 89 and 5  b) The difference between $x$ and $y$

---

- (AxTx25x01x02x013)
NOTAGS
Express the following WITHOUT calculating.

  a) The quotient of $A$ and 2  b) The quotient of an unknown and 7

---

## 4.2 Substitution *(S)*

### 4.2.1 Substitution With Integers *(I)*

- (AxSxIx25x01x01x001)
NOTAGS
Evaluate

  a) $x + 3$ if $x = 5$ b) $x + 3$ if $x = 10$

---

- (AxSxIx25x01x01x002)
NOTAGS
Evaluate

  a) $a + 6b$ if $a = 4,\ b = 8$ b) $a + 6b$ if $a = -7,\ b = -2$

---

- (AxSxIx25x01x01x003)
NOTAGS
Evaluate

  a) $4xy$ if $x = 4,\ y = -3$ b) $x^2 - y^2 + 3$ if $x = 2,\ y = 3$

---

- (AxSxIx25x01x01x004)
NOTAGS
Evaluate

  a) $x^2 - y^2 + 3$ if $x = -2,\ y = 1$ b) $x^2 - y^2 + 3$ if $x = -1,\ y = -3$

---

- (AxSxIx25x02x12x001)
NOTAGS
Evaluate $x^2 - 2y^2 + 14z^3 + 3$ if $x = -3,\ y = -1,\ z = 0$

---

### 4.2.2 Substitution word problems (with integers) *(WI)*

- (AxSxWIx25x01x01x001)
NOTAGS
**Triangle Area Formula:** Area $= \frac{1}{2} \times B \times H$
Find the area of a triangle with base 4 and height 7.

---

- (AxSxWIx25x01x01x002)
NOTAGS
**Triangle Area Formula:** Area $= \frac{1}{2} \times B \times H$
Find the area of a triangle of height 10 and base 3.

---

- (AxSxWIx25x01x01x003)
NOTAGS
**Rectangle Formulas (where L is the length and W the width):**

$$\text{Area} = L \times W \quad , \quad \text{Perimeter} = 2L + 2W$$

**Problem:**

a) Find the *area* of a rectangle with width 5 and length 7.

b) Find the *perimeter* of a rectangle with width 5 and length 7.

---

- (AxSxWIx25x01x01x006)
NOTAGS
**Rectangle Formulas (where L is the length and W the width):**

$$\text{Area} = L \times W \quad , \quad \text{Perimeter} = 2L + 2W$$

**Problem:** A square Manhattan block has side length 50 feet.

a) What is its area?

b) If you walk around the whole square block, how far do you walk?

---

- (AxSxWIx25x01x01x007)
NOTAGS
**Rectangle Formulas (where L is the length and W the width):**

$$\text{Area} = L \times W \quad , \quad \text{Perimeter} = 2L + 2W$$

**Problem:** What is the area of a rectangular playground if its length is 97 meters and its width is 100 meters?

---

- (AxSxWIx25x01x01x009)
NOTAGS
**Rectangle Formulas (where L is the length and W the width):**

$$\text{Area} = L \times W \quad , \quad \text{Perimeter} = 2L + 2W$$

**Problem:** Find a rectangle whose area is equal to its perimeter.

---

- (AxSxWIx25x01x01x010)

NOTAGS

---

NOTAGS

- (AxSxWIx25x01x01x011)
NOTAGS
**Temperature Formulas (where C is Celsius and F is Fahrenheit):**

$$F = C \cdot \frac{9}{5} + 32 \quad , \quad C = \frac{5}{9}(F - 32)$$

**Problem:** In NYC, you hear it is 41°F. Convert this to Celsius so it's understandable in Portugal.

---

- (AxSxWIx25x01x01x012)

NOTAGS

---

NOTAGS
- (AxSxWIx25x01x01x013)
NOTAGS
**Temperature Formulas (where C is Celsius and F is Fahrenheit):**

$$F = C \cdot \frac{9}{5} + 32 \quad , \quad C = \frac{5}{9}(F - 32)$$

**Problem:** 100°F is hot! What is that in Celsius?

---

### 4.2.3 Substitution With Fractions *(F)*

- (AxSxFx25x01x01x005)
NOTAGS
Evaluate

a) $x - \frac{1}{2}$ if $x = \frac{1}{2}$ b) $x - \frac{1}{2}$ if $x = 0.5$

---

- (AxSxFx25x01x01x006)
NOTAGS
Evaluate

a) $x - \frac{1}{2}$ if $x = -\frac{1}{2}$ b) $2a - \frac{b}{6}$ if $a = \frac{1}{12},\ b = 5$

---

- (AxSxFx25x01x01x007)
NOTAGS
Evaluate

a) $2a - \frac{b}{6}$ if $a = \frac{5}{16},\ b = -7$ b) $x^2$ if $x = \frac{1}{4}$

---

- (AxSxFx25x01x01x008)
NOTAGS
Evaluate

a) $x^2$ if $x = 0.25$ b) $x^2$ if $x = -\frac{1}{4}$

---

- (AxSxFx25x01x01x009)
NOTAGS
Evaluate

a) $x^2 + y^2$ if $x = \frac{5}{3},\ y = 0$ b) $x^2 - y^2 - \frac{5}{6}$ if $x = -\frac{1}{3},\ y = \frac{1}{2}$

---

### 4.2.4 Substitution word problems (with fractions) *(WF)*

- (AxSxWFx25x01x01x003)
NOTAGS
**Triangle Area Formula (where B is the base length and H is the height):**

$$\text{Area} = \frac{1}{2} \times B \times H$$

**Problem:** Find the area of a triangle whose base is $\frac{7}{4}$ and height is $\frac{3}{7}$.

---

- (AxSxWFx25x01x01x008)
NOTAGS
**Rectangle Formulas (where L is the length and W the width):**

$$\text{Area} = L \times W \quad , \quad \text{Perimeter} = 2L + 2W$$

**Problem:** What is the perimeter of a rectangle if its length is $\frac{7}{24}$ meters and its width is $\frac{3}{36}$ meters?

---

### 4.2.5 Substitution word problems (with radicals) *(WR)*

- (AxSxWRx25x01x15x006)
NOTAGS
The time (in seconds) that it takes for an object to hit the ground is given by $f(d) = \frac{1}{4}\sqrt{d}$, where $d$ is the distance (in feet) dropped.
If the object is dropped from 64 feet, how long does it take?

---

- (AxSxWRx25x01x15x007)
NOTAGS
The time (in seconds) that it takes for an object to hit the ground is given by $f(d) = \frac{1}{4}\sqrt{d}$.
If the object is dropped from 25 feet, how long does it take?

---

- (AxSxWRx25x01x15x008)
NOTAGS
The time (in seconds) that it takes for an object to hit the ground is given by $f(d) = \frac{1}{4}\sqrt{d}$.
If the object is dropped from 100 feet, how long does it take?

---

## 4.3 Simplifying *(SY)*

- (AxSYx25x01x02x014)
NOTAGS
Distribute:

a) $3(4x + 2) =$ b) $3(4x - 2) =$

---

- (AxSYx25x01x02x015)
NOTAGS
Distribute:

a) $-3(4x + 2) =$ b) $-7(-3y - 1) =$

---

- (AxSYx25x01x02x016)
NOTAGS
Distribute:

a) $(-2w - 2)10 =$ b) $\frac{1}{2}(2x + 4) =$

---

- (AxSYx25x01x02x017)
NOTAGS
Distribute:

a) $(8x + 2)\frac{3}{2} =$

---

- (AxSYx25x01x02x018)

NOTAGS

---

NOTAGS

- (AxSYx25x01x02x019)
NOTAGS
Combine Like Terms:

  a) $35x + 5 - 35x - 11 =$ b) $-3c - 5 - 20c =$

---

- (AxSYx25x01x02x020)
NOTAGS
Combine Like Terms:

  a) $\frac{1}{2} + \frac{1}{3}x + \frac{3}{2} + \frac{2}{3}x =$

---

- (AxSYx25x01x02x021)

NOTAGS

---

NOTAGS
- (AxSYx25x01x02x022)
NOTAGS
Distribute then combine like terms:

  a) $4 - 2(-3x + 2) =$ b) $2(-3u - 1) + 3 =$

---

- (AxSYx25x01x02x023)
NOTAGS
Distribute then combine like terms:

  a) $2(-3u - 1) + 3u =$ b) $2(2 + 3x) + 5(2x + 3) =$

---

- (AxSYx25x01x02x024)

NOTAGS

---

NOTAGS
- (AxSYx25x01x02x025)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $2x + 7x$ for $x = 3$.

---

- (AxSYx25x01x02x026)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $3x + 2 + 4x - 1$ for $x = 2$.

---

- (AxSYx25x01x02x027)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $5x + 2 - 4x - 2$ for $x = -1$.

---

- (AxSYx25x01x02x028)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $3(2y + 2)$ for $y = 2$.

---

- (AxSYx25x01x02x029)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $(y - 3)10$ for $y = 2$.

---

- (AxSYx25x01x02x030)
NOTAGS
For the following, simplify first. Then evaluate BOTH the ORIGINAL expression ...

  a) $(y - 3)9 + y$ for $y = -2$.

---

- (GROUPxAxSYx25x02x07x001)
NOTAGS
Create one algebra "magic trick" like (but not the same as!) what I did in class.
Do the following:

  a) Clearly write down the steps the person should perform (e.g. add 5, then etc...).

  b) Describe what the magic trick does.

  c) Clearly write down the algebra and simplification that justifies your magic trick.

  d) Be ready to perform as a group in front of the class during the next meeting!

---

### 4.3.1 Simplifying with exponents *(E)*

- (AxSYxEx25x01x15x001)
NOTAGS
Simplify.

  a) $x^5x^3 =$

  b) $\dfrac{x^5}{x^3} =$

---

- (AxSYxEx25x02x12x001)
NOTAGS
Simplify.

  a) $x^4x^3 =$

  b) $\dfrac{x^9}{x^5} =$

---

- (AxSYxEx25x01x15x002)
NOTAGS
Simplify.

  a) $(xy)^7 =$

  b) $\dfrac{x^3}{x^9} =$

---

- (AxSYxEx25x01x15x003)
NOTAGS
Simplify.

  a) $\left(\dfrac{x}{y}\right)^3 =$

  b) $(xy)^0 =$

---

- (AxSYxEx25x01x15x004)
NOTAGS
Simplify.

  a) $\left(\dfrac{x^5}{x^3}\right)^0 =$

  b) $\left(\dfrac{x}{y}\right)^{-5} =$

---

- (AxSYxEx25x01x15x005)
NOTAGS
Simplify.

  a) $(x^4)^2 =$

  b) $(x^3)^{-2} =$

---

- (AxSYxEx25x01x15x006)
NOTAGS
Simplify.

a) $\dfrac{10y^9}{4y^5} =$

b) $x^7y^2x^6y^3 =$

---

- (AxSYxEx25x01x15x007)
NOTAGS
Simplify.
$(x^3y^{-6}z^5)(8x^{-3}yz^4) =$

---

- (AxSYxEx25x01x15x008)
NOTAGS
Simplify.
$\dfrac{x^5y^3z^{-2}}{x^{-2}y^5z^{-3}} =$

---

- (AxSYxEx25x01x15x009)
NOTAGS
Simplify.
$\left(\dfrac{12x^2y^{-3}}{4x^{-5}}\right)^{-2} =$

---

- (AxSYxEx25x01x15x010)
NOTAGS
Simplify.
$(x^{-2}y^{-4}z^2)(8x^{-1}y^2z^3) =$

---

- (AxSYxEx25x01x15x011)
NOTAGS
Simplify.
$\dfrac{x^{-5}y^3z^{-2}}{x^2y^3z^{-2}} =$

---

- (AxSYxEx25x01x15x012)
NOTAGS
Simplify.
$\left(\dfrac{3x^4y^{-1}}{9x^4y^{-1}}\right)^2 =$

---

## 4.4 Checking Solutions *(CH)*

- (AxCHx25x01x02x001)
NOTAGS
Determine whether the given value of the variable is a solution of the given equation.

  a) $x = 2$, of $2x + 6 = 10$

  b) $x = 4$, of $2x - 2 = -5$

---

- (AxCHx25x01x02x002)
NOTAGS
Determine whether the given value of the variable is a solution of the given equation.

  a) $x = (-1)$, of $2(x + 4) = 3(x + 3)$

  b) $x = 5$, of $4(x - 2) = 4x - 8$

---

- (AxCHx25x01x02x003)
NOTAGS
Determine whether the given value of the variable is a solution of the given equation.

  a) $x = (-2)$, of $2x + 6 = 3x + 8$

  b) $x = 4$, of $3x + 6 = 5x - 2$

---

- (AxCHx25x01x02x004)
NOTAGS
Determine whether the given value of the variable is a solution of the given equation.

  a) $x = \frac{2}{3}$, of $3x - 2 = 0$

  b) $x = -\frac{2}{3}$, of $3x + 2 = -2$

---

## 4.5 Solving Equations (General Issues) *(SG)*

- (AxSGx25x01x22x001)
NOTAGS
For the following equations, solve them using logic, together with guessing and checking. Do not use any solution techniques.

  a) $10x = 50$

  b) $y^2 = 25$

---

- (AxSGx25x01x22x002)
NOTAGS
For the following equations, solve them using logic, together with guessing and checking. Do not use any solution techniques.

  a) $4x^9 + x^3 + 10 = 4x^9 + x^3 + 10$

  b) $4x^9 + x^3 + 10 = 4x^9 + x^3 + 50$

---

- (AxSGx25x01x22x003)
NOTAGS
For the following equations, solve them using logic, together with guessing and checking. Do not use any solution techniques.

  a) $3^u = 9$

  b) $\dfrac{x}{(x-1)(x+5)} = \dfrac{x}{(x-1)(x+5)}$

---

- (AxSGx25x01x22x004)
NOTAGS
For the following equations, solve them using logic, together with guessing and checking. Do not use any solution techniques.

  a) $\sqrt[4]{x} = 2$

  b) $\sqrt[4]{x} = -2$

---

- (AxSGx25x01x22x005)
NOTAGS
For the following equations, solve them using logic, together with guessing and checking. Do not use any solution techniques.

  a) $\sqrt[3]{x} = -3$

  b) $(y-10)(y+25) = 0$

---

- (AxSGx25x01x22x006)
NOTAGS
For the below equation, solve it using logic, together with guessing and checking. Do not use any solution techniques.

$$A^2 + 27665 = -12939887$$

---

- (AxSGx25x01x22x007)
NOTAGS
Draw the graph of $y = x^3 - 3x^2 + 2x$ using Desmos or some other graphing utility. Just using that graph, find all solutions to $0 = x^3 - 3x^2 + 2x$.

---

## 4.6 Solving Linear Equations *(EL)*

### 4.6.1 Guess and Check *(GC)*

#### 4.6.1.1 Guess and Check With Integer Solutions *(I)*

- (AxGCxIx25x01x02x005)
NOTAGS
Solve each equation by guessing and checking.

a) $y - 5 = 20$ b) $y - 5 = -20$

---

- (AxGCxIx25x01x02x006)
NOTAGS
Solve each equation by guessing and checking.

a) $4x = 8$ b) $\frac{x}{2} = 50$

---

- (AxGCxIx25x01x02x007)
NOTAGS
Solve each equation by guessing and checking.

a) $2x = 2x + 1$ b) $x + 5 = x + 3 + 2$

---

### 4.6.2 Solving Linear Equations With Integer Solutions *(I)*

- (AxELxIx24x12x12x001)
NOTAGS
Solve the equation $2x + 1 = 11$.

---

- (AxELxIx24x12x12x002)
NOTAGS
Solve the equation $3x = 15$.

---

- (AxELxIx24x12x13x003)
NOTAGS
Solve the equation $1 - 3x = 10$.

---

- (AxELxIx25x01x02x008)
NOTAGS
Solve the following equations.

  a) $2x + 7 = 15$

  b) $5x - 4 = 18$

---

- (AxELxIx25x01x02x009)
NOTAGS
Solve the following equations.

  a) $7x - 5 = 12$

  b) $6 - 2x = 14$

---

- (AxELxIx25x01x02x010)
NOTAGS
Solve the following equations.

  a) $-8 - 7x = -1$

  b) $-5x + 7 = 12$

---

- (AxELxIx25x01x02x011)
NOTAGS
Solve the following equations.

  a) $6x - 5 = 2x - 13$

  b) $4x + 2 = 2 + x$

---

- (AxELxIx25x01x02x012)
NOTAGS
Solve the following equations.

  a) $4 = 2 + x$

  b) $\frac{x}{5} + 6 = 9$

---

- (AxELxIx25x01x02x013)
NOTAGS
Solve the following equations.

  a) $\frac{5x}{2} = 15$

  b) $\frac{-4x}{3} = -16$

---

- (AxELxIx25x01x02x014)
NOTAGS
Solve the following equations.

  a) $\frac{-2x}{3} = 8$

  b) $\frac{x}{2} - 4 = 7$

---

- (AxELxIx25x01x02x015)
NOTAGS
Solve the following equations.

  a) $\frac{5x}{3} + 6 = -5$

---

- (AxELxIx25x01x02x019)
NOTAGS
Solve the following equations.

  a) $5(x - 4) = 18$

---

- (AxELxIx25x01x02x020)
NOTAGS
Solve $-(x - 5) + 4 = 12$

---

- (AxELxIx25x01x02x021)
NOTAGS
Solve $2(x + 7) - 4 = 15 - 2(x - 3)$

---

- (AxELxIx25x01x02x022)
NOTAGS
Solve $6 - 2(x + 3) = 14 - 3(x + 1)$

---

- (AxELxIx25x01x02x023)
NOTAGS
Solve $8 - (x + 3) = 9 - (x + 4)$

---

- (AxELxIx25x01x02x024)
NOTAGS
Solve $-5(x + 3) = 4 - 5x$

---

### 4.6.3 Linear Equations: Identity or Contraditiction *(IC)*

- (AxELxICx25x01x02x025)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $x = x + 2$

---

- (AxELxICx25x01x02x026)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $x = 4x$

---

- (AxELxICx25x01x02x027)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $x + 2x + x + 7 = 4x + 7$

---

- (AxELxICx25x01x02x028)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $5(x - 1) + 3 = 5x - 2$

---

- (AxELxICx25x01x02x029)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $-(x - 2) + 4 = 7 - x$

---

- (AxELxICx25x01x02x030)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $2(x + 1) - 1 = 3 - x$

---

- (AxELxICx25x01x02x031)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $6x - 2(x + 3) = -(4x + 6)$

---

- (AxELxICx25x01x02x032)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $2-(x+1)=5-2(x-1)$

---

- (AxELxICx25x01x02x033)
NOTAGS
For each equation determine if it is 1) An identity, i.e. always true, 2) A contradiction, i.e. no solution, or 3) Conditional...

  a) $-2(x-2)+2(x-2)=1$

---

### 4.6.4 Solving Linear Equations With Fractional Solutions *(F)*

- (AxELxFx24x12x12x001)
NOTAGS
Solve the equation $2x=9$.

---

- (AxELxFx24x12x12x002)
NOTAGS
Solve the equation $3x-2=15$.

---

- (AxELxFx24x12x13x003)
NOTAGS
Solve the equation $1-3x=12$.

---

### 4.6.5 Solving Equations For Variable (in a linear way) *(FM)*

- (AxELxFMx25x01x03x001)
NOTAGS
Solve for $y$ in the formula $2x+y=7$

---

- (AxELxFMx25x01x03x002)
NOTAGS
Solve for $b$ in the formula $a+b=h^2$

---

- (AxELxFMx25x01x03x003)
(AxELxFMx25x01x03x004)
Solve for $L$ in $A=L\cdot W$ (i.e., formula for area of a rectangle). Then use that to do the following:

---

10NOTAGS Find the length of a rectangle whose area is 8 and width is 2.

- (AxELxFMx25x01x03x005)
NOTAGS
Find the length of a rectangle whose area is 20 and width is 4.

---

- (AxELxFMx25x01x03x006)
NOTAGS
Find the length of a rectangle whose area is 10 and width is 4.

---

- (AxELxFMx25x01x03x007)
NOTAGS
Solve for $T$ in the formula $PV = nRT$

---

- (AxELxFMx25x01x03x008)
NOTAGS
Solve for $x$ in the formula $2x + 6y = 4$

---

- (AxELxFMx25x01x03x009)
NOTAGS
Solve for $x$ in the formula $-2x + 6y = 4$

---

- (AxELxFMx25x01x03x010)
NOTAGS
Solve for $r$ in the formula $C = 2\pi r$

---

- (AxELxFMx25x01x03x011)
(AxELxFMx25x01x03x012)
Solve for $L$ in $P = 2L + 2W$ (perimeter of a rectangle). Then use that to do the following:

---

10NOTAGS Find the length of a rectangle whose perimeter is 50 and width is 10.

- (AxELxFMx25x01x03x013)
NOTAGS
Find the length of a rectangle whose perimeter is 100 and width is 40.

---

- (AxELxFMx25x01x03x014)
NOTAGS
Find the length of a rectangle whose area is 35 and width is 10.

---

- (AxELxFMx25x01x03x015)
NOTAGS
Solve for $B$ in the formula $A = \dfrac{h(B+b)}{2}$

---

- (AxELxFMx25x01x03x016)
NOTAGS
Solve for $y$ in the formula $3x - 6y = 3$

---

### 4.6.6 Solving Linear Equations With Absolute Value *(EAL)*

- (AxEALx25x01x21x006)
NOTAGS
Solve these absolute value equations:

  a) $|x - 5| = 20$ b) $|4x + 7| = 15$

---

- (AxEALx25x02x05x001)
NOTAGS
Solve.

  a) $|x - 3| = 4$ b) $|x + 2| = 5$

---

- (AxEALx25x02x05x002)
NOTAGS
Solve $|2x + 3| = 9$.

---

- (TESTxxxAxEALx25x03x26x001)
type: TEST
Solve.

  a) $|3x - 2| = 14$ b) $|42x + 7| = -3$

---

### 4.6.7 Solving Linear Equations With Function Symbol *(FL)*

- (AxFLx25x01x23x001)
NOTAGS
Suppose that $f(x) = 7 - 2x$. Solve $f(x) = 11$.

---

- (AxFLx25x01x23x002)
NOTAGS
Suppose that $h(u) = 20u$. For what $u$ does $h(u) = 50$?

---

- (TESTxAxFLx25x02x12x001)
NOTAGS
Suppose that $g(x) = 3x^2$. Solve $g(x) = 12$.

---

## 4.7 Lines *(L)*

- (AxLx25x01x06x001)
NOTAGS
**For each equation determine whether it is linear or non-linear?**

  a) $3x - 2y + x = 7 + y$  b) $3x^4 + 2y^4 = 8$

---

- (AxLx25x01x06x002)
NOTAGS
**For each equation determine whether it is linear or non-linear?**

  a) $3 + x = -x + 2(7 - y)$  b) $3x + x = -2y + 7$

---

- (AxLx25x01x06x003)
NOTAGS
**For each equation determine whether it is linear or non-linear?**

  a) $3x + y^2 = 1$  b) $3(x - y) - 1 = 2y + x$

---

- (AxLx25x01x06x004)
NOTAGS
Is the following equation linear or non-linear? $2(x + x^2) = y + 1$

---

- (AxLx25x01x06x005)
NOTAGS
**For each equation, state which of the following it is: A) horizontal line, B) vertical line, or C) a slanted line.**

  a) $x = 0$  b) $y = \frac{3}{47}$

---

- (AxLx25x01x06x006)
NOTAGS
**For each equation, state which of the following it is: A) horizontal line, B) vertical line, or C) a slanted line.**

  a) $x + 7y = 0$  b) $x + 7 = 0$

---

- (AxLx25x01x06x007)
NOTAGS
**For the following equation, state which of the following it is: A) horizontal line, B) vertical line, or C) a slanted line:** $3x + 7y = 0$

---

- (AxLx25x01x06x008)
NOTAGS
**Find the intercepts of the lines.**

a) $2x + 3y = 6$ b) $-4x + y = 4$

---

- (AxLx25x01x06x009)
NOTAGS
**Find the intercepts of the lines.**

a) $x + 4y = 0$ b) $y = 6$

---

- (AxLx25x01x06x010)
NOTAGS
**Find the intercepts of the lines.**

a) $x = 4$ b) $y = 2x + 5$

---

- (AxLx25x01x06x011)
NOTAGS
**Find the intercepts of the lines.**

a) $y = 6x - 2$ b) $y = 4x + 10$

---

- (AxLx25x01x06x012)
NOTAGS
Find the intercepts of the line: $y = -x + \frac{5}{3}$

---

- (AxLx25x01x06x013)
NOTAGS
**Put each line into Slope-Intercept Form.**

a) $y = 3x - 2$ b) $x = y - 2$

---

- (AxLx25x01x06x014)
NOTAGS
**Put each line into Slope-Intercept Form.**

a) $5x - y = 20$ b) $2y = 4x - 6$

---

- (AxLx25x01x06x015)
NOTAGS
**Put each line into Slope-Intercept Form.**

a) $3y = 4x - 6$ b) $4y + 6x - 10 = 0$

---

- (AxLx25x01x06x016)
NOTAGS
Find the slope and y intercept of the lines.

a) $y = 3x - 2$ b) $y = \frac{3}{5}x + 23$

---

- (AxLx25x01x06x017)
NOTAGS
Find the slope and y intercept of the lines.

a) $y = -\frac{7}{3}x - 9$ b) $3x + y = 4$

---

- (AxLx25x01x06x018)
NOTAGS
Find the slope and y intercept of the lines.

a) $3x - 5y = 4$ b) $y = 2$

---

- (AxLx25x01x06x019)
NOTAGS
Find the slope and y intercept of the lines.

a) $x = 4$ b) $4x - y = 4$

---

- (AxLx25x01x06x020)
NOTAGS
Find the slope and y intercept of the lines.

a) $x - 3y = -3$ b) $x + y = 4$

---

- (AxLx25x01x06x021)
NOTAGS
Find the slope and y intercept of the lines.

a) $y = 2x - 1$ b) $y = 4$

---

- (AxLx25x01x06x022)
NOTAGS
Find the slope and y intercept of the lines.

a) $x = -1$ b) $3x - 7y = -1$

---

- (AxLx25x01x06x023)
NOTAGS
Find the slope of the line from the given points.

a) $(1, 2)$ and $(3, 6)$ b) $(0, 5)$ and $(2, 1)$

---

- (AxLx25x01x06x024)
NOTAGS
Find the slope of the line from the given points.

a) $(15, 17)$ and $(21, 5)$ and $(16, 15)$ b) $(-2, 3)$ and $(2, -9)$ and $(1, -6)$

---

- (AxLx25x01x06x025)
NOTAGS
**Graph the set of solutions of each equation.**

a) $y = 3$ b) $x = 3$

---

- (AxLx25x01x06x026)
NOTAGS
**Graph the set of solutions of each equation.**

a) $y = -2$ b) $x = -2$

---

- (AxLx25x01x06x027)
NOTAGS
**Graph the set of solutions of each equation.**

a) $y = 0$ b) $x = 0$

---

- (AxLx25x01x06x028)
NOTAGS
Graph each equation.

a) $y = 3x + 1$ b) $y = 2x - 1$

---

- (AxLx25x01x06x029)
NOTAGS
Graph each equation.

a) $y = \frac{1}{2}x + 2$ b) $y = -\frac{3}{2}x$

---

- (AxLx25x01x06x030)
NOTAGS
Graph each equation.

a) $y = 4$ b) $y = 4x$

---

- (AxLx25x01x06x031)
NOTAGS
Graph each equation.

a) $x = y + 3$ b) $x = 1$

---

- (AxLx25x01x06x032)
NOTAGS
Graph each equation.

a) $y = -1$ b) $2x = 4y - 8$

---

- (AxLx25x01x06x033)
NOTAGS
Graph the equation: $3x + 4y = 4$

---

- (AxLx25x01x15x015)
NOTAGS
Graph the following function.
$f(x) = 2x + 1$

---

- (AxLx25x01x15x016)
NOTAGS
Graph the following function.
$g(x) = 2x$

---

- (AxLx25x01x15x017)
NOTAGS
Graph the following function.
$h(x) = \frac{1}{2}x + 1$

---

- (AxLx25x01x15x018)
NOTAGS
Graph the following function.
$f(x) = -2x + 3$

---

- (AxLx25x01x15x019)
NOTAGS
Graph the following function.
$f(x) = -2x - 3$

---

- (AxLx25x01x21x009)
NOTAGS
Let $f(x) = 4x - 3$. Find the slope and all intercepts. Then graph $f$.

---

- (AxLx25x01x21x005)
NOTAGS
Consider two linear functions $f(x) = -2x + 1$ and $g(x) = 2x + 1$.

a) Find the intersection point of $f$ and $g$.

b) Give an example of a linear function $h(x)$ that does not intersect $f(x)$. Justify.

---

- (AxLx25x02x12x001)
NOTAGS
Find the intersection point of the two linear functions $f(x) = -5x + 1$ and $h(x) = -5x - 12$.

---

- (AxLx25x02x12x002)
NOTAGS
Find the linear function such that $f(-1) = 4$ and $f(5) = 1$.

---

- (AxLx25x02x12x003)
NOTAGS
Find the linear function such that $f(2) = 0.5$ and $f(0) = 0$.

---

- (AxLx25x02x03x001)
NOTAGS
Find the equation of the line passing through the point $(1, 3)$ that is parallel to the line with equation $3x + 2y = 5$.

---

- (AxLx25x02x12x004)
NOTAGS
Determine whether the lines are parallel, perpendicular, or neither.

a) $y = 3x + 1$

b) $y = 30x + 1$

---

- (AxLx25x02x12x005)
NOTAGS
Determine whether the lines are parallel, perpendicular, or neither.

a) $y + 5x = 1$ b) $y = -5x + 19$

---

- (AxLx25x02x12x105)
NOTAGS
Determine whether the lines are parallel, perpendicular, or neither.

a) $2y + 4x = 1$ b) $8y = 4x + 10$

---

- (AxLx25x02x12x006)
NOTAGS
Consider the linear equation $f(x) = (3/2)x - 6$. Find the slope of a line parallel to $f$. Find the slope of a line perpendicular to $f$.

---

- (AxLx25x02x12x007)
NOTAGS
Consider the linear equation $f(x) = -6x$. Find the slope of a line parallel to $f$. Find the slope of a line perpendicular to $f$.

---

- (AxLx25x02x12x010)
NOTAGS
Write down the linear function whose graph is parallell to the linear function $g(x) = 3x - 1$ and passing through the point $(4, 9)$.

---

- (AxLx25x02x12x011)
NOTAGS
Write down the equation for the line parallel to $h(t) = 3t - 9$ and passing through the point $(0, 0)$.

---

- (AxLx25x02x12x008)
NOTAGS
Write an equation for a line perpendicular to $f(x) = 3x + 4$ and passing through the point $(3, 1)$.

---

- (AxLx25x02x12x009)
NOTAGS
Write an equation for a line perpendicular to $f(x) = -2x + 4$ and passing through the point $(-4, -1)$.

---

- (AxLx25x02x12x012)
NOTAGS
Find the point of intersection between the lines, if it exists:
$2x - 3y = 12$ and $5y + x = 30$.

---

- (AxLx25x02x12x013)
NOTAGS
Find the point of intersection between the lines, if it exists:
$f(x) = x + 1$ and $g(x) = 2x - 1$.

---

- (AxLx25x02x12x015)
NOTAGS
Consider two kinds of gym membership plans.

  - Platinum Plan: \$20 per month + \$3 per fitness class.
  - Gold Plan: \$10 per month + \$6 per fitness class.

  Which plan is better if you attend 2 classes a month? Which plan is better if you attend 5 classes a month?
  How many classes must you attend in a month for the Platinum plan to save you money?

---

- (AxLx25x02x12x016)
NOTAGS
Suppose you want to rent a car, and these are your options.

  - Option 1: \$25 per day + \$0.10 per mile.
  - Option 2: \$35 per day + \$0.05 per mile.

  How many miles do you need to drive in a day for option 2 to be the cheaper?

---

- (AxLx25x02x12x014)
NOTAGS
A cell phone company offers two plans.
Plan A: \$20 per month and \$1 for every one hundred texts.
Plan B: \$50 per month with free unlimited texts.
How many texts would you need to send per month for plan B to save you money?

---

- (AxLx25x02x12x115)
NOTAGS
A cell phone company offers two plans.
Plan A: \$15 per month and \$2 for every 300 texts.
Plan B: \$25 per month and \$0.50 for every 100 texts.
How many texts would you need to send per month for plan B to save you money?

---

- (AxLx25x02x12x017)
NOTAGS
Suppose a city's population has been decreasing at a constant rate of 50 people a year.

  a) If the population is now 400, what will the population be in 2 years?

  b) When will it be a ghost town (that is, when will the population be zero)?

  c) Write down a function $P(t)$ which models the population.

---

- (AxLx25x02x12x018)
NOTAGS
Suppose a city's population in 2020 was 4005, has been increasing at a constant rate of 35 people a year.

  a) Write down a function $P(t)$ which models the population.

  b) Use your model to predict the population in the year 2030.

  c) Use your model to predict the population 7.2 years after 2020? (use a calculator if you have one, or leave your answer unevaluated)

---

- (AxLx25x02x12x019)
NOTAGS
Suppose a child's weight fom ages 1 to age 5 is modeled by the following equation: $W(t) = 11 + 6t$

  a) How much does the child weigh at age 3? And at 3 and a half?

  b) How much weight does the child gain each year?

  c) What does the model say about the child's weight at age 9?

---

- (AxLx25x02x12x020)
NOTAGS
Suppose that the number of people in a town who have a cold grows at a constant rate. In December 100 people have the cold, and a month later, 120 poeple have a cold.

  a) How many people have a cold in March?

  b) Write down an equation which models this situation.

---

- (AxLx25x02x12x021)
NOTAGS
Suppose we have a scatter plot of some data comparing hours spent studying with grade on an exam, where hours (t) is the independent variable. Suppose the best-fit line turns out to be this: $G(t) = 20 + 15t$

  a) According to this best fit line, what will your grade be if you do not study at all?

  b) How much do you need to study to get 100?

---

- (TESTxxxAxLx25x03x26x001)
NOTAGS
Consider the linear function $f(x) = -2x + 4$. Find its slope and all its intercepts. Then graph $f$.

---

- (AxLx25x10x27x001)
type: TEST
Consider the linear function $f(x) = -3x + 2$. Find its slope and all its intercepts. Then graph $f$.

---

- (TESTxxxAxLx25x03x26x002)
NOTAGS
Find the point of intersection between the lines, if it exists:
$f(x) = \frac{2}{3}x - 2$ and $g(x) = -\frac{1}{3}x + 1$.

---

- (TESTxxxAxLx26x03x22x004)
type: TEST
Find the point of intersection between the lines, if it exists:
$h(x) = 7x + 2$ and $g(x) = 4x - 10$.

---

- (TESTxxxAxLx25x03x26x003)
NOTAGS
Consider the linear equation $f(x) = \frac{4}{3}x - 12$.

  a) Write down the equation for any line that does not intersect $f$.

  b) Find the slope of a line perpendicular to $f$.

---

- (TESTxxxAxLx25x04x28x001)
type: TEST
Consider the linear equation $f(x) = 3x - 1$.

  a) Write the equation for a line parallel to $f(x)$ and passing through the point $(-2, 7)$.

  b) Write the equation for a line perpendicular to $f(x)$ and passing through the point $(0, -5)$.

---

- (GROUPxAxLx25x02x12x001)
NOTAGS
The following data from a US Census report indicates for various years, the percentage of the population (who are 25 years or older) with a college degree.

| *Year* | 1990 | 1992 | 1994 | 1996 | 1998 | 2000 |
|---|---|---|---|---|---|---|
| *Graduate %* | 21.3 | 21.4 | 22.2 | 23.6 | 24.4 | 25.6 |

  a) Use google sheets to find the equation of a best-fit line for this data. Make a rough sketch of the scatter diagram along with the best-fit line on top of it.

  b) Use your model to guess what the percent of college graduates was in the year 1995? (this is interpolation ...)

  c) Use your model to guess what the percent of college graduates was in the year 2025? (this is predication ...). Wonder if the prediction is correct??

---

## 4.8 Solving Linear Inequalities *(NL)*

- (AxNLx25x01x04x001)
NOTAGS
**Determine which of the given values of the variable are solutions of the given inequality.**

  a) $x = 1, 3, \quad \text{of} \quad x > -2$

  b) $x = -2, -3, \quad \text{of} \quad x \geq -2$

---

- (AxNLx25x01x04x002)
NOTAGS
**Determine which of the given values of the variable are solutions of the given inequality.**

  a) $x = 2, 5, \quad \text{of} \quad 2x > 6$

  b) $x = -1, 0, \quad \text{of} \quad x + 7 < 9$

---

- (AxNLx25x01x04x003)
NOTAGS
**Determine which of the given values of the variable are solutions of the given inequality.**

a) $x = -7, -1, \quad \text{of} \quad 3x - 1 \leq -22$ b) $x = -10, -4, \quad \text{of} \quad 2(x + 3) \geq 0$

---

- (AxNLx25x01x04x004)
NOTAGS
**Represent the following sets on the real line.**

a) $x > 1$ b) $x \leq 0$

---

- (AxNLx25x01x04x005)
NOTAGS
**Represent the following sets on the real line.**

a) $x > 0$ b) $x < -3$

---

- (AxNLx25x01x04x006)
NOTAGS
**Represent the following sets on the real line.**

a) $x \geq -1.5$ b) $x \leq \frac{2}{3}$

---

- (AxNLx25x01x04x007)
NOTAGS
**Solve the inequality and graph its solution.**
$2x + 7 > 15$

---

- (AxNLx25x01x04x008)
NOTAGS
**Solve the inequality and graph its solution.**
$5x - 4 < 16$

---

- (AxNLx25x01x04x009)
NOTAGS
**Solve the inequality and graph its solution.**
$3x - 5 < 12$

---

- (AxNLx25x01x04x010)
NOTAGS
**Solve the inequality and graph its solution.**
$6 - 2x \leq 14$

---

- (AxNLx25x01x04x011)
NOTAGS
**Solve the inequality and graph its solution.**
$-8 - 7x > -1$

---

- (AxNLx25x01x04x012)
NOTAGS
**Solve the inequality and graph its solution.**
$-5x + 7 > 12$

---

- (AxNLx25x01x04x013)
NOTAGS
**Solve the inequality and graph its solution.**
$6x - 5 < 2x - 13$

---

- (AxNLx25x01x04x014)
NOTAGS
**Solve the inequality and graph its solution.**
$x + 2 \geq 2 + 4x$

---

- (AxNLx25x01x04x015)
NOTAGS
**Solve the inequality and graph its solution.**
$4 \geq 2 + x$

---

- (AxNLx25x01x04x016)
NOTAGS
**Solve the inequality and graph its solution.**
$\frac{x}{5} + 6 < 9$

---

- (AxNLx25x01x04x017)
NOTAGS
**Solve the inequality and graph its solution.**
$\frac{5x}{2} \geq 15$

---

- (AxNLx25x01x04x018)
NOTAGS
**Solve the inequality and graph its solution.**
$\frac{-4x}{3} \leq -16$

---

- (AxNLx25x01x04x019)
NOTAGS
**Solve the inequality and graph its solution.**
$5(x - 1) + 3 \geq 5x - 2$

---

- (AxNLx25x01x04x020)
NOTAGS
**Solve the inequality and graph its solution.**
$-(x-2)+4>7-x$

---

- (AxNLx25x01x04x021)
NOTAGS
**Solve the inequality and graph its solution.**
$2(x+1)-1\geq 3-x$

---

- (AxNLx25x01x04x022)
NOTAGS
**Solve the inequality and graph its solution.**
$6x-2(x+3)\geq -(4x+6)$

---

- (AxNLx25x01x04x023)
NOTAGS
**Solve the inequality and graph its solution.**
$2-(x+1)<5-2(x-1)$

---

- (AxNLx25x01x04x024)
NOTAGS
**Solve the inequality and graph its solution.**
$-2(x-2)+2(x-2)\leq 1$

---

## 4.9 Solving Linear Inequalities With Absolute Value *(NAL)*

### 4.9.1 With integer solutions *(I)*

- (AxNALxIx24x12x28x001)
NOTAGS
Solve the inequality $|3x+2|>-2$

---

- (AxNALx25x01x21x007)
NOTAGS
Solve these absolute value inequalities:

a) $4|3x-5|<40$ b) $|4x+1|+7\geq 12$

---

- (AxNALx25x02x05x001)
NOTAGS
Solve.

  a) $|x-3| \leq 4$ b) $|x+2| \geq 5$

---

- (AxNALx25x02x05x002)
NOTAGS
Solve $|2x+3| > 9$.

---

- (TESTxxxAxNALx25x03x26x001)
type: TEST
Solve.

  a) $|2x-5| \leq 11$ b) $|x+20| \geq 12$

---

- (AxNALx25x10x27x001)
type: TEST
Solve.

  a) $|3x-5| \leq 13$ b) $|x+10| \geq 2$

---

## 4.10 Linear Equations In Two Variables *(TVL)*

### 4.10.1 Checking pairs as solutions *(C)*

- (AxTVLxCx25x01x04x001)
NOTAGS
**For each equation, determine which pairs are solutions.**
$x+y=3$; pairs: $(1,2), (-1,2), (5,-2)$

---

- (AxTVLxCx25x01x04x002)
NOTAGS
**For each equation, determine which pairs are solutions.**
$2x+y=-4$; pairs: $(0,0), (-2,0), (0,-2)$

---

- (AxTVLxCx25x01x04x003)
NOTAGS
**For each equation, determine which pairs are solutions.**
$2x+y-1=0$; pairs: $(-3,7), (1,0), (0,1)$

---

### 4.10.2 Finding other variable value *(OV)*

- (AxTVLxOVx25x01x04x004)
NOTAGS
**Find the solution (should be a pair!) which has the given $x$ value.**

$x + y = 3$ ; $x = 2$ $\qquad$ $2x + y = -4$ ; $x = \frac{1}{2}$

---

- (AxTVLxOVx25x01x04x005)
NOTAGS
**Find the solution (should be a pair!) which has the given $x$ value.**

$2x + y - 1 = 0$ ; $x = 2$

---

### 4.10.3 Finding solutions *(FS)*

- (AxTVLxFSx25x01x04x006)
NOTAGS
**Find 3 solutions of each of the following equations.**

$y = 3x - 2$ $\qquad$ $2x + 5y = 10$

---

- (AxTVLxFSx25x01x04x007)
NOTAGS
**Find 3 solutions of each of the following equations.**

$x + y = 0$ $\qquad$ $-2x + 3y = 5$

---

- (AxTVLxFSx25x01x04x008)
NOTAGS
**Find 3 solutions of each of the following equations.**

$x = 3y - 4$ $\qquad$ $3x + 4y = 5$

---

### 4.10.4 Graphing Linear Equations *(GL)*

- (AxTVLxGLx25x01x04x009)
NOTAGS
Draw $x$ and $y$ axis on the right. Then plot the points (labeling each point).
$(1, 3)$, $(3, 1)$, $(2, -2)$, $(-3, 4)$, $(-1, -2)$, $(0, 2)$, $(1.5, 0)$, $(0, 0)$

---

- (AxTVLxGLx25x01x04x010)
NOTAGS
For each point: If it is on an axis, state which axis it is on. Otherwise, state which quadrant it is in.
$(-4, 6)$, $(8, 10)$, $(0, 12)$, $(-50, -17)$, $(-100, 0)$, $(0.7, -188.9)$

---

- (AxTVLxGLx25x01x04x011)
NOTAGS
On graph paper, graph each equation by plotting some solutions and guessing what the rest looks like. Which of the graphs (if any), look like lines?

$x - y = 0$ $y = x^2$

---

- (AxTVLxGLx25x01x04x012)
NOTAGS
On graph paper, graph each equation by plotting some solutions and guessing what the rest looks like. Which of the graphs (if any), look like lines?

$y + x^2 = 1$ $y = 2x + 1$

---

- (AxTVLxGLx25x01x04x013)
NOTAGS
**Graph the set of solutions of each of the following equations (on separate graph paper).**

$y = 3x - 2$ $2x + 5y = 10$

---

- (AxTVLxGLx25x01x04x014)
NOTAGS
**Graph the set of solutions of each of the following equations (on separate graph paper).**

$x + y = 0$ $-2x + 3y = 5$

---

## 4.11 Polynomial basics *(P)*

- (AxPx25x01x15x001)
NOTAGS
For each expression, is it a polynomial? (Yes or No)

a) $3x + 5y - 6$ b) $x^2 + x + \frac{6}{x}$

---

- (AxPx25x01x15x002)
NOTAGS
For each expression, is it a polynomial? (Yes or No)

a) $x^2 - 5x + 6$ b) $\sqrt{x} + 2x + 6$

---

- (AxPx25x01x15x003)
NOTAGS
For each expression, is it a polynomial? (Yes or No)

a) $\left(\frac{3}{5}\right) x - 2 + 3x - 1$ b) $\frac{3+x}{2+x} + 3x - 5$

---

- (AxPx25x01x15x004)
NOTAGS
Classify the following polynomials as (1) monomials, (2) binomials, (3) trinomials, or (4) none of those.

a) $3x + 5y - 6$ b) $x^2 - 5x + 6$

---

- (AxPx25x01x15x005)
NOTAGS
Classify the following polynomials as (1) monomials, (2) binomials, (3) trinomials, or (4) none of those.

a) $5x + 3$ b) $-3x^3y^2z^3$

---

- (AxPx25x01x15x006)
NOTAGS
Classify the following polynomials as (1) monomials, (2) binomials, (3) trinomials, or (4) none of those.

a) $3x + 5y$ b) $x^2$

---

- (AxPx25x01x15x007)
NOTAGS
Classify the following polynomials as (1) monomials, (2) binomials, (3) trinomials, or (4) none of those.

a) $5x + 3 + 7y - 5xy$ b) $x^3 + 3x - 1$

---

- (AxPx25x01x15x008)
NOTAGS
Circle the terms in each expression, and find the coefficient of each term.

a) $3x + 5y - 6$ b) $x^2 - 5x + 6$

---

- (AxPx25x01x15x009)
NOTAGS
Circle the terms in each expression, and find the coefficient of each term.

a) $4x + 4$ b) $-x^2 + x$

---

- (AxPx25x01x15x010)
NOTAGS
Circle the terms in each expression, and find the coefficient of each term.

a) $x + 2y - 1$ b) $x^3$

---

- (AxPx25x02x18x001)
NOTAGS
Find the leading term and leading coefficient in each polynomial.

a) $-5x^6 + 4x - 30x^3$ b) $5 - x^7 + 3x - 6x^3$

---

- (HWxAxPx25x02x18x002)
type: HW
Find the leading term and leading coefficient in each polynomial.

a) $-x^2 - 30 + 2x$ b) $5 + x + x^2 + x^3 + x^4$

---

- (AxPx25x01x15x011)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(x^2 + 3x - 2) + (3x^2 - 5x - 6)$

---

- (AxPx25x01x15x012)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(3x^3 - 4x - 2x^2) + (3x^2 - 6 + x)$

---

- (AxPx25x01x15x013)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(5x^3 - 2x - 1) + (3x^5 - 5^2)$

---

- (AxPx25x01x15x014)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(4x^2y + 3xy - 2) + (3xy - 5xy^2 + 6)$

---

- (AxPx25x01x15x015)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(3x - 4xy - z + x^2) + (-5x^2 + z - 3x)$

---

- (AxPx25x01x15x016)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(x^2 + 3x - 2) + (-3x^2 - 3x + 2)$

---

- (AxPx25x01x15x017)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(x^2 + 3x - 2) - (3x^2 - 5x - 6)$

---

- (AxPx25x01x15x018)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(3x^3 - 4x - 2x^2) - (3x^2 - 6 + x)$

---

- (AxPx25x01x15x019)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(5x^3 - 2x - 1) - (3x^5 - 5^2)$

---

- (AxPx25x01x15x020)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(4x^2y + 3xy - 2) - (3xy - 5xy^2 + 6)$

---

- (AxPx25x01x15x021)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(3x - 4xy - z + x^2) - (-5x^2 + z - 3x)$

---

- (AxPx25x01x15x022)
NOTAGS
Perform the operation and simplify. Write the answer in descending order of degree.
$(x^2 + 3x - 2) - (-3x^2 - 3x + 2)$

---

- (AxPx25x01x15x023)
NOTAGS
Evaluate the following (by first simplifying, then evaluating).
$f(x) = 5x^2 + 3x - 5x^2 + x + 1$. Evaluate $f(2)$ and $f(1)$.

---

- (AxPx25x01x15x024)
NOTAGS
Evaluate the following (by first simplifying, then evaluating).
$f(y) = 8y^3 - 10y - 9 - 7y^3 + 9y + 10$. Evaluate $f(2)$ and $f(1)$.

---

- (AxPx25x01x15x025)
NOTAGS
Evaluate the following (by first simplifying, then evaluating).
$g(t) = -t^2 - t - 1 + t + 2t^2 + 1$. Evaluate $g(-1)$ and $g(0)$.

---

- (AxPx25x01x15x026)
NOTAGS
Evaluate the following (by first simplifying, then evaluating).
$h(u) = 3 - 30u^2 + 47u + 15u^2 - 30u + 15u^2 - 17u$. Evaluate $h(2024)$ and $g(-3024)$.

---

- (TESTxAxPx25x02x18x001)
NOTAGS
Find the leading term and leading coefficient in each polynomial.

a) $7x^5 - 3x + 2x^3$ b) $5x - x^2 + x^5 - 8x^3$

---

## 4.12 Polynomial Multiplication *(PM)*

- (AxPMx25x01x17x001)
NOTAGS
Multiply.

a) $2x(x + 3) =$ b) $-2x(x - 2) =$

---

- (AxPMx25x01x17x002)
NOTAGS
Multiply.

a) $3x^2(x + 5) =$ b) $2x(x^3 - 7x^2 + 4x + 6) =$

---

- (AxPMx25x01x17x003)
NOTAGS
Multiply.

  a) $-4x^2(-2x^2-5x+6) =$

  b) $2xy(x-2y+xy) =$

---

- (AxPMx25x02x12x003)
NOTAGS
Multiply. $-3x^2(-2x^2+3x-1) =$

---

- (AxPMx25x01x17x004)
NOTAGS
Multiply.

  a) $(3-3x^2+6x)\cdot x^2 =$

  b) $-4xy^2(-2x^2y-5) =$

---

- (AxPMx25x01x17x005)
NOTAGS
Multiply.
$(x+3)(x+2) =$

---

- (AxPMx25x01x17x006)
NOTAGS
Multiply.
$(x+5)(x-3) =$

---

- (AxPMx25x01x17x007)
NOTAGS
Multiply.
$(2x-3)(5x-6) =$

---

- (AxPMx25x02x12x001)
NOTAGS
Multiply.
$(x-3)(4x-5) =$

---

- (AxPMx25x01x17x008)
NOTAGS
Multiply.
$(x^2-5)(-x-3) =$

---

- (AxPMx25x01x17x009)
NOTAGS
Multiply.
$(x+3)(x^2+3x+1) =$

---

- (AxPMx25x01x17x010)
NOTAGS
Multiply.
$(x-3)(x^2-6x-4) =$

---

- (AxPMx25x01x17x011)
NOTAGS
Multiply.
$(x^2+x+3)(x^2-7) =$

---

- (AxPMx25x01x17x012)
NOTAGS
Multiply.
$(x^2-6x-4)(x-5) =$

---

- (AxPMx25x01x17x013)
NOTAGS
Multiply.
$(x^2-2x-3)(x^2-x-4) =$

---

- (AxPMx25x01x17x014)
NOTAGS
Multiply.
$(-3x^2-2x+1)(-x^3-5x+6) =$

---

- (AxPMx25x01x17x015)
NOTAGS
Multiply.
$x(x+3)(x+2) =$

---

- (AxPMx25x01x17x016)
NOTAGS
Multiply.
$(x+1)(x+2)(x+3) =$

---

- (AxPMx25x01x17x017)
NOTAGS
Multiply.
$(x+3)^2 =$

---

- (AxPMx25x01x17x018)
NOTAGS
Multiply.
$(x-3)^2 =$

---

- (AxPMx25x01x17x019)
NOTAGS
Multiply.
$(x-9)^2 =$

---

- (AxPMx25x01x17x020)
NOTAGS
Multiply.
$(x+9)^2 =$

---

- (AxPMx25x01x17x021)
NOTAGS
Multiply.
$(x+1)^2 =$

---

- (AxPMx25x01x17x022)
NOTAGS
Multiply.
$(x-1)^2 =$

---

- (AxPMx25x01x17x023)
NOTAGS
Multiply.
$(x-7)^2 =$

---

- (AxPMx25x01x17x024)
NOTAGS
Multiply.
$(x+7)^2 =$

---

- (AxPMx25x01x17x025)
NOTAGS
Multiply.
$(x-4)(x+4) =$

---

- (AxPMx25x01x17x026)
NOTAGS
Multiply.
$(x-10)(x+10) =$

---

- (AxPMx25x01x17x027)
NOTAGS
Multiply.
$(x+1)(x-1) =$

---

- (AxPMx25x01x17x028)
NOTAGS
Multiply.
$(x+3)(x-3) =$

---

- (AxPMx25x01x17x029)
NOTAGS
Multiply.
$(a+b)^2 =$

---

- (AxPMx25x01x17x030)
NOTAGS
Multiply.
$(a-b)^2 =$

---

- (AxPMx25x01x17x031)
NOTAGS
Multiply.
$(a+b)(a-b) =$

---

- (AxPMx25x01x18x001)
NOTAGS
Expand and simplify.
$5(3x+4)$

---

- (AxPMx25x01x18x002)
NOTAGS
Expand and simplify.
$x(3x+4)$

---

- (AxPMx25x01x18x003)
NOTAGS
Expand and simplify.
$(3x+4)x$

---

- (AxPMx25x01x18x004)
NOTAGS
Expand and simplify.
$5x(3x+4)$

---

- (AxPMx25x01x18x005)
NOTAGS
Expand and simplify.
$5x^2(3x+4)$

---

- (AxPMx25x01x18x006)
NOTAGS
Expand and simplify.
$(3x-4)(-5x^2)$

---

- (AxPMx25x01x18x007)
NOTAGS
Expand and simplify.
$(3x-4x^6)(-5x^2)$

---

- (AxPMx25x01x18x008)
NOTAGS
Expand and simplify.
$(3x+2)(4x+1)$

---

- (AxPMx25x01x18x009)
NOTAGS
Expand and simplify.
$(3x+2)(4x+1)$

---

- (AxPMx25x01x18x010)
NOTAGS
Expand and simplify.
$(3x+2)(4x-1)$

---

- (AxPMx25x01x18x011)
NOTAGS
Expand and simplify.
$(3x-2)(4x-1)$

---

- (AxPMx25x01x18x012)
NOTAGS
Expand and simplify.
$(3x+2)^2$

---

- (AxPMx25x01x18x013)
NOTAGS
Expand and simplify.
$(3x-2)^2$

---

- (AxPMx25x01x18x014)
NOTAGS
Expand and simplify.
$(3x-2)(3x+2)$

---

- (AxPMx25x01x18x015)
NOTAGS
Expand and simplify.
$(x-2)(x+2)$

---

- (AxPMx25x01x18x016)
NOTAGS
Expand and simplify.
$(3x^2+2x+2)(x+4)$

---

- (AxPMx25x01x18x017)
NOTAGS
Expand and simplify.
$(3x^2-2x+2)(x-4)$

---

## 4.13 Factoring *(F)*

### 4.13.1 By Factoring Out GCF *(GCF)*

- (AxFxGCFx25x02x12x001)
NOTAGS
Factor $20xy^4 - 15y^3 =$

---

- (AxFxGCFx25x01x17x001)
NOTAGS
Factor out the GCF.

a) $20A + 30yA =$ b) $30xA - 5A =$

---

- (AxFxGCFx25x01x17x002)
NOTAGS
Factor out the GCF.

a) $x(x+1) - 5(x+1) =$ b) $9(y-1) - 5x(y-1) =$

---

- (AxFxGCFx25x01x17x003)
NOTAGS
Factor out the GCF.

a) $x^2(x^2+4) - 4(x^2+4) =$ b) $x^2(x^2+4) - 4(x^2+4) =$

---

- (AxFxGCFx25x01x17x004)
NOTAGS
"Shift the negatives".

  a) $2(-x+4) = -2(x-4)$ (example)  b) $3(-y+7) =$

---

- (AxFxGCFx25x01x17x005)
NOTAGS
"Shift the negatives".

  a) $-3y(-y+7) =$  b) $(-x-5)7 =$

---

- (AxFxGCFx25x01x17x006)
NOTAGS
"Shift the negatives".

  a) $(u^2-7)(-34y) =$  b) $3x(-y+7) =$

---

### 4.13.2 By Grouping *(GP)*

- (AxFxGPx25x01x17x007)
NOTAGS
Factor $cd + 6c + 4d + 24 =$

---

- (AxFxGPx25x01x17x008)
NOTAGS
Factor by Grouping. $6y^2 + 7y + 24y + 28 =$

---

- (AxFxGPx25x01x17x009)
NOTAGS
Factor by Grouping.
$x^2 - x + 4x - 4 =$

---

- (AxFxGPx25x01x17x010)
NOTAGS
Factor by Grouping.
$9p^2 + 12p - 15p - 20 =$

---

- (AxFxGPx25x01x17x011)
NOTAGS
Factor by Grouping.
$mn - 6m - 4n + 24 =$

---

- (AxFxGPx25x01x17x012)
NOTAGS
Factor by Grouping.
$x^3 + x^2 + x + 1 =$

---

- (AxFxGPx25x01x17x013)
NOTAGS
Factor by Grouping.
$5x^3 - 3x^2 + 5x - 3 =$

---

- (AxFxGPx25x01x17x014)
NOTAGS
Factor by Grouping.
$3x^2 - x + 6xy - 2y =$

---

### 4.13.3 By Difference Of Squares *(DS)*

- (AxFxDSx25x01x17x015)
NOTAGS
Factor using the GCF and/or using difference of squares.
$x^2 + 9x =$

---

- (AxFxDSx25x01x17x016)
NOTAGS
Factor using the GCF and/or using difference of squares.
$x^2 - 9 =$

---

- (AxFxDSx25x01x17x017)
NOTAGS
Factor using the GCF and/or using difference of squares.
$x^2 - 36 =$

---

- (AxFxDSx25x01x17x018)
NOTAGS
Factor using the GCF and/or using difference of squares.
$15x^4 - 20x^2 =$

---

- (AxFxDSx25x01x17x019)
NOTAGS
Factor using the GCF and/or using difference of squares.
$9x^2 - 1 =$

---

- (AxFxDSx25x01x17x020)
NOTAGS
Factor using the GCF and/or using difference of squares.
$9x^2 - 4y^2 =$

---

- (AxFxDSx25x01x17x021)
NOTAGS
Factor using the GCF and/or using difference of squares.
$4x^4y^3 - 8x^2y + 4xy =$

---

- (AxFxDSx25x01x17x022)
NOTAGS
Factor using the GCF and/or using difference of squares.
$9 - 25x^2 =$

---

- (AxFxDSx25x02x12x001)
NOTAGS
Factor. $25 - 9y^2 =$

---

- (AxFxDSx25x01x17x023)
NOTAGS
Factor using the GCF and/or using difference of squares.
$5x^4 - 15x^3 + 25x^2 =$

---

- (AxFxDSx25x01x17x024)
NOTAGS
Factor using the GCF and/or using difference of squares.
$64x^2 - 49y^2 =$

---

### 4.13.4 By Trinomials *(TRI)*

- (AxFxTRIx25x01x17x001)
NOTAGS
Factor the trinomial.
$x^2 - 6x + 8 =$

---

- (AxFxTRIx25x01x17x002)
NOTAGS
Factor the trinomial.
$x^2 + 9x + 18 =$

---

- (AxFxTRIx25x01x17x003)
NOTAGS
Factor the trinomial.
$x^2 - 5x + 6 =$

---

- (AxFxTRIx25x01x17x004)
NOTAGS
Factor the trinomial.
$x^2 - 3x - 10 =$

---

- (AxFxTRIx25x01x17x005)
NOTAGS
Factor $x^2 + 4x - 12 =$

---

- (AxFxTRIx25x01x17x006)
NOTAGS
Factor the trinomial.
$x^2 - 7x - 8 =$

---

- (AxFxTRIx25x01x17x007)
NOTAGS
Factor the trinomial.
$x^2 + 9x + 14 =$

---

- (AxFxTRIx25x01x17x008)
NOTAGS
Factor the trinomial.
$x^2 + 9x + 8 =$

---

- (AxFxTRIx25x01x17x009)
NOTAGS
Factor the trinomial.
$x^2 - x - 6 =$

---

- (AxFxTRIx25x01x17x010)
NOTAGS
Factor the trinomial.
$x^2 + x - 2 =$

---

- (AxFxTRIx25x01x17x011)
NOTAGS
Factor the trinomial.
$x^2 + 10x + 16 =$

---

- (AxFxTRIx25x01x17x012)
NOTAGS
Factor the trinomial.
$x^2 + 15x + 36 =$

---

- (AxFxTRIx25x01x17x013)
NOTAGS
Factor the trinomial.
$5x^2 + 21x + 4 =$

---

- (AxFxTRIx25x01x17x014)
NOTAGS
Factor the trinomial.
$8x^2 + 25x + 3 =$

---

- (AxFxTRIx25x01x17x015)
NOTAGS
Factor the trinomial.
$6x^2 + x - 15 =$

---

- (AxFxTRIx25x01x17x016)
NOTAGS
Factor the trinomial.
$2x^2 - 27x - 45 =$

---

- (AxFxTRIx25x01x17x017)
NOTAGS
Factor the trinomial.
$5x^2 - 9x + 4 =$

---

- (AxFxTRIx25x01x17x018)
NOTAGS
Factor the trinomial.
$4x^2 + 16x + 15 =$

---

- (AxFxTRIx25x01x17x019)
NOTAGS
Factor the trinomial.
$9x^2 + 6x + 1 =$

---

- (AxFxTRIx25x01x17x020)
NOTAGS
Factor the trinomial.
$9 + 30x + 25x^2 =$

---

- (GROUPxAxFxTRIx25x02x10x001)
NOTAGS

a) Choose 3 products of the form $(x + E)(x + F)$, where $E$ and $F$ are numbers (some of your choices for $E$ and $F$ should include negative numbers).

b) For each choice, multiply it out to get an expression of the form $x^2 + Sx + P$, where $S$ and $P$ are numbers.

c) Based on your 3 examples, how do $S$ and $P$ relate to $E$ and $F$?

---

### 4.13.5 By Some Method *(ANY)*

- (AxFxANYx25x01x17x001)
NOTAGS
Factor. $x^5 + 4x^3 =$

---

- (AxFxANYx25x01x17x002)
NOTAGS
Factor. $75m^3 + 12m =$

---

- (AxFxANYx25x01x17x003)
NOTAGS
Factor. $y^2 - 25 =$

---

- (AxFxANYx25x01x17x004)
NOTAGS
Factor. $25 - y^2 =$

---

- (AxFxANYx25x01x17x005)
NOTAGS
Factor. $p^2 + 11p + 30 =$

---

- (AxFxANYx25x01x17x006)
NOTAGS
Factor. $a^5 + 9a^3 =$

---

- (AxFxANYx25x01x17x007)
NOTAGS
Factor. $a^5 - 9a^3 =$

---

- (AxFxANYx25x01x17x008)
NOTAGS
Factor. $4x^2 + 40x + 84 =$

---

- (AxFxANYx25x01x17x009)
NOTAGS
Factor. $12ab - 6a + 10b - 5 =$

---

- (AxFxANYx25x01x17x010)
NOTAGS
Factor. $y^2 - 5y + 6 =$

---

- (AxFxANYx25x01x17x011)
NOTAGS
Factor. $2n^2 + 13n - 7 =$

---

- (AxFxANYx25x01x17x012)
NOTAGS
Factor. $4x^5y - 32x^2y =$

---

- (AxFxANYx25x01x17x013)
NOTAGS
Factor. $7b^2 + 7b - 42 =$

---

- (AxFxANYx25x01x17x014)
NOTAGS
Factor. $36q^2 - 100 =$

---

- (AxFxANYx25x02x12x001)
NOTAGS
Factor. $18x^2y - 2y =$

---

- (AxFxANYx25x01x17x015)
NOTAGS
Factor. $36x^2y + 15xy - 6y =$

---

- (AxFxANYx25x01x17x016)
NOTAGS
Factor. $-11 - 10x + x^2 =$

---

- (AxFxANYx25x01x17x017)
NOTAGS
Factor. $6n - 7 + n^2 =$

---

- (AxFxANYx25x01x17x018)
NOTAGS
Factor. $8x^2 - 10x - 3 =$

---

- (AxFxANYx25x01x17x019)
NOTAGS
Factor. $16x^2 - 24xy + 9y^2 - 64 =$

---

- (AxFxANYx25x01x17x020)
NOTAGS
Factor. $9 + 30x + 25x^2 =$

---

- (AxFxANYx25x01x17x021)
NOTAGS
Factor. $49b^2 - 112b + 64 =$

---

- (AxFxANYx25x01x17x022)
NOTAGS
Factor. $75x^3 + 12x =$

---

- (AxFxANYx25x01x17x023)
NOTAGS
Factor. $x^4 - 81 =$

---

- (AxFxANYx25x01x17x024)
NOTAGS
Factor. $3x^4 - 48 =$

---

- (AxFxANYx25x01x17x025)
NOTAGS
Factor. $y^6 - 1 =$

---

- (AxFxDSx25x01x17x025)
NOTAGS
Factor.
$12x^4y^3 - 18x^3y^2 - 6x^4y^6 =$

---

- (AxFxDSx25x01x17x026)
NOTAGS
Factor.
$12x^4y^3z - 18x^3y^2z^3 - 9x^4z^2 =$

---

- (TESTxxxAxFxDSx25x04x28x001)
NOTAGS
Factor.

a) $3x^3y - 21x^2y + 30xy$

b) $x^3y^2 - x^3y + 4x^3y - 4x^3$

---

## 4.14 Polynomial Division (Synthetic Division, etc) *(PDIV)*

- (AxPDIVx25x01x22x003)
NOTAGS
Is $(x - 1)$ a factor of $x^5 + 3x^4 - x + 2$?

---

- (AxPDIVx25x02x24x001)
NOTAGS
Is $(x + 1)$ a factor of $x^5 - 2x^4 - x + 2$?

---

- (AxPDIVx25x01x22x004)
NOTAGS
Use long division to find $(4x^2 - 10x + 6) \div (4x + 2)$.

---

- (AxPDIVx25x02x03x001)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{x^3 - 2x^2 - 5x + 6}{x + 2}$

---

- (AxPDIVx25x02x03x004)
NOTAGS
Divide $\dfrac{x^4 - 2x^2 - 5x + 6}{x - 3}$ using long division. Write your answer in the form $N(x) = d(x)q(x) + r(x)$, where $N$ is the divident, $d$ is the divisor, $q$ is the quotient, and $r$ is the remainder.

---

- (AxPDIVx25x02x03x002)
NOTAGS
Divide $\dfrac{3x^4 - 2x^3 - 7x^2 + x - 2}{x^2 - 2x + 3}$ using long division. Write your answer in the form $N(x) = d(x)q(x) + r(x)$, where $N$ is the numerator, $d$ is the denominator, $q$ is the quotient, and $r$ is the remainder.

---

- (AxPDIVx25x02x03x003)
NOTAGS
Divide $\dfrac{x^5 + x^4 - x^3 - x^2 + 3x - 1}{x^2 + x + 1}$ using long division. Write your answer in the form $\dfrac{N(x)}{d(x)} = q(x) + \dfrac{r(x)}{d(x)}$, where $N$ is the numerator, $d$ is the denominator, $q$ is the quotient, and $r$ is the remainder.

---

- (AxPDIVx25x02x03x006)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{x^7 - 1}{x - 1}$

---

- (AxPDIVx25x02x03x008)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{-2x^3 - 7x^2 + x - 2}{x + 1}$

---

- (AxPDIVx25x02x03x009)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{x^4 - x^3 + x - 1}{x - 2}$

---

- (AxPDIVx25x02x03x010)
NOTAGS
Solve the equation $2x^3 - 3x^2 - 11x + 6 = 0$ given that $-2$ is a zero.

---

- (AxPDIVx25x02x03x011)
NOTAGS
Solve the equation $3x^3 + 7x^2 - 22x - 8 = 0$ given that $-\frac{1}{3}$ is a root.

---

- (AxPDIVx25x02x03x013)
NOTAGS
The remainder from dividing a polynomial $p(x)$ by $\left(x - \frac{1}{2}\right)$ is $\frac{11}{17}$. How much is $p\left(\frac{1}{2}\right)$? Which theorem are you using?

---

- (AxPDIVx25x02x03x113)
NOTAGS
The remainder from dividing a polynomial $f(x)$ by $(x + 5)$ is $\frac{6}{7}$. How much is $f(-5)$? Which theorem are you using?

---

- (AxPDIVx25x02x03x014)
NOTAGS
What is the remainder you would get it you divide the polynomial $f(x) = x^{103}+x^{50}+2$ by $(x - 1)$?

---

- (HWxAxPDIVx25x02x03x005)
type: HW
Divide $\dfrac{3x^6 - 2x^3 - 7x^2 - 2}{x^2 - x + 2}$ using long division. Write your answer in the form $\dfrac{N(x)}{d(x)} = q(x) + \dfrac{r(x)}{d(x)}$, where $N$ is the divident, $d$ is the divisor, $q$ is the quotient, and $r$ is the remainder.

---

- (TESTxAxPDIVx25x02x24x001)
NOTAGS

a) Is $(x - 2)$ a factor of $x^3 - 2x - x + 2$?

b) The remainder from dividing a polynomial $p(x)$ by $(x + 4)$ is 3. How much is $p(-4)$? Which theorem are you using?

---

- (TESTxAxPDIVx25x02x03x007)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{x^3 - 2x^2 - 5x + 6}{x - 3}$

---

- (TESTxAxPDIVx25x03x28x001)
NOTAGS
Find the quotient and remainder that result from the following division: $\dfrac{4x^3 + x^2 - 10x + 20}{x + 2}$

---

- (TESTxxxAxPDIVx25x03x27x001)
NOTAGS
Solve the equation $x^3 - 7x^2 + 7x + 15 = 0$ given that 3 is a zero.

---

## 4.15 Solving Polynomial Equations *(SP)*

### 4.15.1 Already Factored *(FA)*

- (AxSPxFAx01x17x001)
NOTAGS
Solve.

a) $(x + 2)(x + 3) = 0$ b) $(x - 2)(x - 3) = 0$

---

- (AxSPxFAx01x17x002)
NOTAGS
Solve.

a) $(x + 2)(x - 3) = 0$ b) $(x - 2)(x + 3) = 0$

---

- (AxSPxFAx01x17x003)
NOTAGS
Solve.

a) $(x)(x + 8) = 0$ b) $(x + 2)(x - 10)(x + 31) = 0$

---

- (AxSPxFAx01x17x004)
NOTAGS
Solve.

a) $(x + 7)(x + 7) = 0$ b) $(x + 2)(x + 2)(4x) = 0$

---

- (AxSPxFAx01x17x005)
NOTAGS
Solve.

a) $(x + 7)(2x - 1) = 0$ b) $(7x - 21)(2x) = 0$

---

- (AxSPxFAx01x17x006)
NOTAGS
Solve.
$x(x-1)(5x+1)(x-3)=0$

---

- (AxSPxFAx01x17x013)
NOTAGS
Solve.
$(x+2)(x-3)(3x-5)=0$

---

### 4.15.2 Solve by Factoring *(F)*

- (AxSPxFx01x17x008)
NOTAGS
Solve.
$x^2+9x+18=0$

---

- (AxSPxFx01x17x009)
NOTAGS
Solve.
$x^2-5x+6=0$

---

- (AxSPxFx01x17x010)
NOTAGS
Solve.
$x^2-10=3x$

---

- (AxSPxFx01x17x011)
NOTAGS
Solve.
$x^2+4x=12$

---

- (AxSPxFx01x17x012)
NOTAGS
Solve.
$x^2=7x+8$

---

- (AxSPxFx01x17x014)
NOTAGS
Solve.
$x^2=9$

---

- (AxSPxFx01x17x015)
NOTAGS
Solve.
$4x^4-20x^3+24x^2=0$

---

- (AxSPxFx01x17x016)
NOTAGS
Solve.
$12x^3 + 12x^2 = 24x$

---

- (AxSPxFx01x17x017)
NOTAGS
Solve.
$100x^2 = 9$

---

- (AxSPxFx01x17x018)
NOTAGS
Solve.
$x^2 + 15x + 36 = 0$

---

- (AxSPxFx01x17x019)
NOTAGS
Solve.
$5x^2 + 21x + 4 = 0$

---

- (AxSPxFx01x17x020)
NOTAGS
Solve.
$8x^2 + 80x + 200 = 0$

---

- (AxSPxFx01x17x021)
NOTAGS
Solve.
$6x^3 = 24x$

---

- (AxSPxFx01x17x022)
NOTAGS
Solve (only allow real solutions).
$5x^2 + 4 = 9x$

---

- (AxSPxFx25x02x26x001)
NOTAGS
Solve (complex solutions allowed): $x^2 + 12 = -7x$

---

- (TESTxAxSPxFx01x17x007)
NOTAGS
Solve.
$x^2 - 6x + 8 = 0$

---

- (TESTxAxSPxFx25x01x20x023)
NOTAGS
Solve the equation.
$-5x - 2 = -3x^2$

---

- (TESTxAxSPxFx25x04x28x001)
NOTAGS
Solve. $7x^2 - 8x = -1$

---

### 4.15.3 Solve by Quadratic Equation *(QE)*

- (AxSPxQEx25x01x20x005)
NOTAGS
Solve each equation (complex solutions are NOT allowed):

  a) $2x^2 - 2x = 6$ b) $x^2 + x + 2 = 0$

---

- (AxSPxQEx25x01x20x027)
NOTAGS
Solve (complex solutions are allowed):

  a) $x^2 + 9x = -14$

  b) $y^2 = 2y + 35$

---

- (AxSPxQEx25x01x20x030)
NOTAGS
Solve.

  a) $3u^2 - 6u = -2$

  b) $y^2 = 2y - 2$

---

- (AxSPxQEx25x02x26x001)
NOTAGS
Solve (complex solutions allowed): $x^2 - 4x - 7 = 0$.

---

- (AxSPxQEx25x02x26x101)
NOTAGS
Solve (complex solutions allowed): $-2x + 1 = -2x^2$.

---

- (TESTxxxAxSPxQEx25x04x28x001)
NOTAGS
Solve. $3x^2 + 1 = 2x$

---

### 4.15.4 Polynomial Intercepts with Theory *(PT)*

- (AxSPxPTx25x02x18x005)
NOTAGS
Suppose a polynomial function has degree 5. What is the maximum number of intercepts it can have?

---

- (AxSPxPTx25x02x18x106)
NOTAGS
Suppose a polynomial function has degree 7. If it is completely factored, what is the maximum number of factors?

---

- (AxSPxPTx25x02x18x007)
NOTAGS
Suppose a polynomial function has degree 7. If it is completely factored, and complex numbers are allowed, how many factors are there?

---

- (AxSPxPTx25x02x18x001)
NOTAGS
Suppose a polynomial function has exactly 2 roots. What is the least possible degree of the function?

---

- (AxSPxPTx25x02x18x002)
NOTAGS
Suppose a polynomial function has exactly 5 intercepts. What is the least possible degree of the function?

---

- (AxSPxPTx25x02x18x003)
NOTAGS
Draw a polynomial function which has to have degree at least 3 (just a rough sketch; no equation required).

---

- (AxSPxPTx25x02x18x004)
NOTAGS
Draw a polynomial function which has to have degree at least 5 (just a rough sketch; no equation required).

---

- (AxSPxPTx25x02x18x105)
NOTAGS
Show that the polynomial function $f(x) = x^3 - 9x$ has a zero between $x = -4$ and $x = -2$.

---

- (AxSPxPTx25x02x18x006)
NOTAGS
Show that the polynomial function $g(x) = x^4 - 7x^3 - 5$ has a zero between $x = -1$ and $x = 2$.

---

- (TESTxAxSPxPTx25x02x18x001)
NOTAGS
Show that the polynomial function $g(x) = x^5 - 3x$ has a zero between $x = 1$ and $x = 2$.

---

### 4.15.5 Solve Using Rational Zeroes Theorem, etc *(RZ)*

- (AxSPxRZx25x02x03x005)
NOTAGS
Find the possible rational zeros of the polynomial $4x^3 + 5x^2 - 3x + 6$

---

- (AxSPxRZx25x02x03x006)
NOTAGS
Find the possible rational zeros of the polynomial $6x^4 + 3x^2 + 4x - 15$

---

- (AxSPxRZx25x02x03x001)
NOTAGS
Solve. $x^3 - 4x^2 - 7x + 10 = 0$. Then use that result to easily factor $x^3 - 4x^2 - 7x + 10$.

---

- (AxSPxRZx25x02x03x002)
NOTAGS
Solve. $3x^3 - 8x^2 - 8x + 8 = 0$

---

- (AxSPxRZx25x02x03x003)
NOTAGS
Solve. $x^4 + 3x^3 - 20x^2 + 24x - 8 = 0$. Then use that result to easily factor $x^4 + 3x^3 - 20x^2 + 24x - 8$.

---

- (AxSPxRZx25x02x03x004)
NOTAGS
Solve. $x^4 - x^3 + 2x^2 - 4x - 8 = 0$

---

- (TESTxxxAxSPxRZx25x03x27x001)
NOTAGS
Find the possible rational zeros of the polynomial $4x^5 + 6x^2 + 4x - 10$ (all fractional answers should be simplified, and do not repeat the same answer twice).

---

## 4.16 Graphing Polynomials *(PG)*

- (AxPGx25x02x18x001)
NOTAGS
Graph $h(x) = 4(x + 3)(x - 1)$. Find its zeroes and the multiplicity of each.

---

- (AxPGx25x02x18x002)
NOTAGS
Graph $f(x) = (x+3)^2(x-1)$. Find its zeroes and the multiplicity of each.

---

- (AxPGx25x02x18x003)
NOTAGS
Graph $g(x) = 2(x-5)^3(x+1)^2$. Find its zeroes and the multiplicity of each.

---

- (AxPGx25x02x18x004)
NOTAGS
Graph $f(x) = (x-3)^3(x+3)^2$. Find its zeroes and describe its end behavior.

---

- (AxPGx25x02x18x005)
NOTAGS
Graph $f(x) = (x+7)(x+2)^4$. Find its zeroes and describe its end behavior.

---

- (AxPGx26x05x01x002)
type: TEST
Let $g(x) = -3(x-12)^5(x+1)^2(x-7)$. Find its zeroes and the multiplicity of each.

---

- (AxPGx25x02x18x006)
NOTAGS
Graph a polynomial function of degree 3 with zeros at $-3, -2$, and 1 and $y$ intercept at $(0, 5)$.

---

- (AxPGx25x02x18x007)
NOTAGS
Graph a polynomial function of degree 5 with a root of multiplicity 2 at $-3$, a root of multiplicity 2 at 2, and a final root at $-2$. It has a $y$-intercept at $(0, -3)$.

---

- (TESTxAxPGx25x02x18x001)
NOTAGS
Graph $f(x) = 3(x+2)^2(x-3)^3$ roughly: show the x-intercepts and the end behavior. Also find its zeroes and the multiplicity of each.

---

- (TESTxAxPGx25x02x18x002)
NOTAGS
Graph $f(x) = (x+2)(x-4)^2$ roughly: show the x-intercepts and the end behavior. Also, describe its end behavior using limit notation.

---

- (TESTxAxPGx25x02x18x003)
NOTAGS
Graph a polynomial function of degree 4 with a root of multiplicity 2 at $-3$, and roots at 2 and 4. It has a $y$-intercept at $(0, 3)$.

---

## 4.17 Radicals and Rational Exponents

### 4.17.1 Simplifying Radicals *(SR)*

- (AxSRx25x01x17x001)
NOTAGS
Simplify.

  a) $\sqrt{25}$ b) $\sqrt{-25}$

---

- (AxSRx25x01x17x002)
NOTAGS
Simplify.

  a) $\sqrt{1}$ b) $\sqrt{0}$

---

- (AxSRx25x01x17x003)
NOTAGS
Simplify.

  a) $\sqrt{20}$ b) $\sqrt{10}$

---

- (AxSRx25x01x17x004)
NOTAGS
Simplify.

  a) $\sqrt{98}$ b) $\sqrt{50}$

---

- (AxSRx25x01x17x005)
NOTAGS
Simplify.

  a) $\sqrt[3]{8}$ b) $\sqrt[3]{27}$

---

- (AxSRx25x01x17x006)
NOTAGS
Simplify.

  a) $\sqrt[3]{125}$ b) $\sqrt[3]{40}$

---

- (AxSRx25x01x17x007)
NOTAGS
Simplify.

a) $\sqrt[3]{108}$ b) $\sqrt[3]{x^9}$

---

- (AxSRx25x01x17x008)
NOTAGS
Simplify.

a) $\sqrt[7]{x^{21}}$ b) $\sqrt{x^4}$

---

- (AxSRx25x01x17x009)
NOTAGS
Simplify.

a) $\sqrt[4]{x^{32}}$ b) $\sqrt[9]{z^{36}}$

---

- (AxSRx25x01x17x010)
NOTAGS
Simplify.

a) $\sqrt[11]{z^{22}}$ b) $\sqrt[4]{81x^{12}}$

---

- (AxSRx25x01x17x011)
NOTAGS
Simplify.

a) $\sqrt[7]{128\,y^{35}}$ b) $\sqrt[3]{125\,z^{35}}$

---

- (AxSRx25x01x17x012)
NOTAGS
Simplify.

a) $\sqrt{9\,x^{32}\,y^{18}}$ b) $\sqrt{36\,x^2\,y^4\,z^6}$

---

- (AxSRx25x01x17x013)
NOTAGS
Simplify.

a) $\sqrt{-9}$ b) $\sqrt{-16}$

---

- (AxSRx25x01x17x014)
NOTAGS
Simplify.

  a) $\sqrt[6]{-64}$

  b) $\sqrt[3]{-8}$

---

- (AxSRx25x01x22x007)
NOTAGS
For each expression, evaluate or state it is undefined (no complex numbers allowed):

  a) $\sqrt{-16}$

  b) $\sqrt[3]{-8}$

  c) $\sqrt[4]{-16}$

  d) $\sqrt[5]{-1}$

---

- (AxSRx25x01x17x015)
NOTAGS
Simplify.

  a) $\sqrt{x^9}$

  b) $\sqrt{y^5}$

---

- (AxSRx25x01x17x016)
NOTAGS
Simplify.

  a) $\sqrt{z^{21}}$

  b) $\sqrt{18\,x^5}$

---

- (AxSRx25x01x17x017)
NOTAGS
Simplify.

  a) $\sqrt{12\,x^{13}}$

  b) $\sqrt{50\,x^7\,y^4}$

---

- (AxSRx25x01x17x018)
NOTAGS
Simplify.

  a) $\sqrt{15\,x^3\,y^9\,z^5}$

  b) $\sqrt{20\,x^5\,y^9}$

---

- (AxSRx25x01x17x019)
NOTAGS
Simplify.

  a) $\sqrt[3]{x^4}$

  b) $\sqrt[3]{y^7}$

---

- (AxSRx25x01x17x020)
NOTAGS
Simplify.

a) $\sqrt[3]{z^5}$ b) $\sqrt[3]{x^{25}}$

---

- (AxSRx25x01x17x021)
NOTAGS
Simplify.

a) $\sqrt[4]{32\,x^9}$ b) $\sqrt[3]{24\,x^5\,y^{10}}$

---

- (AxSRx25x01x17x022)
NOTAGS
Simplify.
$\sqrt[3]{4000\,x^{91}}$

---

- (AxSRx25x01x17x023)
NOTAGS
Simplify.

a) $\sqrt{\frac{45}{80}}$ b) $\sqrt[3]{\frac{16}{27}}$

---

- (AxSRx25x01x17x024)
NOTAGS
Simplify.

a) $\sqrt[3]{\frac{24}{81}}$ b) $\sqrt[4]{\frac{32}{162}}$

---

- (AxSRx25x01x17x025)
NOTAGS
Simplify.

a) $\sqrt{\frac{x^{10}}{x^6}}$ b) $\sqrt[3]{\frac{y^{11}}{y^2}}$

---

- (AxSRx25x01x17x026)
NOTAGS
Simplify.

a) $\sqrt[5]{\frac{x^{12}}{x^7}}$ b) $\sqrt[6]{\frac{y^{30}}{y^{12}}}$

---

- (AxSRx25x01x17x027)
NOTAGS
Simplify.

a) $\sqrt{\frac{50\,x^7}{98\,x^3}}$ b) $\sqrt[3]{\frac{15\,x^5}{40\,x^3}}$

---

- (AxSRx25x01x17x028)
NOTAGS
Simplify.

a) $\sqrt{\frac{54\,x^5}{450\,x}}$ b) $\sqrt[4]{\frac{10\,x^9}{32\,x}}$

---

- (TESTxxxAxSRx25x03x25x001)
NOTAGS
Simplify.

a) $5 \cdot \sqrt[3]{48}$ b) $\sqrt{8x^4y^5}$

---

- (TESTxxxAxSRx25x03x25x101)
NOTAGS
Simplify.

a) $\sqrt{\frac{20x^8}{x^2}}$ b) $\frac{2}{\sqrt{6}}$

---

### 4.17.2 Radicals versus Rational Exponents *(RRE)*

- (AxRREx25x01x17x001)
NOTAGS
Rewrite using radicals.

a) $x^{\frac{2}{3}}$ b) $x^{\frac{6}{7}}$

---

- (AxRRExx25x01x17x002)
NOTAGS
Rewrite using radicals.

a) $y^{\frac{9}{2}}$ b) $x^{\frac{2}{5}}$

---

- (AxRRExx25x01x17x003)
NOTAGS
Rewrite using radicals.

a) $y^{\frac{1}{3}}$ b) $u^{\frac{1}{2}}$

---

- (AxRRExx25x01x17x004)
NOTAGS
Rewrite using radicals.

a) $x^{\frac{1}{4}}$ b) $y^{\frac{1}{5}}$

---

- (AxRRExx25x01x17x005)
NOTAGS
Rewrite using radicals.

a) $y^{-\frac{1}{3}}$ b) $x^{-\frac{1}{2}}$

---

- (AxRRExx25x01x17x006)
NOTAGS
Rewrite using radicals.

a) $y^{-\frac{1}{4}}$ b) $u^{-\frac{1}{5}}$

---

- (AxRRExx25x01x17x007)
NOTAGS
Rewrite using radicals.

a) $x^{-\frac{7}{3}}$ b) $y^{-\frac{4}{5}}$

---

- (AxRRExx25x01x17x008)
NOTAGS
Rewrite using radicals.

a) $u^{-\frac{3}{2}}$ b) $x^{\frac{2}{3}}$

---

- (AxRRExx25x01x17x009)
NOTAGS
Rewrite using rational exponents.

  a) $\sqrt{x^3}$ b) $\sqrt{x^5}$

---

- (AxRRExx25x01x17x010)
NOTAGS
Rewrite using rational exponents.

  a) $\sqrt[3]{x^5}$ b) $\sqrt[5]{x^3}$

---

- (AxRRExx25x01x17x011)
NOTAGS
Rewrite using rational exponents.

  a) $\sqrt{x}$ b) $\sqrt[3]{x}$

---

- (AxRRExx25x01x17x012)
NOTAGS
Rewrite using rational exponents.

  a) $\sqrt[5]{x}$ b) $\sqrt[7]{x}$

---

- (AxRRExx25x01x17x013)
NOTAGS
Rewrite using rational exponents.

  a) $(\sqrt{x})^3$ b) $(\sqrt[3]{x})^5$

---

- (AxRRExx25x01x17x014)
NOTAGS
Rewrite using rational exponents.

  a) $(\sqrt[3]{x})^{-5}$ b) $\sqrt[5]{x^{-3}}$

---

- (AxRRExx25x01x17x015)
NOTAGS
Rewrite using rational exponents.

  a) $\left(\frac{1}{\sqrt{x}}\right)^3$ b) $\frac{1}{(\sqrt[6]{x})^5}$

---

- (AxRREx25x01x17x016)
NOTAGS
Rewrite using rational exponents.

  a) $\frac{1}{(\sqrt[3]{x})^4}$ b) $\left(\frac{1}{\sqrt[5]{x}}\right)^{-2}$

---

- (AxRREx25x01x20x007)
NOTAGS
For each expression, evaluate/simplify or state if undefined (and why).

  a) $1000^{-\frac{1}{3}}$

  b) $\sqrt[4]{\frac{81}{16}}$

---

- (TESTxxxAxRREx25x03x25x001)
NOTAGS
Rewrite using rational exponents.

  a) Rewrite $(\sqrt[5]{x})^3$ using exponents b) Rewrite $y^{-(\frac{3}{4})}$ using radicals.

---

### 4.17.3 Simplifying Rational Exponents *(SRE)*

- (AxSREx25x01x17x017)
NOTAGS
Simplify (so no radical or exponents are left).

  a) $8^{\frac{2}{3}}$ b) $8^{\frac{1}{3}}$

---

- (AxSREx25x01x17x018)
NOTAGS
Simplify (so no radical or exponents are left).

  a) $9^{\frac{1}{2}}$ b) $25^{-\frac{1}{2}}$

---

- (AxSREx25x01x17x019)
NOTAGS
Simplify (so no radical or exponents are left).

  a) $16^{\frac{1}{4}}$ b) $16^0$

---

- (AxSREx25x01x17x020)
NOTAGS
Simplify (so no radical or exponents are left).

  a) $16^{\frac{3}{4}}$ b) $16^{-\frac{3}{4}}$

---

- (AxSREx25x01x17x021)
NOTAGS
Simplify (so no radical or exponents are left).

  a) $27^{\frac{2}{3}}$ b) $27^{-\frac{4}{3}}$

---

- (AxSREx25x01x20x007)
NOTAGS
Simplify (express results without exponents):

  a) $27^{-\frac{2}{3}}$

  b) $\sqrt[3]{\frac{64}{27}}$

---

- (AxSREx25x01x20x017)
NOTAGS
Simplify (express results with positive exponents only and rationalized denominators):

  a) $1000^{-\frac{2}{3}}$

  b) $\left(4x^3y^{-5}\right)^2$

---

- (AxSREx25x01x20x018)
NOTAGS
Simplify.

  a) $\sqrt{50}$

  b) $\sqrt{\frac{5}{8}}$

  c) $\dfrac{9\sqrt{20}}{\sqrt{3}}$

---

- (AxSREx25x01x20x012)
NOTAGS
Simplify the following expression.
$\left(\frac{8x^2y^{-3}u^2}{27x^2y^6u^{-1}}\right)^{-\frac{2}{3}}$

---

- (AxSREx25x01x20x013)
NOTAGS

Simplify (express results with positive exponents only and rationalized denominators):

a) $1000^{-2/3}$ b) $(4x^3y^{-6})^{-3/2}$

---

- (TESTxxxAxSREx25x03x26x001)
NOTAGS
Simplify.

a) $16^{\frac{3}{2}}$ b) $8^{-\frac{1}{3}}$

---

### 4.17.4 Multiplying: Radicals and Rational Exponents *(MR)*

- (AxMRx25x01x17x022)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $x^{\frac{2}{3}} \cdot x^{\frac{7}{3}}$ b) $x^{\frac{6}{7}} \cdot x^{\frac{2}{7}}$

---

- (AxMRx25x01x17x023)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $x^{\frac{3}{7}}$ b) $x^{\frac{2}{5}} \cdot x^{\frac{2}{5}}$

---

- (AxMRx25x01x17x024)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $x^{\frac{1}{6}} \cdot x^{\frac{1}{4}}$ b) $x^{\frac{1}{2}} \cdot x^{-\frac{3}{5}}$

---

- (AxMRx25x01x17x025)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $x^{-\frac{5}{6}} \cdot x^{-\frac{3}{8}}$ b) $x^{-\frac{1}{5}} \cdot x^{\frac{1}{5}}$

---

- (AxMRx25x01x17x026)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $\sqrt{x^3} \cdot \sqrt{x}$ b) $\sqrt[3]{x} \cdot \sqrt[3]{x^5}$

---

- (AxMRx25x01x17x027)
NOTAGS
Multiply and simplify. Write the answer using radicals.

a) $\sqrt[4]{x^3} \cdot \sqrt[6]{x^5}$ b) $\sqrt[5]{x^3} \cdot \sqrt[3]{x^2}$

---

- (AxMRx25x01x17x028)
NOTAGS
Multiply and simplify.

a) $\sqrt{x^3} \cdot \sqrt{x}$ b) $\sqrt[3]{x} \cdot \sqrt[3]{x^5}$

---

- (AxMRx25x01x17x029)
NOTAGS
Multiply and simplify.

a) $\sqrt[4]{x^3} \cdot \sqrt[4]{x^3}$ b) $\sqrt[5]{x^3} \cdot \sqrt[5]{x^6}$

---

- (AxMRx25x01x17x030)
NOTAGS
Multiply and simplify.

a) $\sqrt{6x^2} \cdot \sqrt{2x}$ b) $\sqrt[3]{3x^4y} \cdot \sqrt[3]{10x^5y^2}$

---

- (AxMRx25x01x17x031)
NOTAGS
Multiply and simplify.

a) $\sqrt[4]{20x^3y^5} \cdot \sqrt[4]{12x^3y}$ b) $\sqrt[3]{4x^2y} \cdot \sqrt[3]{12x^5y^3}$

---

- (AxMRx25x01x17x032)
NOTAGS
Multiply and simplify.

a) $\sqrt{6x^3y} \cdot \sqrt{2x^2y}$ b) $\sqrt[3]{25xy^2} \cdot \sqrt[3]{10x^5y}$

---

- (AxMRx25x01x17x033)
NOTAGS
Multiply and simplify.

a) $\sqrt[4]{9x^2y} \cdot \sqrt[4]{18x^3y^2}$ b) $\sqrt[3]{49x^2y^2} \cdot \sqrt[3]{7xy^3}$

---

- (AxMRx25x01x20x008)
NOTAGS
Perform the indicated operation and simplify (express results without negative exponents; positive rational exponents are ok):

a) $4^9 \cdot 4^{50}$

b) $7^{\frac{3}{2}} \cdot 7^{\frac{5}{3}}$

---

- (TESTxxxAxMRx25x02x28x001)
NOTAGS
Simplify (express without negative exponents; positive exponents are ok).

a) $B^{\frac{2}{3}} \cdot B^2$ b) $(x^3y^{-6})^{-\frac{5}{3}}$

---

### 4.17.5 Dividing: Radicals and Rational Exponents *(DR)*

- (AxDRx25x01x17x001)
NOTAGS
Perform the operation and simplify:

$$\sqrt{\frac{8y^5x^6}{z^{12}x^4}}$$

---

- (AxDRx25x01x17x002)
NOTAGS
Perform the operation and simplify: $\dfrac{5}{\sqrt{3y}}$

---

- (AxDRx25x01x22x011)
NOTAGS
Simplify $\dfrac{10ab}{\sqrt{8ab}}$

---

- (AxDRx25x01x20x009)
NOTAGS
Perform the indicated operation and simplify:

  a) $\dfrac{10^5}{10^{\frac{7}{2}}}$

  b) $\dfrac{\pi^{\frac{3}{4}}}{\pi^{\frac{7}{3}}}$

---

- (AxDRx25x01x20x013)
NOTAGS
Simplify.

  a) $\dfrac{25ab^2c^3}{\sqrt{10ab}}$

  b) $\sqrt{\dfrac{a^3b^9c^7}{60\,a^7b}}$

---

- (AxDRx25x01x20x014)
NOTAGS
Simplify.

  a) $\sqrt{\dfrac{5}{8}}$

  b) $\dfrac{2\sqrt{20x^2y^3}}{\sqrt{3}}$

---

- (AxDRx25x01x20x015)
NOTAGS
Simplify $\dfrac{2\sqrt{10}}{1+\sqrt{6}}$

---

- (TESTxxxAxDRx25x04x28x001)
NOTAGS
Simplify

  a) $\dfrac{2\sqrt{27y^3}}{\sqrt{3x}}$

  b) $\dfrac{6}{1+\sqrt{10}}$

---

### 4.17.6 Adding and Subtracting: Radicals *(ASR)*

- (AxASRx25x01x22x029)
NOTAGS
Simplify $\sqrt{13} + 9\sqrt{13} - 21\sqrt{13}$.

---

- (AxASRx25x01x17x003)
NOTAGS
Perform the operation and simplify: $3\sqrt{7} + \sqrt{5} - 6\sqrt{7} + 2\sqrt{5}$

---

- (AxASRx25x01x17x004)
NOTAGS
Perform the operation and simplify: $\sqrt{8} + \sqrt{18} + \sqrt{12}$

---

- (AxASRx25x01x17x005)
NOTAGS
Perform the operation and simplify: $3\sqrt{8} + 5\sqrt{18} + 2\sqrt{12}$

---

- (AxASRx25x01x20x019)
NOTAGS
Perform the indicated operations and simplify:

  a) $\sqrt{20} - \sqrt{45}$

  b) $5\sqrt{8} - 3\sqrt{12} + \sqrt{50}$

---

- (AxASRx25x01x20x020)
NOTAGS
Perform the indicated operations and simplify.

  a) $5\sqrt{8} + 3\sqrt{12} + \sqrt{50}$

  b) $10\sqrt[3]{16} - 3\sqrt[3]{54}$

---

- (AxASRx25x01x31x001)
NOTAGS
Simplify.

  a) $5\sqrt{2} - 3\sqrt{2}$

  b) $6\sqrt[3]{6} + 7\sqrt[3]{6}$

---

- (AxASRx25x01x31x002)
NOTAGS
Simplify.

  a) $-12\sqrt[5]{18} + 7\sqrt[5]{18}$

  b) $4\sqrt[3]{5} - 6\sqrt{5}$

---

- (AxASRx25x01x31x003)
NOTAGS
Simplify.

a) $3\sqrt{2x} + 5\sqrt{2x}$ b) $-5\sqrt[3]{5x^2} + 7\sqrt[3]{5x^2}$

---

- (AxASRx25x01x31x004)
NOTAGS
Simplify.

a) $-4\sqrt[4]{7xy^2} + 5\sqrt[4]{7xy^2}$ b) $4\sqrt[3]{10x^2} - 6\sqrt[3]{10x}$

---

- (AxASRx25x01x31x005)
NOTAGS
Simplify.

a) $\sqrt{27} + \sqrt{75}$ b) $\sqrt[3]{24} - \sqrt[3]{81}$

---

- (AxASRx25x01x31x006)
NOTAGS
Simplify.

a) $\sqrt{72} + \sqrt{98}$ b) $4\sqrt{45} - \sqrt{80}$

---

- (AxASRx25x01x31x007)
NOTAGS
Simplify.

a) $\sqrt{72x^5} - \sqrt{50x^5}$ b) $\sqrt[3]{64x^6} - 3\sqrt[3]{125x^6}$

---

- (AxASRx25x01x31x008)
NOTAGS
Simplify.

a) $\sqrt{48x^5} + \sqrt{75x^5}$ b) $4\sqrt[4]{243y^6} + 2\sqrt[4]{3y^6}$

---

### 4.17.7 Radicals: Combining Arithmetic Operations *(CR)*

- (AxCRx25x01x17x006)
NOTAGS
Perform the operation and simplify: $(\sqrt{2})^2 + (\sqrt{6})^2$

---

- (AxCRx25x01x17x007)
NOTAGS
Perform the operation and simplify: $(\sqrt{2}+\sqrt{6})^2$

---

- (AxCRx25x01x20x020)
NOTAGS
Perform the indicated operations and simplify:

  a) $7\sqrt[3]{16}-3\sqrt[3]{54}$

  b) $3\sqrt{8}\left(2\sqrt{10}-\sqrt{27}\right)$

---

- (AxCRx25x01x20x021)
NOTAGS
Perform the indicated operations and simplify:

  a) $(\sqrt{2}+\sqrt{3})^2$

  b) $\dfrac{6}{2+\sqrt{5}}$

---

- (AxCRx25x01x20x022)
NOTAGS
Perform the indicated operations and simplify:

  a) $\dfrac{10AB}{\sqrt{15AB}}$

  b) $\sqrt{20xy^9}$

---

- (AxCRx25x01x20x023)
NOTAGS
Perform the indicated operations and simplify:

  a) $\sqrt{\dfrac{24a^5b^6}{100a^{10}c^3}}$

  b) $\sqrt{y^{1001}}$

---

- (AxCRx25x01x20x012)
NOTAGS
Simplify the following expression.
$5\sqrt{15}-2\sqrt{3}+(3\sqrt{6}+\sqrt{8})(\sqrt{8}-2\sqrt{10})$

---

- (AxCRx25x01x31x001)
NOTAGS
Simplify.

a) $\sqrt{7}(5+2\sqrt{7})$ b) $\sqrt[3]{6}(4+2\sqrt[3]{18})$

---

- (AxCRx25x01x31x002)
NOTAGS
Simplify.

a) $\sqrt[4]{2}(\sqrt[4]{12}+\sqrt[4]{24})$ b) $\sqrt{2}(-5+9\sqrt{2}$

---

- (AxCRx25x01x31x003)
NOTAGS
Simplify.

a) $(7+\sqrt{3})(9-\sqrt{3})$ b) $(7+\sqrt[3]{6})(4+2\sqrt[3]{18})$

---

- (AxCRx25x01x31x004)
NOTAGS
Simplify.

a) $(\sqrt{3}+\sqrt{10})(\sqrt{3}+2\sqrt{10})$ b) $(2\sqrt{7}-5\sqrt{11})(4\sqrt{7}+9\sqrt{11})$

---

- (AxCRx25x01x31x005)
NOTAGS
Simplify.

a) $(\sqrt{11}+\sqrt{5})(\sqrt{11}+6\sqrt{5})$ b) $(\sqrt{3}+\sqrt{5})^2$

---

- (AxCRx25x01x31x006)
NOTAGS
Simplify.

a) $(\sqrt{2}-7)^2$ b) $(\sqrt{6}+\sqrt{2})^2$

---

- (AxCRx25x01x31x007)
NOTAGS
Simplify.

a) $(\sqrt{3}+\sqrt{5})(\sqrt{3}-\sqrt{5})$ b) $(\sqrt{2}-\sqrt{7})(\sqrt{2}+\sqrt{7})$

---

- (AxCRx25x01x31x008)
NOTAGS
Simplify.

a) $(2+\sqrt{5})(2-\sqrt{5})$ b) $(2\sqrt{6}+3\sqrt{2})(2\sqrt{6}-3\sqrt{2})$

---

- (AxCRx25x01x31x009)
NOTAGS
Simplify.

a) $(3\sqrt{3}-2\sqrt{2})(3\sqrt{2}+2\sqrt{2})$ b) $(2+3\sqrt{7})(2-3\sqrt{7})$

---

- (TESTxxxAxCRx25x04x28x001)
NOTAGS
Simplify.

a) $3\sqrt{20}+4\sqrt{5}$ b) $(3-\sqrt{6})(4+2\sqrt{2})$

---

### 4.17.8 Solving Radical equations *(SR)*

- (AxSRx25x01x17x101)
NOTAGS
Solve the equation.
$\sqrt{5x-6}=8$

---

- (AxSRx25x01x17x102)
NOTAGS
Solve the equation.
$\sqrt[3]{4x-1}=3$

---

- (AxSRx25x01x17x103)
NOTAGS
Solve the equation.
$\sqrt{3x-2}=5$

---

- (AxSRx25x01x17x104)
NOTAGS
Solve the equation.
$\sqrt{x+3}=-5$

---

- (AxSRx25x01x17x105)
NOTAGS
Solve the equation.
$\sqrt{2x-5}=7$

---

- (AxSRx25x01x17x106)
NOTAGS
Solve the equation.
$\sqrt[3]{2x+1} = 1$

---

- (AxSRx25x01x17x107)
NOTAGS
Solve the equation.
$\sqrt{x+5} + 10 = -1$

---

- (AxSRx25x01x17x108)
NOTAGS
Solve the equation.
$\sqrt{6x+1} + 4 = 8$

---

- (AxSRx25x01x17x109)
NOTAGS
Solve the equation.
$\sqrt{x+10} + 2 = x$

---

- (AxSRx25x01x17x110)
NOTAGS
Solve the equation.
$\sqrt{x-6} = x - 6$

---

- (AxSRx25x01x17x111)
NOTAGS
Solve the equation.
$\sqrt{x+4} = x - 2$

---

- (AxSRx25x01x17x112)
NOTAGS
Solve the equation.
$\sqrt{x+1} = x - 1$

---

- (AxSRx25x01x17x113)
NOTAGS
Solve the equation.
$\sqrt{x+25} - x = -5$

---

- (AxSRx25x01x17x114)
NOTAGS
Solve the equation.
$\sqrt{x+3} + x = 9$

---

- (AxSRx25x01x20x017)
NOTAGS
Solve the following equations (no complex solutions unless allowed):

  a) $\sqrt{x} = 0$

  b) $\sqrt{x} = -2$

---

- (AxSRx25x01x20x022)
NOTAGS
Solve these equations (without guess/check; no complex solutions allowed):

  a) $3\sqrt{x} - 1 = 0$

  b) $\sqrt{n+5} = 3$

---

- (AxSRx25x01x17x115)
NOTAGS
Solve the equation.
$\sqrt{2x+6} - x + 1 = 0$

---

- (AxSRx25x01x17x116)
NOTAGS
Solve the equation.
$\sqrt{3x+7} = \sqrt{5x+1}$

---

- (AxSRx25x01x17x117)
NOTAGS
Solve the equation.
$\sqrt[3]{8x-5} = \sqrt[3]{3x+5}$

---

- (AxSRx25x01x17x118)
NOTAGS
Solve the equation.
$\sqrt{x+1} = \sqrt{x-1}$

---

- (AxSRx25x01x17x119)
NOTAGS
Solve the equation.
$\sqrt{x} + 2 = \sqrt{x+4}$

---

- (AxSRx25x01x17x120)
NOTAGS
Solve the equation.
$\sqrt{2x+1} = 1 + \sqrt{x}$

---

- (AxSRx25x01x17x121)
NOTAGS
Solve the equation.
$\sqrt{2x-1} - \sqrt{x-1} = 1$

---

- (AxSRx25x01x20x023)
NOTAGS
Solve.

  a) $\sqrt{2x+4} - 4 = 0$

  b) $\sqrt{2x+4} + 4 = 0$

---

- (AxSRx25x01x20x024)
NOTAGS
Solve each equation:

  a) $\sqrt{6u+1} - 1 = 2u$

  b) $\sqrt{2y+7} + 4 = y$

---

- (TESTxxxAxSRx25x04x28x001)
NOTAGS
Solve the equation: $\sqrt{3-2x} - x = 0$

---

### 4.17.9 Complex Numbers (Imaginary Numbers) *(IM)*

- (AxIMx25x01x20x003)
NOTAGS
Simplify the following expressions.

  a) $\mathbf{i}^9$

  b) $\sqrt{-24}$

---

- (AxIMx25x01x20x008)
NOTAGS
Evaluate/simplify or state if undefined (and why):
$\sqrt[7]{-1}$

---

- (AxIMx25x01x20x005)
NOTAGS
Perform the operation and simplify.

  a) $(2 - 4\mathbf{i}) \cdot (-3\mathbf{i})$

  b) $(5 - 2\mathbf{i})(2 - 3\mathbf{i})$

---

- (AxIMx25x01x20x006)
NOTAGS
Simplify.

  a) $\dfrac{6}{9\mathbf{i}}$

  b) $\dfrac{4}{1 - \mathbf{i}}$

---

- (AxIMx25x01x20x001)
NOTAGS
Simplify each expression:

  a) $-\mathbf{i} + 12 - \mathbf{i}^2 + 5\mathbf{i} - 3\mathbf{i}^4 + 4\mathbf{i}^3$

  b) $(-2 + 9\mathbf{i}) - (11 - 5\mathbf{i})$

---

- (AxIMx25x01x20x002)
NOTAGS

  Simplify each expression:

  a) $4\mathbf{i}(3 - 10\mathbf{i})$

  b) $(2 - 3\mathbf{i})^2$

---

- (AxIMx25x01x22x001)
NOTAGS
Solve the following (*complex solutions allowed*).

  a) $x^2 - 9 = 0$

  b) $x^2 + 9 = 0$

---

- (AxIMx25x01x22x002)
NOTAGS
Simplify.

  a) $\mathbf{i}^7$

  b) $\mathbf{i}^{82}$

---

- (AxIMx25x01x22x003)
NOTAGS
Simplify.

  a) $\sqrt{-16}$ b) $\sqrt{-20}$

---

- (AxIMx25x01x22x004)
NOTAGS
Perform the operation and simplify.

  a) $14\mathbf{i} + 1 + (-3\mathbf{i}) + 17$ b) $(2 - 3\mathbf{i}) - (1 - 4\mathbf{i})$

---

- (AxIMx25x01x22x005)
NOTAGS
Simplify.
$-\mathbf{i} + 7 - 5\mathbf{i}^3 + 3\mathbf{i} + 4\mathbf{i}^2$

---

- (AxIMx25x01x22x006)
NOTAGS
Perform the operation and simplify.

  a) $(1 + 5\mathbf{i})(-2\mathbf{i})$ b) $(-3 - \mathbf{i})(2 - 4\mathbf{i})$

---

- (AxIMx25x01x22x007)
NOTAGS
Perform the operation and simplify.

  a) $\dfrac{21}{7\mathbf{i}}$ b) $\dfrac{15}{2 + 3\mathbf{i}}$

---

- (TESTxAxIMx25x01x20x004)
NOTAGS

  Simplify $-3\mathbf{i} + 7 - 5\mathbf{i}^3 - (3\mathbf{i} + 4\mathbf{i}^2)$

---

- (TESTxxxAxIMx25x01x20x103)
NOTAGS
Simplify each expression:

  a) $\sqrt{-40}$ b) $\dfrac{5}{5 - 3\mathbf{i}}$

---

- (TESTxxxAxIMx25x04x28x001)
NOTAGS
Perform the operation and simplify.

a) $(3\mathbf{i})(2-7\mathbf{i})$ b) $(1+2\mathbf{i})(2-3\mathbf{i})$

---

## 4.18 Rational Expressions *(RE)*

### 4.18.1 Simplifying Rational Expressions *(SP)*

- (AxRExSPx25x01x30x001)
NOTAGS
Simplify.

a) $\dfrac{(x+2)(x-3)}{(x+2)(x-5)}$ b) $\dfrac{(x+4)(x^2+1))}{(x^2+1)(x-5)}$

---

- (AxRExSPx25x01x30x002)
NOTAGS
Simplify.

a) $\dfrac{16(x+5)^2(x-3)}{12(x+2)^2(x+5)}$ b) $\dfrac{x^3-2x^2-25x+50}{x^2-5x}$

---

- (AxRExSPx25x01x30x003)
NOTAGS
Simplify.

a) $\dfrac{x^2+8x+15}{x^2-9}$ b) $\dfrac{x^2+2x-15}{x^2+6x+5}$

---

- (AxRExSPx25x01x30x004)
NOTAGS
Simplify.

a) $\dfrac{49-x^2}{x^2+8x+7}$ b) $\dfrac{x-7}{7-x}$

---

- (AxRExSPx25x01x30x005)
NOTAGS
Simplify.

a) $\dfrac{x^2+7x+12}{x^2+3x+2}$ b) $\dfrac{-5x^2-10x}{-10x^2+30x+100}$

---

- (HWxxxAxRExSPx25x03x16x001)
type: HW
Simplify.

a) $\dfrac{(x+8)(x+2)}{(x-2)(x+8)}$

b) $\dfrac{20(x+4)(2x+1)(x-1)(7x)}{30(x)(x-1)(2x+1)(2x-1)}$

---

- (TESTxxxAxRExSPx25x03x25x001)
NOTAGS
Simplify.

a) $\dfrac{6(x-4)^2}{8(x+2)(x-4)}$

b) $\dfrac{x^2-7x+10}{x^2-25}$

---

### 4.18.2 When Undefined or Equal to Zero *(UZ)*

- (AxRExUZx25x01x17x011)
NOTAGS
For what $x$ is the expression $\dfrac{x+98}{x-5}$ undefined?

---

- (AxRExUZx25x01x17x012)
NOTAGS
For what $x$ is the expression $\dfrac{x+98}{x-5}$ equal to 0?

---

- (AxRExUZx25x01x17x013)
NOTAGS
When is the expression $\dfrac{x+98}{(x-5)(x+8)}$ undefined?

---

- (AxRExUZx25x01x17x014)
NOTAGS
When is the expression $\dfrac{3x+9}{x^2+3x-10}$ undefined?

---

- (AxRExUZx25x01x17x015)
NOTAGS
When is the expression $\dfrac{x^2+8}{x^2+x-20}$ undefined?

---

- (AxRExUZx25x01x17x016)
NOTAGS
When is the following expression undefined? And when is it equal to 0?

$$\frac{x^2 - 25}{x^2 - x - 20}$$

---

- (TESTxxxAxRExUZx25x03x25x001)
NOTAGS
Consider the expression $\frac{4(x-3)}{3x^2 - 21x}$.
Which $x$ make it undefined? Which $x$ make it equal to 0?

---

### 4.18.3 Multiplying Rational Expressions *(MR)*

- (AxRExMRx25x01x17x002)
NOTAGS
Perform the operation and simplify:

$$\frac{20x^2y^7}{zw^2} \cdot \frac{12z^3w^2}{15xy^6}$$

---

- (AxRExMRx25x01x17x003)
NOTAGS
Simplify:

$$\frac{x(2x-7)(x+1)}{3(x+3)} \cdot \frac{12}{x(2x-7)(x+1)}$$

---

- (AxRExMRx25x01x30x001)
NOTAGS
Multiply and simplify.

a) $\frac{x+2}{x-3} \cdot \frac{x+1}{x+2}$

b) $\frac{(x+4)(x+3)}{(x-2)(x-5)} \cdot \frac{(x+4)(x-5)}{(x+3)(x+5)}$

---

- (AxRExMRx25x01x30x002)
NOTAGS
Multiply and simplify.

a) $\frac{x^2+5x+6}{x^2+2x-3} \cdot \frac{x^2+7x+12}{x+2}$

b) $\frac{x^2+3x}{x^2-3x-4} \cdot \frac{(x-4)(x-5)}{x^2}$

---

- (AxRExMRx25x01x30x003)
NOTAGS
Multiply and simplify.

a) $\dfrac{72x-12x^2}{8x+32}\cdot\dfrac{x^2+10x+24}{36x^2-1}$ b) $\dfrac{3x^2+15x}{x^2+10x+25}\cdot\dfrac{1}{6x^2+30x}$

---

- (TESTxxxAxRExMRx25x03x25x001)
NOTAGS
Multiply and simplify. $\dfrac{x^x-4x}{x+1}\cdot\dfrac{x^2-2x-3}{x^2-16}$

---

### 4.18.4 Dividing Rational Expressions *(DR)*

- (AxRExDRx25x01x22x013)
NOTAGS
Simplify $\dfrac{\left(\frac{y^4}{8x^3}\right)}{\left(\frac{y^9}{6x}\right)}$

---

- (AxRExDRx25x01x17x004)
NOTAGS
Perform the operation:

$$\frac{5x^3y^2}{6z^2w^{10}}\div\frac{15x^2y}{12z^5w^9}$$

---

- (AxRExDRx25x01x17x005)
NOTAGS
Calculate the following:

$$\frac{3x+15}{8(x^2+5x-14)}\div\frac{x^2-5}{6x^2-12x}$$

---

- (AxRExDRx25x01x20x024)
NOTAGS
Perform the indicated operations and simplify:

a) $\dfrac{m^3n}{2mn}\cdot\dfrac{6mn^2}{m^3n}\div\dfrac{3mn}{5m^2n}$

b) $\dfrac{D^2-3D-18}{16D-96}\div\dfrac{D^2-9}{20D}$

---

- (AxRExDRx25x01x20x014)
NOTAGS
Simplify.
$\dfrac{y^2-9}{2y^2-6y} \div \dfrac{2y^2+5y-3}{4y^2-1}$

---

- (AxRExDRx25x01x20x015)
NOTAGS
Simplify. $\dfrac{\dfrac{12}{y}}{\dfrac{3}{xy}-\dfrac{9}{y}}$

---

- (AxRExDRx25x01x20x026)
NOTAGS
Simplify:

a) $\dfrac{\left(\frac{x^3}{12}\right)}{\left(\frac{x^5}{18}\right)}$

b) $\dfrac{\left(\frac{m-3n}{4m}\right)}{\left(\frac{m^2-3mn}{8n}\right)}$

---

- (AxRExDRx25x01x30x001)
NOTAGS
Divide and simplify.

a) $\dfrac{x+2}{x-1} \div \dfrac{x+2}{x+5}$

b) $\dfrac{(x-3)(x+3)}{(x-7)(x-6)} \div \dfrac{(x-3)(x-5)}{(x-7)(x+5)}$

---

- (AxRExDRx25x01x30x002)
NOTAGS
Divide and simplify.

a) $\dfrac{x^2+5x+6}{x^2+7x+12} \div \dfrac{x+2}{x^2+2x-3}$

b) $\dfrac{\dfrac{x^2-4x}{x^2-3x-10}}{\dfrac{(x-4)}{x^2(x-5)}}$

---

- (AxRExDRx25x01x30x003)
NOTAGS
Divide and simplify.

a) $\dfrac{\dfrac{12x-72x^2}{8x-32}}{\dfrac{x^2-36}{x^2-10x+24}}$

b) $\dfrac{\dfrac{1}{x^2-10x+25}}{\dfrac{1}{5x^3-25x^2}}$

---

- (TESTxxxAxRExDRx25x03x25x001)
NOTAGS
Perform the operation and simplify. $\dfrac{x-2}{10x^2-30x} \div \dfrac{x^2+6x-16}{5x}$

---

- (TESTxxxAxRExDRx25x03x25x002)
NOTAGS
Simplify. $\dfrac{\dfrac{6}{x}}{\dfrac{2}{x}-\dfrac{4}{xy}}$

---

### 4.18.5 Adding and Subtracting Rational Expressions *(ASR)*

- (AxRExASRx25x01x30x001)
NOTAGS
Simplify. $\dfrac{5x}{x+1}+\dfrac{2x-2}{x+1}$

---

- (AxRExASRx25x01x30x002)
NOTAGS
Simplify. $\dfrac{5x}{x-3}-\dfrac{15}{x-3}$

---

- (AxRExASRx25x01x30x003)
NOTAGS
Simplify. $\dfrac{3x}{x^2+4x+3}-\dfrac{x-2}{x^2+4x+3}$

---

- (AxRExASRx25x01x20x025)
NOTAGS
Perform the indicated operations and simplify.
$\dfrac{5x-12}{x^2-8x+15} - \dfrac{3x-2}{x^2-8x+15}$

---

- (AxRExASRx25x01x20x014)
NOTAGS
Simplify.
$\dfrac{5}{x^2+5x+6} + \dfrac{2}{x+2}$

---

- (AxRExASRx25x01x22x014)
NOTAGS
Perform the operation and simplify: $\dfrac{5x}{x+5} + \dfrac{6x}{x^2+4x-5}$.

---

- (AxRExASRx25x01x30x004)
NOTAGS
Simplify. $\dfrac{5}{x+1} + \dfrac{2}{x-3}$

---

- (AxRExASRx25x01x30x005)
NOTAGS
Simplify. $\dfrac{1}{x} - \dfrac{3x-2}{x+3}$

---

- (AxRExASRx25x01x30x006)
NOTAGS
Simplify. $\dfrac{x-2}{x+3} + \dfrac{x+1}{x-2}$

---

- (AxRExASRx25x01x30x007)
NOTAGS
Simplify. $\dfrac{1}{x-1} - \dfrac{1}{x+1}$

---

- (AxRExASRx25x01x30x008)
NOTAGS
Simplify. $\dfrac{5}{x+4} + \dfrac{3x-12}{x^2-16}$

---

- (AxRExASRx25x01x30x009)
NOTAGS
Simplify. $\dfrac{2}{x+3} + 2$

---

- (AxRExASRx25x01x30x010)
NOTAGS
Simplify. $\dfrac{x}{x-1}+\dfrac{1}{x+1}-\dfrac{2}{x^2-1}$

---

- (AxRExASRx25x01x30x011)
NOTAGS
Simplify. $\dfrac{x+2}{x^2+4x-5}+\dfrac{x-12}{x^2-25}$

---

- (AxRExASRx25x01x30x012)
NOTAGS
Simplify. $\dfrac{2}{x-3}-\dfrac{4}{3-x}$

---

- (TESTxxxAxRExASRx25x03x30x001)
NOTAGS
Simplify. $\dfrac{2}{x+7}+\dfrac{-x+1}{x^2+6x-7}$

---

### 4.18.6 Rational Expressions - Complex *(CPX)*

- (AxRExCPXx25x01x30x001)
NOTAGS
Simplify. $\dfrac{\dfrac{1}{x}+\dfrac{1}{x^2}}{\dfrac{1}{x}-\dfrac{1}{x^2}}$

---

- (AxRExCPXx25x01x30x002)
NOTAGS
Simplify. $\dfrac{\dfrac{2}{x}+\dfrac{2}{y}}{\dfrac{1}{x^2}-\dfrac{1}{y^2}}$

---

- (AxRExCPXx25x01x30x003)
NOTAGS
Simplify. $\dfrac{4+\dfrac{4}{x-5}}{\dfrac{1}{x-5}+\dfrac{x}{4}}$

---

- (AxRExCPXx25x01x30x004)
NOTAGS
Simplify. $\dfrac{\dfrac{1}{x}+\dfrac{x}{y}}{\dfrac{y}{x}-\dfrac{1}{y}}$

---

- (AxRExCPXx25x01x30x005)
NOTAGS
Simplify. $\dfrac{x-\dfrac{2x}{x+3}}{\dfrac{1}{x+3}+\dfrac{1}{x-3}}$

---

### 4.18.7 Solving Rational Equations *(S)*

- (AxRExSx25x01x20x018)
NOTAGS
Solve the following equations (no complex solutions unless allowed):

a) $(x-8)(x-8)(x)(2x+1)=0$

b) $\dfrac{3}{x-10}=\dfrac{-3}{x-10}$

---

- (AxRExSx25x01x20x028)
NOTAGS
Solve.

a) $\dfrac{x}{4}=\dfrac{x}{12}+\dfrac{1}{2}$

b) $\dfrac{5}{4x}-\dfrac{1}{2}=\dfrac{1}{2x}$

---

- (AxRExSx25x01x20x029)
NOTAGS
Solve.

a) $\dfrac{9}{x}+2=\dfrac{2x}{x+3}$

b) $\dfrac{11}{x+2}-\dfrac{5}{x^2-x-6}=\dfrac{1}{x-3}$

---

- (AxRExSx25x01x20x035)
NOTAGS
Solve the equation. $\dfrac{7}{x}-\dfrac{2}{x-3}=\dfrac{6}{x}$

---

- (AxRExSx25x01x20x008)
NOTAGS
Solve each equation:

a) $\frac{3}{2} - \frac{2}{2x-4} = \frac{1}{x-2}$ b) $\frac{3y}{2+y} - \frac{2}{y+3} = \frac{36}{y^2+5y+6}$

---

- (AxRExSx25x01x30x001)
NOTAGS
Solve. $1 - \frac{2}{x} = \frac{15}{x^2}$

---

- (AxRExSx25x01x30x002)
NOTAGS
Solve. $\frac{2}{x+1} + \frac{1}{x-1} = \frac{1}{x^2-1}$

---

- (AxRExSx25x01x30x003)
NOTAGS
Solve. $\frac{x-6}{x^2+3x-4} = \frac{2}{x+4} + \frac{7}{x-1}$

---

- (AxRExSx25x01x30x004)
NOTAGS
Solve. $\frac{x}{x+4} = \frac{32}{x^2-16} + 5$

---

- (AxRExSx25x01x30x005)
NOTAGS
Solve. $\frac{2}{x+7} - \frac{3}{x-3} = 1$

---

- (AxRExSx25x01x30x006)
NOTAGS
Solve. $\frac{x-10}{x^2-5x+4} = \frac{3}{x-1} - \frac{6}{x-4}$

---

- (AxRExSx25x01x30x007)
NOTAGS
Solve. $\frac{15}{x^2+x-6} - \frac{3}{x-2} = \frac{2}{x+3}$

---

- (AxRExSx25x01x30x008)
NOTAGS
Solve. $\frac{5}{x^2+2x-3} - \frac{3}{x^2+x-2} = \frac{1}{x^2+5x+6}$

---

- (AxRExSx25x01x30x009)
NOTAGS
Solve. $\dfrac{x}{5x-10} - \dfrac{5}{3x+6} = \dfrac{2x^2-19x+54}{15x^2-60}$

---

- (TESTxxxAxRExSx25x03x25x001)
NOTAGS
Solve. $\dfrac{6}{x-2} = 3$

---

- (TESTxxxAxRExSx25x03x25x002)
NOTAGS
Solve. $\dfrac{x+4}{2x-6} - \dfrac{1}{x-3} = \dfrac{2}{3}$

---

### 4.18.8 Rational Functions : Limits and Asymptotes *(ASYM)*

- (AxRExASYMx25x02x24x001)
NOTAGS
Consider $f(x) = \frac{3}{x+4}$. Find its vertical and horizontal asymptotes, use arrow notation for end behavior and near the vertical asymptote.

---

## 4.19 Log and Exponentials *(LE)*

- (GROUPxxxFxLEx25x04x02x001)
NOTAGS
SIMULATION: SPREAD OF RUMOR OR DISEASE

  To start, if YES is written in round 0, do nothing; otherwise write NO in round 0. On each round, walk around the room till the instructor says "Stop". At that point, get into pairs and look at just eachother's paper. If there is a YES anywhere on the other person's paper, enter a YES on your card in the current round. If you have a YES in any of your earlier rounds, enter a YES on your card in the current round. Otherwise (if all they have is NO's and all you have is NO's), enter a NO on your card in the current round. You should never pair up with someone more than once.

  - Round 0:
  - Round 1:
  - Round 2:
  - Round 3:
  - Round 4:
  - Round 5:

---

### 4.19.1 Graphing Log and Exponential *(G)*

- (FxLExGx25x01x20x016)
NOTAGS
Sketch $f(x) = \left(\frac{1}{2}\right)^x$ and $g(x) = 2^x$ on the same axes.
  - What is the relationship between the two graphs?
  - Find any two points that are symmetric across the $y$-axis.

---

### 4.19.2 Solving Log and Exponential Equations *(SLE)*

- (AxLExSLEx25x01x20x019)
NOTAGS
Solve.

  a) $10^y = 1000000$

  b) $555^x = 1$

---

- (AxLExSLEx25x01x20x020)
NOTAGS
Solve.

  a) $5^u = -5$

  b) $x^{1000} = 1$

---

- (AxLExSLEx25x01x20x021)
NOTAGS
Solve. $x^{1000} = -1$

---

- (AxLExSLEx25x01x20x024)
NOTAGS
Solve.

  a) $4^{x+5} = 16$

  b) $\left(\frac{1}{3}\right)^{2x-1} = 9$

---

- (AxLExSLEx25x01x20x025)
NOTAGS
Solve.

  a) $\log_3(x) = -2$

  b) $\log_4(x^2) = 2$

---

- (AxLExSLEx25x01x20x027)
NOTAGS
Solve each equation:

  a) $7^{-x^2-3x} = 49$

  b) $\log_{10}(1000) = 5x + 1$

---

- (AxLExSLEx25x02x04x001)
NOTAGS
Solve. $4^x = 32$

---

- (AxLExSLEx25x02x04x002)
NOTAGS
Solve. $27^x = 81$

---

- (AxLExSLEx25x02x04x004)
NOTAGS
Solve. $5^{2-x} = \dfrac{1}{125}$

---

- (AxLExSLEx25x02x04x007)
NOTAGS
Solve (expressing your answer using logarithms). $5e^x = 7$

---

- (AxLExSLEx25x02x04x008)
NOTAGS
Solve (expressing your answer using logarithms). $3^{\frac{x}{7}} = 0.2$

---

- (AxLExSLEx25x02x04x010)
NOTAGS
Solve. $\log_5 x = 3$

---

- (AxLExSLEx25x02x04x011)
NOTAGS
Solve. $\log_4(x - 7) = 3$

---

- (AxLExSLEx25x02x04x012)
NOTAGS
Solve. $5 \ln 2x = 20$

---

- (AxLExSLEx25x02x04x013)
NOTAGS
Solve each.

a) $\log_5(x - 2) = \log_5 3$

b) $(\log_5 x) - 2 = \log_5 3$

---

- (AxLExSLEx25x02x04x014)
NOTAGS
Solve. $2 \log_5 x = 4$

---

- (AxLExSLEx25x02x04x017)
NOTAGS
Solve. $3 \log x = \log 125$

---

- (AxLExSLEx25x02x04x018)
NOTAGS
Solve. $\log_2 \sqrt{x + 4} = 1$

---

- (AxLExSLEx25x02x04x020)
NOTAGS
Solve. $\log(x + 7) - \log 3 = \log(7x - 1)$

---

- (AxLExSLEx25x02x04x021)
NOTAGS
Solve. $\log(x + 3) + \log(x - 2) = \log 14$

---

- (HWxAxLExSLEx25x02x04x005)
type: HW
Solve. $7^{\frac{x-2}{6}} = \sqrt{7}$

---

- (HWxAxLExSLEx25x02x04x006)
type: HW
Solve. $8^{1-x} = 4^{x+2}$

---

- (HWxAxLExSLEx25x02x04x009)
type: HW
Solve (expressing your answer using logarithms). $7^{2x-1} = 3^{x+2}$

---

- (TESTxxxAxLExSLEx25x02x04x019)
NOTAGS
Solve. $\log_2(x-1) + \log_2(x+1) = 3$

---

- (TESTxxxAxLExSLEx25x02x04x003)
NOTAGS
Solve. $2^{2x-1} = 32$

---

- (AxLExSLEx25x11x17x001)
type: TEST
Solve. $3^{5x-4} = 9$

---

- (TESTxxxAxLExSLEx25x04x28x001)
NOTAGS
Solve (expressing your answer using logarithms). $e^{5x} = 2^{x-3}$

---

- (TESTxxxAxLExSLEx25x01x20x026)
NOTAGS
Solve. $\log_5\left(5^{2x-7}\right) = 3x$

---

## 4.20 Word problems (algebra) *(WP)*

### 4.20.1 Word problems using linear equations *(L)*

- (AxWPxLx25x01x02x016)
NOTAGS
Solve: $3(x+3) = 2x + 15$
The solution is the number of NBA championships won by Michael Jordan. How many did he win?

---

- (AxWPxLx25x01x02x017)
NOTAGS
Solve the equation: $5.2x - 7 = 2.2x + 8$
The solution is the age Jennifer Lopez began dancing and singing lessons. How old was she?

---

- (AxWPxLx25x01x02x018)
NOTAGS
Solve: $2x - 3 + 5x = 4 + 4x + 1$
3 times the solution is the number of siblings Michael Jackson had. How many siblings did Michael Jackson have?

---

- (AxWPxLx25x01x03x008)
NOTAGS
**Write the sentence as an equation, and then solve.**
$x$ added to 5 is 8.

---

- (AxWPxLx25x01x03x009)
NOTAGS
**Write the sentence as an equation, and then solve.**
Adding $u$ and 6 is -1.

---

- (AxWPxLx25x01x03x010)
NOTAGS
**Write the sentence as an equation, and then solve.**
Two times this number is 1.

---

- (AxWPxLx25x01x03x011)
NOTAGS
**Write the sentence as an equation, and then solve.**
$y$ equals 4 times $-2$.

---

- (AxWPxLx25x01x03x012)
NOTAGS
**Write the sentence as an equation, and then solve.**
$x$ divided by 10 is 5.

---

- (AxWPxLx25x01x03x013)
NOTAGS
**Write the sentence as an equation, and then solve.**
$\frac{1}{3}$ of $x$ is 6.

---

- (AxWPxLx25x01x03x014)
NOTAGS
**Write the sentence as an equation, and then solve.**
A rectangle has an area of 40 and a length of 5. What is its width?

---

- (AxWPxLx25x01x03x015)
NOTAGS
**Write the sentence as an equation, and then solve.**
A triangle has an area of 40 and a base of 5. What is its height?

---

- (AxWPxLx25x01x03x016)
NOTAGS
**Write the sentence as an equation, and then solve.**
A square has a perimeter of 40. What is its width?

---

- (AxWPxLx25x01x03x017)
NOTAGS
**Write the sentence as an equation, and then solve.**
A rectangle has a perimeter of 40 and a length of 5. What is its width?

---

- (AxWPxLx25x01x03x018)
NOTAGS
**Write the word problem as an equation, and then solve.**
Mary buys a box of apples and oranges that weigh 39 ounces. Each apple weighs 5 ounces and each orange weighs 4 ounces. If there are 3 apples in the box, how many oranges are there?

---

- (AxWPxLx25x01x03x019)
NOTAGS
**Write the word problem as an equation, and then solve.**
According to the International Table Tennis Federation, a ping-pong ball should weigh about 3 grams. If Vic has 7 ping-pong balls, how much total weight is he carrying?

---

- (AxWPxLx25x01x03x020)
NOTAGS
**Write the word problem as an equation, and then solve.**
Suppose Mary has 42 grams in ping-pong balls. How many ping-pong balls does she have?

---

- (AxWPxLx25x01x03x021)
NOTAGS
**Write the word problem as an equation, and then solve.**
A basketball weighs about 600 grams. How much heavier is a basketball than a ping-pong ball?

---

- (AxWPxLx25x01x03x022)
NOTAGS
**Write the word problem as an equation, and then solve.**
Suppose Vic can carry 2400 grams in his bag. How many basketballs can he carry?

---

- (AxWPxLx25x01x03x023)
NOTAGS
**Write the word problem as an equation, and then solve.**
Suppose Mary has 18 kilograms in weight, in a mystery ball. If she has 9 mystery balls, then how much does each mystery ball weigh?

---

- (AxWPxLx25x01x03x024)
NOTAGS
**Write the word problem as an equation, and then solve.**
Vic bought 2 stamps that cost 10 cents and 3 stamps that cost 25 cents. If Vic paid $1, how much change did he get?

---

- (AxWPxLx25x01x03x025)
NOTAGS
**Write the word problem as an equation, and then solve.**
Vic bought 7 stamps that cost 10 cents or 25 cents. Altogether he paid $1.30. How many 10 cent stamps did he buy? How many 25 cent stamps did he buy?

---

- (AxWPxLx25x01x03x026)
NOTAGS
**Write the word problem as an equation, and then solve.**
Mary bought 10 stamps that cost $1 or $1.20. Altogether he paid $11. How many $1 stamps did he buy? How many $1.20 stamps did he buy?

---

- (AxWPxLx25x01x03x027)
NOTAGS
**Write the word problem as an equation, and then solve.**
In the 2012-2013 NBA season Dwayne Wade and LeBron James combined to make 711 free throws. If James made 95 more free throws than Wade, determine how many free throws each player made.

---

- (AxWPxLx25x01x03x028)
NOTAGS
**Write the word problem as an equation, and then solve.**
As of 2014, the number of Grammy nominations JayZ had was $\frac{3}{2}$ of what Eminem had. If they combine for a total of 100 nominations, determine how many nominations each artist had.

---

- (AxWPxLx25x01x03x029)
NOTAGS
**Write the word problem as an equation, and then solve.**
In the 2012-2013 NBA season Jason Kidd had 4 fewer steals than LeBron James. If the *average* of their total steals is 127, find the total number of steals each player had in the season.

---

- (AxWPxLx25x02x12x001)
NOTAGS
A city's population has been growing linearly. In the year 2005, the population was 50,000, and the population has been growing by 1000 people each year. Write an equation, for the population $t$ years after 2005.

---

### 4.20.2 Word problems with linear IN-equalities *(NL)*

- (AxWPxNLx25x01x04x025)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
$x$ plus 7 is larger than 10.

---

- (AxWPxNLx25x01x04x026)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
$x$ plus 7 is larger than or equal to 10.

---

- (AxWPxNLx25x01x04x027)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
3 times a number is larger than 6.

---

- (AxWPxNLx25x01x04x028)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
10 more than a number is at least 100.

---

- (AxWPxNLx25x01x04x029)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
$y$ minus 2 is positive.

---

- (AxWPxNLx25x01x04x030)
NOTAGS
**Translate the expression into an inequality. Then solve it.**
$y$ times 157 is negative.

---

### 4.20.3 Word problems using polynomial equations *(P)*

- (AxWPxPx02x18x001)
NOTAGS
An oil slick is expanding as a circle. The radius of the circle is increasing at the rate of 20 meters per day. Express the area of the circle as a function of the number of days elapsed.

---

- (AxWPxPx02x18x002)
NOTAGS
A cube has an edge of 3 feet. The edge is increasing at the rate of 2 feet per minute. Express the volume of the cube as a function of the number of minutes elapsed.

---

- (AxWPxPx02x18x003)
NOTAGS
An open box is to be constructed by cutting out square corners of $x$ inch sides from a piece of cardboard 8 inches by 8 inches and then folding up the sides. Express the volume of the box as a function of $x$.

---

- (AxWPxPx01x17x023)
NOTAGS

If you square this positive number you get 100. What is the number?

---

- (AxWPxPx01x17x024)
NOTAGS

If you first subtract 1 from this positive number, then square it, you get 25. What is the number?

---

- (AxWPxPx01x17x025)
NOTAGS

The product of two consecutive positive integers is 56. Find the integers.

---

- (AxWPxPx01x17x026)
NOTAGS

The product of two consecutive odd integers is 143. Find the integers.

---

- (AxWPxPx01x17x027)
NOTAGS

The product of two consecutive even integers is 168. Find the integers.

---

- (AxWPxPx01x17x028)
NOTAGS

A rectangular carpet has area 28 square feet. The length is three feet more than the width. Find the length and width.

---

- (AxWPxPx01x17x029)
NOTAGS

A rectangular retaining wall has area 15 square feet. The height is two feet less than its length. Find the height and the length.

---

- (AxWPx25x01x20x031)
NOTAGS
Answer using algebra, (guess-and-check not allowed):

  a) The sum of a number and its reciprocal is 3. Find that number.

  b) The length of a rectangle is 3 times its width, and its area is 22 sq. ft. Find the width.

---

- (AxWPx25x01x20x032)
NOTAGS
If 5 is added to the product of two consecutive odd integers, the result is 104. Find such integers.

---

- (AxWPx25x01x20x033)
NOTAGS
A projectile is launched from the ground on planet A with initial velocity 400 ft/s. Its height $y$ satisfies $y = -6t^2 + v_0 t$. If we want $y = 2506$, how many seconds until it reaches that height?

---

# 5 Trigonmetry *(T)*

## 5.1 Angle Conversions *(C)*

- (TxCx24x12x29x001)
NOTAGS

Convert 45° to radians. **Answer:** [ ]

---

- (TxCx24x12x29x002)
NOTAGS

Convert 0° to radians. **Answer:** $\square$

---

- (TxCx24x12x29x003)
NOTAGS

Convert $5\pi/6$ radians to degrees.. **Answer:** $\square$

---

- (TxCx24x12x29x004)
NOTAGS

Convert $2\pi$ radians to degrees.. **Answer:** $\square$

---

- (TxCx25x01x20x018)
NOTAGS
Convert from degrees to radians.

a) $180°$ b) $-180°$

---

- (TxCx25x01x20x019)
NOTAGS
Convert from degrees to radians.

a) $270°$ b) $120°$

---

- (TxCx25x01x20x020)
NOTAGS
Convert the angle measure $-405°$ to radians.

---

- (TxCx25x01x20x023)
NOTAGS
Convert the following angle measures (in radians) to degrees.

a) $\pi$ b) $-3\pi$

---

- (TxCx25x01x20x024)
NOTAGS
Convert the following angle measures (in radians) to degrees.

a) $\pi/3$ b) $-\frac{5\pi}{4}$

---

## 5.2 Arclength *(A)*

- (TxAx24x12x29x001)
NOTAGS
Suppose a circle with radius 6 has a central angle of $180^o$. How long is the arc of the circle that corresponds to this central angle? **Answer:** $\square$

---

- (TxAx24x12x29x002)
NOTAGS
Suppose a circle of radius 4 has a central angle which subtends an arc of length $6\pi$. Find the measure of the central angle. **Answer:** $\square$

---

- (TxAx24x12x29x003)
NOTAGS
In a circle, suppose a central angle of 60 degrees subtends an arc of length $2\pi$. Find the *circumference* of the circle. **Answer:** $\square$

---

- (TxAx25x01x20x025)
NOTAGS
Suppose a circle has a central angle of $50^\circ$ which subtends an arc of length 10.

  a) Find the radius of the circle.  b) Find the circumference of the circle.

---

- (GROUPxTxAx25x02x07x001)
NOTAGS
The radius of the earth is about 4000 miles. Right now we are in NYC; pick a city to travel to. In taking this journey we travel along the arc of a circle. Use a map program (like Google Maps) to determine the arclength. Now use the formula to determine the central angle you would span in going from NYC to your city of choice. Use a calculator as needed and round to the nearest tenth.

---

## 5.3 Sector *(SEC)*

- (TxSECx25x01x20x026)
NOTAGS
Suppose a sector of a circle has a central angle of $\pi/7$ and its area is 20. Find the radius of the circle.

---

- (TxSECx25x01x22x001)
NOTAGS
Suppose a sector of a circle has a central angle of $\frac{2\pi}{3}$ and radius 3. Find the area of the sector.

---

## 5.4 Trig Functions On Triangle *(FT)*

- (TxFTx25x04x04x001)
NOTAGS
Consider the right triangle with legs of length 7 and 24, and a hypotenuse of length 25. Find all six trigonometric functions of the angle that includes the side of length 7.

---

- (TxFTx24x12x29x001)
NOTAGS
Consider the right triangle with a hypotenuse of length 13 and one leg of length 12. Find all six trigonometric functions of the angle that includes the side of length 12.

---

- (TxFTx25x01x20x027)
NOTAGS
Consider the right triangle whose 2 legs are length 5 and 12. Find all six trigonometric functions of the angle that includes the side of length 5.

---

- (TxFTx25x04x04x002)
NOTAGS
Consider the right triangle with legs of length 2 and 4. Find all six trigonometric functions of the angle that includes the side of length 4 (give your answers as simplified radicals).

---

- (TxFTx25x04x04x003)
NOTAGS
Consider the right triangle with a leg of length 1 and a hypotenuse of length 9. Find all six trigonometric functions of the angle that includes the side of length 1 (give your answers as simplified radicals).

---

- (TxFTx25x04x04x009)
NOTAGS
Suppose $\sin(\theta) = \dfrac{7}{241}$. What is $\csc(\theta)$?

---

- (TxFTx25x04x04x010)
NOTAGS
Suppose $\sec(\theta) = \dfrac{2025}{13}$. What is $\cos(\theta)$?

---

- (TxFTx25x04x04x004)
NOTAGS
Suppose $\cos(X) = \frac{3}{5}$. Evaluate the other five trig functions at $X$.

---

- (TxFTx25x04x04x008)
NOTAGS
Suppose $\tan(A) = \frac{4}{3}$. What is $\cot(A)$? What is $\sec(A)$?

---

- (TxFTx25x01x20x028)
NOTAGS
Suppose $\sin(\theta) = \frac{2}{5}$. Evaluate the other five trig functions at $\theta$.

---

- (TxFTx25x01x22x004)
NOTAGS
Suppose $\tan(\alpha) = 3$. Evaluate the other five trigonometric functions at $\alpha$.

---

- (TxFTx25x04x04x006)
NOTAGS
Draw the $45° - 45° - 90°$ right triangle, and use it to evaluate $\sin(45°)$, $\cos(45°)$, and $\cos(45°)$.

---

- (TxFTx25x04x04x007)
NOTAGS
Draw the $45° - 45° - 90°$ right triangle, and use it to evaluate $\sin(45°)$, $\csc(45°)$, $\cos(45°)$, and $\sec(45°)$.

---

- (TxFTx25x01x20x011)
NOTAGS
Draw the $30° - 60° - 90°$ right triangle, and use it to evaluate $\cos(60°)$, $\sin(30°)$, $\csc(60°)$, and $\sec(30°)$.

---

- (TxFTx25x01x20x012)
NOTAGS
Draw the $30° - 60° - 90°$ right triangle, and use it to evaluate $\tan(60°)$, $\tan(30°)$, $\cot(60°)$, and $\cot(30°)$.

---

- (TxFTx25x01x20x029)
NOTAGS
Draw the $45° - 45° - 90°$ right triangle, and use it to evaluate $\sin(45°)$, $\cos(\frac{\pi}{4})$, and $\csc(\frac{\pi}{4})$.

---

- (TxFTx25x01x20x030)
NOTAGS
Draw the $30° - 60° - 90°$ right triangle, and use it to evaluate $\cos(60°)$, $\sin(30°)$, $\csc(\frac{\pi}{3})$, $\sec(\frac{\pi}{6})$, and $\tan(\frac{\pi}{3})$.

---

- (TxFTx25x01x22x005)
NOTAGS
Evaluate all six trigonometric functions at the angles $\frac{\pi}{6}$, $\frac{\pi}{4}$, and $\frac{\pi}{3}$.

---

- (TESTxxxTxFTx25x04x04x001)
NOTAGS
Consider the right triangle with legs of length 1 and 7. Find all six trigonometric functions of the angle that includes the side of length 7.

---

- (TESTxxxTxFTx25x04x04x005)
NOTAGS
Suppose $\tan(X) = \frac{3}{2}$. Evaluate $\cot(X)$ and $\sin(X)$.

---

## 5.5 Finding Missing Side Of Triangle *(M)*

### 5.5.1 Using Just Pythagorean Theorem *(PY)*

- (TxMxPYx25x04x04x001)
NOTAGS
Suppose a right triangle has legs of length 3 and 4. What is the length of the hypotenuse?

---

- (TxMxPYx25x04x04x002)
NOTAGS
Suppose a right triangle has legs of length 9 and 12. What is the length of the hypotenuse?

---

- (TxMxPYx25x04x04x003)
NOTAGS
Suppose a right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

---

- (TxMxPYx25x04x04x004)
NOTAGS
Suppose a right triangle has a leg of length 3 and its hypotenuse is length 5. What is the length of the other leg?

---

- (TxMxPYx25x04x04x005)
NOTAGS
Suppose a right triangle has a leg of length 12 and its hypotenuse is length 15. What is the length of the other leg?

---

- (TxMxPYx25x04x04x006)
NOTAGS
Suppose a right triangle has a leg of length 15 and its hypotenuse is length 17. What is the length of the other leg?

---

- (TxMxPYx25x04x04x007)
NOTAGS
Suppose a right triangle has legs of length 2 and 4. What is the length of the hypotenuse? (give your answer as a simplified radical)

---

- (TxMxPYx25x04x04x008)
NOTAGS
Suppose a right triangle has legs of length 1 and 4. What is the length of the hypotenuse? (give your answer as a simplified radical)

---

- (TxMxPYx25x04x04x009)
NOTAGS
Suppose a right triangle has legs of length 3 and 6. What is the length of the hypotenuse? (give your answer as a simplified radical)

---

- (TxMxPYx25x04x04x010)
NOTAGS
Suppose a right triangle has a leg of length 2 and its hypotenuse is length 3. What is the length of the other leg? (give your answer as a simplified radical)

---

- (TxMxPYx25x04x04x110)
NOTAGS
Suppose a right triangle has a leg of length 2 and its hypotenuse is length 8. What is the length of the other leg? (give your answer as a simplified radical)

---

### 5.5.2 Using Trig *(TR)*

- (TxMxTRx25x01x20x031)
NOTAGS
Suppose a right triangle has a hypotenuse of length 100 and an angle of 50°. Find the length of the side opposite the 50° angle.

---

- (TxMxTRx25x01x22x005)
NOTAGS
Solve the right triangle $ABC$ using the given information:
$m\angle C = 90°$, $m\angle B = 45°$, $c = 50$.

---

- (TxMxTRx25x01x22x006)
NOTAGS
Solve the right triangle $ABC$ using the given information:
$m\angle C = 90°$, $m\angle A = 35°$, $b = 30$.

---

### 5.5.3 Trig Right Angle Word Problems *(W)*

- (TxMxWx25x04x04x001)
NOTAGS
The angle of elevation of the top of a building is $40^o$ from an observation point 100 ft. from the base of the building. Find the height of the building (in terms of a trig function).

---

- (TxMxWx25x04x26x001)
NOTAGS
The angle of elevation of the top of a building is $30^o$ from an observation point 100 ft. from the base of the building. Find the height of the building (exactly, with no trig function in the final answer).

---

- (TxMxWx25x04x26x002)
NOTAGS
A 20 meter ladder leans against a building so that the angle between the ground and the ladder is $37^o$. How high does the ladder reach up the side of the building?

---

- (TxMxWx25x04x26x004)
NOTAGS
Suppose that a 440-foot tall redwood tree grows vertically. You walk some distance from the tree and measure the angle of elevation to the top of the tree to be $45^o$. How far from the base of the tree are you?

---

- (TxMxWx25x04x26x005)
NOTAGS
Suppose that one end of a 35 foot cord is nailed into a wall at some height above the ground. The other end of the cord is spiked into the ground. The cord makes a $20^o$ angle with the wall. How far is it from the base of the wall up to the nail attaching the cord to the wall.

---

- (TESTxxxTxMxWx25x04x26x003)
NOTAGS
A 40 foot pole leans against a wall so that the angle between the ground and the pole is $60^o$. How high does the pole reach up the side of the wall?

---

## 5.6 Angles and Standard Position *(ST)*

- (TxSTx25x02x04x001)
NOTAGS
Draw the angles in standard position.

  a) $\frac{5\pi}{4}$ b) $\frac{11\pi}{6}$

---

- (TxSTx25x02x04x002)
NOTAGS
Draw the angles in standard position.

  a) $\frac{2\pi}{3}$ b) $\frac{13\pi}{4}$

---

- (TxSTx25x02x04x003)
NOTAGS
Draw the angles in standard position.

  a) $\frac{5\pi}{2}$ b) $\frac{7\pi}{6}$

---

- (TxSTx25x02x04x004)
NOTAGS
Draw the angles in standard position.

  a) $\frac{7\pi}{3}$ b) $\frac{5\pi}{3}$

---

- (TxSTx25x01x22x001)
NOTAGS
Draw the angles in standard position: $-270°$ and $\frac{3\pi}{4}$. Find an angle that is co-terminal with $-270°$.

---

- (TxSTx25x02x05x001)
NOTAGS
Find an angle (in degrees) between $0°$ and $360°$ that is coterminal with the following angles:

  a) $525°$ b) $1025°$

---

- (TxSTx25x02x05x002)
NOTAGS
Find an angle (in degrees) between $0°$ and $360°$ that is coterminal with the following angles:

a) $-90°$ b) $-540°$

---

- (TxSTx25x02x05x003)
NOTAGS
Find an angle (in radians) between 0 and $2\pi$ that is coterminal with the following angles:

a) $9\pi$ b) $\frac{7\pi}{2}$

---

- (TxSTx25x02x05x004)
NOTAGS
Find an angle (in radians) between 0 and $2\pi$ that is coterminal with the following angles:

a) $\frac{15\pi}{4}$ b) $\frac{-3\pi}{2}$

---

- (TxSTx25x02x05x005)
NOTAGS
Find the reference angle of the following angles (in degrees).

a) $125°$ b) $247°$

---

- (TxSTx25x02x05x006)
NOTAGS
Find the reference angle of the following angles (in degrees).

a) $312°$ b) $-120°$

---

- (TxSTx25x02x05x007)
NOTAGS
Find the reference angle of the following angles (in radians).

a) $\frac{5\pi}{6}$ b) $\frac{7\pi}{4}$

---

- (TxSTx25x02x05x008)
NOTAGS
Find the reference angle of the following angles (in radians).

a) $\frac{4\pi}{3}$ b) $-\frac{\pi}{6}$

---

## 5.7 Evaluating Trig Functions Beyond Triangle *(F)*

- (TxFx24x12x29x001)
NOTAGS
Evaluate sin and cos at the following angles: $3\pi/2,\ 5\pi/4,\ -\pi/3$.

**Answer:**

---

- (TxFx24x12x29x002)
NOTAGS
Evaluate all six trigonometric functions at the following angles: $\pi/6,\ \pi/4,\ \pi/3$.

---

- (TxFx25x01x22x002)
NOTAGS
Evaluate $\tan(\theta)$, $\sin(\theta)$, and $\cos(\theta)$ where $(12, -5)$ is a point on the terminal side of $\theta$.

---

- (TxFx25x01x22x003)
NOTAGS
Evaluate $\sin(-60°)$ and $\cos(-60°)$ by finding a point on the terminal side of the angle and using a special triangle.

---

- (TxFx25x01x22x004)
NOTAGS
Evaluate $\tan\left(\frac{5\pi}{4}\right)$ and $\cot\left(\frac{5\pi}{4}\right)$ using the reference angle approach.

---

- (TxFx25x01x22x005)
NOTAGS
Suppose $\theta$ is in the fourth quadrant and $\tan\ \theta = -2$. Find the exact values of $\sin\ \theta$ and $\cos\ \theta$.

---

- (TxFx25x01x22x006)
NOTAGS
Evaluate each expression:

a) $\sin\frac{\pi}{6}+\cos\frac{2\pi}{3}$

b) $1+\tan^2\frac{5\pi}{6}$

---

- (TxFx25x02x05x001)
NOTAGS
Given that $\cos\alpha=-\frac{2}{5}$, and that $\alpha$ is in Quadrant II, find the exact value of $\sin\alpha$, $\tan\alpha$, $\sec\alpha$, $\csc\alpha$, $\cot\alpha$.

---

- (TxFx25x02x05x002)
NOTAGS
Given that $\tan\alpha=-\frac{2}{5}$, and that $\alpha$ is in Quadrant IV, find the exact value of $\sin\alpha$, $\cos\alpha$, $\sec\alpha$, $\csc\alpha$, $\cot\alpha$.

---

- (TxFx25x02x05x003)
NOTAGS
Given that $\sin\alpha=\frac{2}{3}$, and that $\alpha$ is in Quadrant II, find the exact value of $\cos\alpha$, $\tan\alpha$, $\sec\alpha$, $\csc\alpha$, $\cot\alpha$.

---

## 5.8 Graphing Trig Functions *(G)*

- (TxGx25x01x22x001)
NOTAGS
Graph $y=2\cos(x)$. What is its amplitude and period?

---

- (TxGx25x01x22x002)
NOTAGS
Graph $y=-4\cos(x)$. What is its amplitude and period?

---

- (TxGx25x04x26x001)
NOTAGS
For the following sinusoidal functions, find the amplitude, the period, and the phase shift. You do not have to graph the functions.

a) $f(x)=(\frac{1}{2})\sin\left(4(x-\frac{\pi}{6})\right)$

b) $g(x)=-5\cos\left(10(x-\pi)\right)$

---

- (TxGx25x02x05x002)
NOTAGS
For the following sinusoidal functions, find the amplitude, the period, and the phase shift. You do not have to graph the functions.

a) $f(x) = \frac{2}{3}\cos\left(\pi x - \frac{\pi}{2}\right)$ b) $f(x) = 4\sin\left(3x + \frac{3\pi}{4}\right)$

---

- (TxGx25x04x26x002)
NOTAGS
Graph $f(x) = \sin(x)$ and $g(x) = 2 + \sin(x - \pi)$ on the same axes (graph $g(x)$ by using graph transformations - describe them).

---

- (TxGx25x04x26x003)
NOTAGS
Graph $f(x) = \cos(x)$ and $g(x) = -2 + \cos(4x)$ on the same axes (graph $g(x)$ by using graph transformations - describe them).

---

- (TxGx25x04x26x004)
NOTAGS
Graph $f(x) = \sin(x)$ and $g(x) = 5 - 2\sin(2x)$ on the same axes (graph $g(x)$ by using graph transformations - describe them).

---

- (TxGx25x04x26x005)
NOTAGS

a) Consider sin with some phase shift, i.e. $f(x) = \sin(x - k)$. For what choice of $k$ does $f(x) = \cos(x)$?

b) Consider cos with some phase shift, i.e. $f(x) = \cos(x - k)$. For what choice of $k$ does $f(x) = \sin(x)$?

---

- (TxGx25x02x05x006)
NOTAGS
Consider the function $f(x) = 3\cos\left(2(x - \frac{\pi}{2})\right)$. Find the a) amplitude, b) the period, and c) the phase shift, of $f(x)$. Then use that information to graph $f(x)$ (indicate how (a), (b), and (c) are used to make the graph).

---

- (TxGx25x02x05x007)
NOTAGS
Consider the function $h(x) = -4\sin(x - 2\pi)$. Find the a) amplitude, b) the period, and c) the phase shift, of $h(x)$. Then use that information to graph $h(x)$ (indicate how (a), (b), and (c) are used to make the graph).

---

- (TxGx25x02x05x008)
NOTAGS
Consider the function $h(x) = \sin(4x - 4\pi)$. Find the a) amplitude, b) the period, and c) the phase shift, of $h(x)$. Then use that information to graph $h(x)$ (indicate how (a), (b), and (c) are used to make the graph).

---

- (TxGx25x02x05x009)
NOTAGS
Consider the function $h(x) = \tan\left(x + \frac{\pi}{2}\right)$. Find the the period of $h(x)$. Then graph $h(x)$, showing the asymptotes.

---

- (TxGx25x02x05x010)
NOTAGS
Consider the function $f(x) = \tan(2x)$. Find the the period of $f(x)$. Then graph $f(x)$, showing the asymptotes.

---

- (TxGx25x02x05x011)
NOTAGS
Consider the function $f(x) = \tan\left(2(x - \frac{\pi}{2})\right)$. Find the the period of $f(x)$. Then graph $f(x)$, showing the asymptotes.

---

- (TxGx25x02x05x012)
NOTAGS
Consider the function $f(x) = -\tan(3x)$. Find the the period of $f(x)$. Then graph $f(x)$, showing the asymptotes.

---

- (TESTxxxTxGx25x02x05x001)
NOTAGS
For the following sinusoidal functions, find the amplitude, the period, and the phase shift. You do not have to graph the functions.

a) $f(x) = 3\sin\left(2x - \frac{2\pi}{3}\right)$

b) $f(x) = -5\sin\left(5x - \frac{\pi}{3}\right)$

---

## 5.9 Trig Inverses *(INV)*

- (TxINVx25x02x05x001)
NOTAGS
Evaluate.

a) $\sin^{-1}(\frac{\sqrt{2}}{2})$

b) $\sin^{-1}(-\frac{\sqrt{3}}{2})$

---

- (TxINVx25x02x05x002)
NOTAGS
Evaluate.

a) $\sin^{-1}(-\frac{\sqrt{1}}{2})$ b) $\sin^{-1}(-1)$

---

- (TxINVx25x02x05x003)
NOTAGS
Evaluate.

a) $\sin^{-1}(1)$ $\cos^{-1}(-\frac{\sqrt{1}}{2})$

---

## 5.10 Trig Identities *(ID)*

- (TxIDx25x02x05x001)
NOTAGS
Use the fundamental identities of trigonometric functions to write the following expressions using only sine and cosine and simplify them as much as possible.

a) $\sin x \cos x \sec x$ $\tan x \sin x + \sec x \cos^2 x$

---

- (TxIDx25x02x05x002)
NOTAGS
Use the fundamental identities of trigonometric functions to write the following expressions using only sine and cosine and simplify them as much as possible.

a) $\csc x - \cos x \cot x$ b) $3 \sin^3 x \csc x + 3 \cos^2 x$

---

- (TxIDx25x02x05x003)
NOTAGS
Use the fundamental identities of trigonometric functions to write

$$\frac{1 - \cos^2 x}{\tan^2 x} + \sin^2 x$$

using only sine and cosine and simplify them as much as possible.

---

- (TxIDx25x02x05x004)
NOTAGS
Use the fundamental identities of trigonometric functions to write

$$\frac{1+\tan^2 x}{\csc^2 x}+\sin^2 x+\frac{1}{\sec^2 x}$$

using only sine and cosine and simplify them as much as possible.

---

- (TxIDx25x01x22x001)
NOTAGS
Show that $\tan(x)\ \csc(x)\ \cos(x)=1$.

---

- (TxIDx25x01x22x002)
NOTAGS
Prove that $\tan(\theta)+\cot(\theta)=\sec(\theta)\ \csc(\theta)$.

---

- (TxIDx25x01x22x003)
NOTAGS
Prove that $\dfrac{\tan^2(\theta)+2}{1+\tan^2(\theta)}=1+\cos^2(\theta)$.

---

- (TxIDx25x02x05x005)
NOTAGS
Prove the following trigonometric identity. $\cos x\,(\tan x-\sec x)=\sin x-1$

---

- (TxIDx25x02x05x007)
NOTAGS
Prove the following trigonometric identity. $\sec^2 x(1+\cos^2 x)=\tan^2 x$

---

- (TxIDx25x02x05x008)
NOTAGS
Prove the following trigonometric identity. $\sin x\,(\cot x+\csc x)=\cos x+1$

---

- (TxIDx25x02x05x009)
NOTAGS
Prove the following trigonometric identity. $\cos^2 x(1+\tan^2 x)=1$

---

- (TxIDx25x02x05x010)
NOTAGS
Prove the following trigonometric identity. $\sin x\ \tan x=\sec x-\cos x$

---

- (TxIDx25x02x05x011)
NOTAGS
Prove the following trigonometric identity. $\sec x-\cos x=\tan x\ \sin x$

---

- (TxIDx25x02x05x012)
NOTAGS
Prove the following trigonometric identity. $\sec x \csc x = \tan x + \cot x$.

---

- (TxIDx25x02x05x013)
NOTAGS
Prove the following trigonometric identity. $\dfrac{\cos x \sec x}{\cot x} = \tan x$

---

- (TESTxxxTxIDx25x02x05x014)
NOTAGS
Prove the following trigonometric identity. $\tan x = \dfrac{\cos x \sec x}{\cot x}$

---

- (TESTxxxTxIDx25x02x05x006)
NOTAGS
Prove the following trigonometric identity. $\cos x - \cos^3 x = \cos x \sin^2 x$

---

## 5.11 Trig Equations *(EQ)*

- (TxEQx25x02x05x001)
NOTAGS
Solve the equation $2\sin x = -\sqrt{2}$ for $x$ in the interval $0 \leq x < 2\pi$.

---

- (TxEQx25x02x05x002)
NOTAGS
Solve the equation $4\sin^2 x - 2 = 0$ for $x$ in the interval $0 \leq x < 2\pi$.

---

- (TxEQx25x02x05x003)
NOTAGS
Solve the equation $2\sin x = -\sqrt{3}$ for $x$ in the interval $0 \leq x < 2\pi$.

---

- (TxEQx25x02x05x004)
NOTAGS
Solve the equation $2\cos x - \sqrt{3} = 0$ for $x$ in the interval $0 \leq x < 2\pi$.

---

- (TxEQx25x02x05x005)
NOTAGS
Solve the equation $3\cos x + \sqrt{3} = \cos x$ for $x$ in the interval $0 \leq x < 2\pi$.

---

- (TxEQx25x02x05x006)
NOTAGS
Solve the equation $3\sin x - \sqrt{3} = \sin x$ for $x$ in the interval $0 \leq x < 2\pi$.

---

### 5.12 Trig Function Concepts *(CON)*

- (TxCONx25x01x22x003)
NOTAGS
For angles between 0 and 90°, what values are possible for the trigonometric functions tan and cot?

---

# 6 Discrete Math *(DISC)*

- (Cx26x05x18x001)
type: TEST , topics: DISC GRAPHTHEORY
Answer the following questions for the graph below.

  a) On the graph, label the degree of each vertex.

  b) What is the graph's degree sequence?

  c) How many vertices does the graph have?

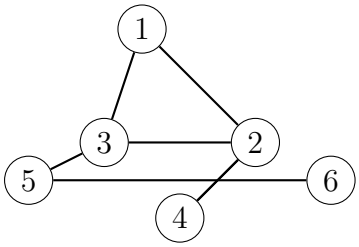
  d) How many edges does the graph have?

  e) What is the graph's diameter?

  f) What are the neighbors of vertex 3?

  g) What is the distance between vertex 2 and vertex 5?

  h) Draw the induced subgraph that has vertices 1,2, 3. What are two common names for this subgraph?




---

- (Cx26x05x18x002)
type: TEST , topics: DISC GRAPHTHEORY
The adjacency list representation of a graph is given below.

  a) Draw the graph.

  b) Find its adjacency matrix.

  c) Is there a matching from $\{W, X\}$ to $\{Z, Y, V\}$?

  d) Is there a matching from $\{Z, Y, V\}$ to $\{W, X\}$?

| | |
|---|---|
| $W$ | $Y, Z$ |
| $X$ | $Y, V$ |
| $Y$ | $W, X$ |
| $Z$ | $W$ |
| $V$ | $X$ |

---

- (Cx26x05x18x003)
type: TEST topics: DISC RECURSIVE
Recursively define $g$ as follows: $g(1) = 1$, $g(2) = 3$ and $g(n) = g(n-1) + 3 \cdot g(n-2)$.

  a) Evaluate $g(3)$ and $g(5)$.

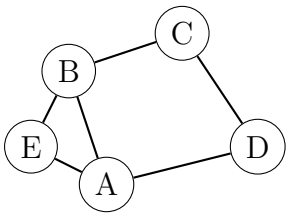
  b) Write a recursive Python function that computes $g$ (you may assume the function input is a positive integer).

---

- (Cx26x05x18x004)
type: TEST , topics: DISC GRAPHTHEORY
Consider the graph below.

  a) Does it have an Eulerian circuit? Does it have an Eulerian path which is not a circuit?

  b) Does it have a Hamiltonian circuit? Does it have a Hamiltonian path which is not a circuit?




---

- (Cx26x05x18x005)
type: TEST , topics: DISC PROOF INDUCTION
Prove that every amount of postage of 7 cents or more can be formed using just 2-cent and 5-cent stamps. Do the proof using strong induction, following these steps:
  - Let $P(n)$ be "postage of $n$ cents can be formed using just 2-cent and 5-cent stamps."
  - State and prove all required base cases.
  - Do the inductive step.

---

## 6.1 Propositional Logic

- (DISCx25x08x15x001)
topics: DISC PROPLOGIC
Which of the following statements are propositions? What are the truth values of those that are propositions?

  a) Do not cheat!
  b) When is class over?
  c) There is a roach in this room.
  d) Bronx Community College.
  e) Honey never rots.

---

- (DISCx25x08x15x002)
topics: DISC PROPLOGIC
Which of the following statements are propositions? What are the truth values of those that are propositions?

  a) $x + y = 4$
  b) $3 + 1 = 4$
  c) $6 - 8 > 0$
  d) $x = 9$

---

- (DISCx25x08x15x003)
topics: DISC PROPLOGIC
Which of the following statements are propositions? What are the truth values of those that are propositions?

  a) All integers are positive.
  b) All numbers are interesting.
  c) A negative integer is less than a positive integers.
  d) There is a largest integer.

---

- (DISCx25x08x15x005)
topics: DISC PROPLOGIC
Let $P$ and $Q$ be the propositions:

$P$: Vic lives in California

$Q$: Vic likes beer

Express each of these propositions as an English sentence.

a) $\neg P$

b) $P \rightarrow Q$

c) $Q \leftrightarrow P$

d) $P \vee \neg Q$

---

- (DISCx25x08x15x006)
topics: DISC PROPLOGIC
Let $P$ and $Q$ be the propositions:

$B$: I am a student at BCC.

$M$: I live in Manhattan.

Express each of these propositions as an English sentence and state whether it is true or false *for you*.

a) $B$

b) $B \wedge M$

c) $B \vee M$

d) $\neg B \vee M$

---

- (DISCx25x08x15x004)
topics: DISC PROPLOGIC
Let $P$ and $Q$ be the propositions:

$P$: I am a student at BCC.

$Q$: I am a student at CUNY.

Express each of these propositions as an English sentence and state whether it is true or false *for anyone saying it (not just you)*.

a) $P \wedge Q$

b) $P \rightarrow Q$

c) $Q \rightarrow P$

d) $P \vee \neg P$

---

- (DISCx25x08x15x007)
topics: DISC PROPLOGIC
Let $P$, $Q$, and $R$ be the following propositions

$P$: You do all the homework.

$Q$: You attend every class.

$R$: You get an A in this class.

Write these propositions using $P, Q$, and $R$, along with the logical connectives.

a) You do all the homework, but you do not attend every class.

b) You get an A in the class, attend every class, and do all the homework.

c) If you do all the homework, and attend every class, then you will get an A in this class.

---

- (DISCx25x08x15x008)
topics: DISC PROPLOGIC
Determine whether each of these conditional statements is true or false.

a) If $1 + 1 = 2$, then Trump is presient of the USA.

b) If $1 + 1 = 3$, then Rihanna is president of the USA.

c) If $1 + 1 < 5$, then $2 + 2 = 5$.

d) If $1 + 1$ is negative, then $2 + 2 = 5$.

---

- (DISCx25x08x28x001)
type: HW , topics: DISC PROPLOGIC
Determine whether each of these statements is true or false.

a) If 7 is even, then Mickey Mouse is presient of the USA.

b) If $1 - 1 < -1$ or $2 + 3 > 5$.

c) If 2 is prime, then The Bronx is not in NYC.

d) $128 + 188$ is positive and $2 + 2 = 4$.

---

### 6.1.1 Truth Tables

- (DISCx25x08x15x009)
topics: DISC PROPLOGIC
How many rows appear in a truth table for each of these compound propositions?

a) $X \vee \neg X$

b) $A \to (A \wedge \neg B)$

c) $(P \vee \neg Q) \leftrightarrow ((P \wedge Q) \to \neg P)$

d) $(A \vee B) \to (A \vee B \wedge \neg C)$

---

- (DISCx25x08x15x010)
topics: DISC PROPLOGIC
Construct a truth table for the following proposition: $X \vee \neg X$

---

- (DISCx25x08x15x011)
topics: DISC PROPLOGIC
Construct a truth table for the following proposition: $\neg A \wedge B$

---

- (DISCx25x08x15x012)
topics: DISC PROPLOGIC
Construct a truth table for the following proposition: $\neg A \rightarrow (A \vee \neg B)$

---

- (DISCx25x08x29x001)
type: HW , topics: DISC PROPLOGIC
Construct a truth table for the following proposition: $(A \wedge \neg B) \rightarrow (\neg A \wedge B)$

---

### 6.1.2 Tautologies and Contradictions

- (DISCx25x08x15x013)
topics: DISC PROPLOGIC
Show that the following proposition is a tautology (using a truth table):

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$$

---

- (DISCx25x08x15x014)
topics: DISC PROPLOGIC
Show that the following proposition is a tautology (just giving an argument in English without using a truth table): $A \rightarrow (A \vee B)$

---

- (DISCx25x08x15x023)
topics: DISC PROPLOGIC
Show that the following proposition is a contradiction (just giving an argument in English without using a truth table): $P \leftrightarrow \neg P$

---

- (DISCx25x08x15x019)
topics: DISC PROPLOGIC
Find an example of a conditional statement in propositional logic which is a tautology.

---

- (DISCx25x08x28x002)
type: HW , topics: DISC PROPLOGIC
Construct a truth table for the following proposition: $(\neg A \wedge A) \rightarrow B$.
Also, given an explanation (from the meaning of the proposition) for why it is a tautology.

---

- (DISCx25x09x11x001)
type: TEST , topics: DISC PROPLOGIC
Construct a truth table for the following proposition and state whether it is a tautology, contradiction, or contingency: $(X \wedge \neg Y) \rightarrow (\neg Y)$

---

- (DISCx25x12x10x001)
type: TEST , topics: DISC PROPLOGIC
Construct a truth table for the following proposition and determine whether it is a tautology, contradiction, or contingency: $(X \vee Y) \leftrightarrow (\neg X \wedge \neg Y)$

---

### 6.1.3 Propositional Logic Equivalence

- (DISCx25x08x15x024)
topics: DISC PROPLOGIC
Show that the following 2 propositions are **not** logically equivalent: $(A \rightarrow B) \rightarrow C$ and $A \rightarrow (B \rightarrow C)$

---

- (DISCx25x08x15x025)
topics: DISC PROPLOGIC
Consider the following proposition: $A \vee B \wedge C$. Make 2 different choices of placing parentheses, so that you get 2 propositions which are **not** logically equivalent.

---

- (DISCx25x08x15x020)
topics: DISC PROPLOGIC
We know that $\neg\neg P$ is equivalent to $P$. More generally, if we build up a proposition using only the letter $P$ and the negation symbol, which of these propositions are equivalent to $P$?

---

- (DISCx25x08x15x021)
topics: DISC PROPLOGIC
Use De Morgan's laws to find the negation of each of the following statements.

  a) Jose is smart and rich.

  b) Mary takes the subway to BCC or walks to BCC

---

- (DISCx25x08x15x022)
topics: DISC PROPLOGIC
Use De Morgan's laws to find the negation of each of the following statements.

  a) $A \vee \neg B$

  b) Mary takes the subway to BCC or walks to BCC

---

- (DISCx25x08x29x002)
type: HW , topics: DISC PROPLOGIC
Show that the following 2 propositions are logically equivalent:
$A \leftrightarrow B$ and $(A \rightarrow B) \wedge (B \rightarrow A)$

---

- (DISCx25x08x28x003)
type: HW , topics: DISC PROPLOGIC
Use De Morgan's laws to find the negation of each of the following statements.

  a) $\neg A \wedge B$

  b) BCC is in CUNY or BCC is not in NYC

---

- (DISCx25x09x11x002)
type: TEST , topics: DISC PROPLOGIC
Show that the following 2 propositions are logically equivalent: $\neg X$ and $\neg\neg\neg X$

---

### 6.1.4 Propositional Logic Applications

- (DISCx25x08x15x015)
topics: DISC PROPLOGIC
Is this system specification consistent?

  – Whenever the system software is being upgraded, users cannot access the file system.

  – If users can access the file system, then they can save new files.

  – If users cannot save new files, then the system software is not being upgraded.

---

- (DISCx25x08x15x016)
topics: DISC PROPLOGIC
Is this system specification consistent?

  – The system is in multiuser state if and only if it is operating normally.

  – If the system is operating normally, the kernel is functioning.

  – The kernel is not functioning or the system is in interrupt mode.

  – If the system is not in multiuser state, then it is in interrupt mode.

  – The system is not in interrupt mode.

---

- (DISCx25x08x15x017)
topics: DISC PROPLOGIC
Construct a circuit that produces the output $x \vee (\neg y \wedge z)$ from input bits $x, y$, and $z$ (only 3 standard gate types allowed: AND, OR, NOT).

---

- (DISCx25x08x15x018)
topics: DISC PROPLOGIC
Construct a circuit that produces the output $(\neg a) \rightarrow (b \wedge c)$ from input bits $a, b$, and $c$ (only 3 standard gate types allowed: AND, OR, NOT).

---

- (DISCx25x08x29x004)
type: HW , topics: DISC PROPLOGIC
Is this system specification consistent?
  - If students are being graded, professors cannot teach.
  - If professors can teach, then students can learn.
  - If students can learn, then students are not being graded.

---

- (DISCx25x09x11x003)
type: TEST topics: DISC PROPLOGIC
Is this system specification consistent?
  - Laughing is forbidden.
  - Crying is not allowed.
  - If laughing is forbidden, then crying is allowed.

---

- (PROPx26x01x19x001)
topics: DISC PYTHON
Use Python (showing your code) to determine if the propositional logic statement

$$\neg(A \wedge B) \vee (\neg C \wedge D)$$

is true or false under the various assignments to propositional variables.

a) $A = T, B = F, C = T, D = F$ b) $A = F, B = T, C = F, D = F$

---

- (PROPx26x01x19x002)
topics: DISC PYTHON
Use Python (showing your code) to find two different assignments to the variables that make the following the propositional logic statement true:

$$((\neg X \wedge Y \wedge \neg Z) \vee (X \wedge \neg Y)) \wedge (Y \vee Z)$$

---

## 6.2 Algorithms and Programming

- (ALGx25x08x17x001)
topics: DISC PROG
Write some code that assigns 34 to the variable $y$.

---

- (ALGx25x08x17x002)
topics: DISC PROG
Write some code that first assigns 9 to the variable $x$, then increments $x$ by 2.

---

- (ALGx25x08x17x003)
topics: DISC PROG
Write some code that assigns the variable $y$ to be 3 times the value of $x$.

---

- (ALGx25x08x17x005)
topics: DISC PROG
Write a statement that defines the list $[3, 4, 2, 3]$.

---

- (ALGx25x08x17x006)
topics: DISC PROG
Write a statement that sets variable $x$ to be the third entry in list $L$ (list indexing starts at 0).

---

- (ALGx25x08x17x004)
topics: DISC PROG
Write conditional statment that depends on the value of variable $x$. If $x$ is positive, then $x$ is set to 1, while if $x$ is negative $x$ is set to $-1$.
Side question: What does your statement do if $x$ is zero?

---

- (ALGx25x12x10x001)
topics: DISC PROG
Suppose two variables $x$ and $y$ are already set to integer values. Write Python code that sets both $x$ and $y$ to the smaller of the two values. If they are equal, the values should be unchanged. *For example:* If $x = 7$ and $y = 2$ then after running your code, both $x$ and $y$ should be equal to 2.

---

- (ALGx25x08x17x007)
topics: DISC PROG
Write a statement that sets variable $W$ to be the first entry in a list if the current value W is even, and to the second entry of the list if the current value of $W$ is odd (list indexing starts at 0).

---

- (ALGx25x08x17x008)
topics: DISC PROG
Write a for-loop that sums the numbers from 1 to 100.

---

- (ALGx25x08x17x009)
topics: DISC PROG
Write a while-loop that sums the numbers from 1 to 100.

---

- (ALGx25x11x05x001)
type: TEST topics: DISC PROG
Write a while-loop that sums the numbers from 10 to 200.

---

- (ALGx25x08x17x010)
topics: DISC PROG
Write a for-loop that sums the numbers in a list.

---

- (ALGx25x08x17x011)
topics: DISC PROG
Write a while-loop that sums the numbers in a list.

---

- (ALGx25x08x17x012)
topics: DISC PROG
Write the code for a function that takes one input and returns double that input.

---

- (ALGx25x08x17x013)
topics: DISC PROG
Write the code for a function that takes one input and returns double that input if the input is positive, and otherwise just returns the given input.
Also - graph this function on the plane.

---

- (ALGx26x01x20x001)
topics: DISC PROG
Using a while loop and a boolean flag, write a program that finds the first integer divisible by 10, between some positive integers $x$ and $y$ (assume $x < y$).

---

- (ALGx26x01x20x002)
topics: DISC PROG
Using a while loop and a boolean flag and the *isPrime* function, write a program that finds the first prime, between some positive integers $x$ and $y$ (assume $x < y$).

---

- (ALGx26x01x20x003)
topics: DISC PROG
Suppose $x$ and $y$ are integers where $x < y$. Use nested for-loops to print out all the pairs of even integers $(a, b)$ where $x \leq a, b, \leq y$.

---

- (ALGx26x01x20x004)
type: LECTURE , topics: DISC PROG
Suppose $x$ and $y$ are integers where $x < y$. Use nested for-loops to print out all the pairs of integers $(a, b)$ where $x \leq a, b, \leq y$.

---

- (ALGx25x08x17x014)
topics: DISC PROG
Write the definition of a function which takes a positive integer as input and returns the largest integer that divides it evenly.

---

- (ALGx25x08x17x015)
topics: DISC PROG
Write program to test the claim: The sum of odds is even (test all pairs of evens up to 100)

---

- (ALGx25x08x17x016)
topics: DISC PROG
What do you think the sum of an even and an odd will be? Write program to test your hypothesis.

---

- (ALGx25x08x25x001)
type: HW , topics: DISC PROG
Suppose variables $x$ and $y$ already have values assigned to them. Write code that swaps the values between $x$ and $y$ (using an extra variable besides $x$ and $y$).

---

- (ALGx25x08x25x002)
type: HW , topics: DISC PROG
Define a Python function which takes one numerical input. If the input is negative, it returns 0; otherwise it just returns back the same input. Graph this function.
Note: This function is famous in the AI neural net work; called the "ReLU" function.

---

- (ALGx25x08x25x003)
type: HW , topics: DISC PROG
Suppose $L$ is a list of positive integers, and $b$ is a boolean variable. Write code that sets $b$ to True if every entry in the list is even, and False otherwise.

---

- (ALGx25x08x25x004)
type: HW , topics: DISC PROG
Some day we will prove that for all positive integers $n$, $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$.
Write a program that verifies this claim for $n$ such that $1 \leq n \leq 1000$.

---

- (ALGx25x08x25x005)
type: HW , topics: DISC PROG
Write a program that finds the sum of the first $n$ odd positive integers; for example, if $n = 2$, the program should calculate 4, since $1 + 3 = 4$. Make conjecture about the value of this sum (no need to prove it! however, use data from your programs as evidence of your conjecture).

---

- (ALGx25x09x18x001)
type: TEST , topics: DISC PROG
In Python, write the definition of a function $F$ taking in an input (expected to be an integer). If the input is odd, then that same number is returned. If the input is even, then half of the number is returned.
Example: $F(20)$ should return 10, while $F(25)$ should return 25

---

- (ALGx25x09x18x002)
type: TEST , topics: DISC PROG
It seems that for $n \geq 4$ the following is true: $2^n < n!$ (recall that $n! = n \cdot (n-1) \cdots 2 \cdot 1$)
Write a Python program that verifies that inequality for $n$ such that $4 \leq n \leq 1000$.

---

## 6.3 Quantifiers and Predicates

- (QUANTx25x08x18x001)
topics: DISC QUANT
Let $P(x)$ be the predicate: $x < 7$. What are these truth values?

a) $P(4)$
b) $P(58)$
c) $P(-7)$
d) $P(7)$

---

- (QUANTx25x08x18x002)
topics: DISC QUANT
Let $C(x, y)$ be the statement: "City $x$ is the capital of state $y$". What are the truth values of the following? (if you have no shame, look on your phone ...)

a) C(Manhattan, New York)
b) C(Little Rock, Arkansas)
c) C(Trenton, New Jersey)
d) C(Los Angeles, California)

---

- (QUANTx25x08x18x003)
topics: DISC QUANT
Let $Q(x, y)$ denote the statement: "$x^2 < y$ or $y = 0$". What are the truth values of the following?

a) $Q(2, 7)$

b) $Q(7, 2)$

c) $Q(-3, 0)$

d) $Q(0, 5)$

---

- (QUANTx25x09x17x001)
type: TEST topics: DISC QUANT
Let $P(x, y)$ denote the statement: "$x < y$ and $x^2 < 10$". What are the truth values of the following?

a) $P(4, 6)$

b) $P(5, 4)$

c) $P(3, 3)$

d) $P(-2, 9)$

---

- (QUANTx25x08x18x004)
topics: DISC QUANT
Suppose the domain is all *positive* integers. Translate to each to English. Is each true or false? (remember the domain: positive integers)

a) $\forall x\ x > 0$

b) $\forall y\ y \leq 100$

c) $\exists y\ y \leq 100$

d) $\exists x\ x + x = 3 \cdot x$

---

- (QUANTx25x09x17x002)
type: TEST topics: DISC QUANT
Suppose the domain is all integers. For each statement, is it true or false?

a) $\forall x\ x^2 \geq 0$

b) $\exists y\ y^3 = -8$

c) $\exists x \forall y\ y > x$

d) $\forall x \exists y\ y = x + 3$

---

- (QUANTx25x12x10x001)
type: TEST topics: DISC QUANT
Let $P(x, y)$ be the predicate: "$x^2 < y$". Suppose the domain is all integers. For each statement, is it true or false?

a) $P(3, 3)$

b) $\exists x\ P(x, 0)$

c) $\forall x\ \exists y\ P(x, y)$

---

- (QUANTx25x08x18x005)
topics: DISC QUANT
Let $B(x)$ be the predicate: "$x$ is a student at BCC".
Let $C(x)$ be the predicate: "$x$ lives in The Bronx".
Express each of the following statements in terms of $B(x)$ and $C(x)$, quantifiers, and logical connectives. Let the domain consist of residents of NYC.

  a) Someone lives in The Bronx, and is also a student at BCC.

  b) There is a student at BCC who does not live in The Bronx.

  c) Every student at BCC lives in The Bronx.

---

- (QUANTx25x08x18x006)
topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all integers.

  a) $\exists x\ x^2 = 2x$ b) $\forall x\ 3x \geq x$

---

- (QUANTx25x08x18x008)
topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all real numbers.

  a) $\exists x\ x - 2 > x$ b) $\forall x\ x^4 \geq 0$

---

- (QUANTx25x08x18x007)
topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all rational numbers. In both cases, what if we changed the domain to be just the integers?

  a) $\forall x \exists y\ x \cdot y = 1$ b) $\forall x(x > 0 \rightarrow (x - 3) > 0)$

---

- (QUANTx25x08x18x009)
topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all integers

  a) $\forall x \exists y\ x^2 < y$ b) $\forall v \exists w\ v + w = 17$

---

- (QUANTx25x08x18x010)
topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all integers

a) $\forall x \exists u \exists w\ u < x < w$ b) $\exists u \exists w \forall x\ u < x < w$

---

- (QUANTx26x02x04x001)
type: HW , topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all integers

a) $\exists u \forall x\ x < \sqrt{u}$ b) $\forall x \exists u\ x < \sqrt{u}$

---

- (QUANTx25x08x30x001)
type: HW , topics: DISC QUANT
Determine the truth value of each of these statements if the domain consists of all integers

a) $\forall x\ (x < 2x)$ b) $\exists u\ (3u + 7 = 34)$

---

- (QUANTx25x08x30x002)
type: HW , topics: DISC QUANT
Let $N(x)$ be the predicate: "$x$ lives in NY".
Let $B(y)$ be the predicate: "$y$ lives in The Bronx".
Express each of the following statements in terms of $N$ and $B$, quantifiers, and logical connectives. Let the domain consist of residents of humans.

a) Someone lives in NY.

b) Everyone who lives in The Bronx lives in NY.

---

## 6.4 Sets

- (SETx25x08x21x005)
topics: DISC SET
If $A$ and $B$ are sets in Python, write commands to calculate the following

a) $A \cup B$ b) $B \cap A$ c) $(A \cup B) - (B \cap A)$

---

- (SETx25x08x21x006)
topics: DISC SET
If $A = \{4, 8, 2, 9\}$ and $B = \{9, 4, 1\}$ are sets in Python, determine the values of the following Python expressions, then check by typing in the appropriate commands.

  a) A.intersection(B)  b) (A.union(B)).difference(A)

---

- (SETx25x08x30x001)
type: HW , topics: DISC SET
List the members of these sets.

  a) $\{x \mid x$ is a negative integer larger than $5\}$

  b) $\{y \mid y$ is a an integer such that $y^2 = -y\}$

---

- (SETx25x08x30x002)
type: HW , topics: DISC SET
List the members of these sets.

  a) $\{x \mid x$ is a negative integer larger than $5\}$

  b) $\{y \mid y$ is an integer such that $y^2 = -y\}$

---

- (SETx25x09x17x001)
type: TEST , topics: DISC SET
List the members of these sets.

  a) $\{x \mid x$ is a postive integer less than $6\}$  b) $\{y \mid y$ is an integer such that $y = -y\}$

---

- (SETx25x08x30x003)
type: HW , topics: DISC SET
For each pair of sets, determine if one is a subset of the other (say which) or if neither is a subset of the other.

  a) $A =$ set of positve integers divisble by 10. $B =$ set of positive even integers.

  b) $X =$ students below the age of 20. $Y =$ students in college.

---

- (SETx25x08x30x004)
type: HW , topics: DISC SET
Find the cardinality of each set. If it is infinite, just say infinite.

  a) $\{1, 4, 2, 2\}$

  b) $\{x \mid x \text{ is an integer and } (x-2)(x+2)(x-\frac{1}{2}) = 0\}$

  c) The even integers larger than 100.

  d) $\{a, \{2, 3\}, \{\{4.8\}, x\}, z\}$

---

- (SETx25x08x30x005)
type: HW , topics: DISC SET
Suppose $A = \{1, 2, 3, 5, 6, 9\}$ and $B = \{6, 7, 8, 9\}$. Find the following.

  a) $A \cup B$

  b) $A \cap B$

  c) $A - B$

  d) $B - A$

---

- (SETx25x09x17x002)
type: TEST , topics: DISC SET
Suppose $A = \{3, 4, 5, 9, 15\}$ and $B = \{1, 2, 5, 9, 11\}$. Find the following (just give the answer).

  a) $A \cup B$

  b) $A \cap B$

  c) $A - B$

  d) $B - A$

---

- (SETx25x08x30x006)
type: HW , topics: DISC SET
Suppose $A = \{x, y, z\}$ and $B = \{2, 6\}$. Find the following.

  a) $A \times B$

  b) $B \times A$

  c) $B \times B$

---

- (SETx25x08x30x007)
type: HW , topics: DISC SET
Suppose $X = \{1, 2, 5, 6, 7\}$, $Y = \{2, 3, 6, 7\}$, and $Z = \{2, 6, 9\}$. Find the following.

  a) $(X \cap Y) \cup Z$

  b) $X \cap (Y \cup Z)$

---

- (SETx25x08x30x008)
type: HW , topics: DISC SET
Suppose we are in the universe $U = \{x \mid x \text{ is an integer such that } 1 \leq x \leq 9\}$. Suppose $A = \{2, 4, 6, 8\}$ and $B = \{1, 3, 7\}$. Find the following and draw Venn Diagrams for each.

  a) $\overline{A}$ b) $\overline{B}$

---

- (SETx25x09x17x003)
type: TEST , topics: DISC SET
Suppose the universal set is $U = \{0, 2, 4, 6, 8, 10, 12\}$. Suppose $A = \{2, 4, 8\}$ and $B = \{0, 4, 8, 12\}$. Find the following.

  a) $\overline{B}$ b) $\overline{A} \cup B$

---

- (SETx25x10x12x001)
type: HW , topics: DISC SET
Suppose we are in the universal set $U = \{x \in Z \mid 1 \leq x \leq 15\}$. Let $E$ be the even integers from $U$. Let $P = \{4, 5, 8, 9, 12, 13\}$. Let $Q = \{5, 6, 9, 10, 13, 15\}$. Find the following.

  a) $\overline{E} \cap P \cap \overline{Q}$ b) $(E \cup \overline{Q}) \cap \overline{P \cup Q}$

---

- (SETx25x12x10x001)
type: TEST , topics: DISC SET
Suppose $A = \{2, 7, 8\}$, $B = \{2, 6\}$, and $C = \{6, 7, 8, 9\}$. Find the following.

  a) $A \times B$ b) $A \cap C$ c) $(A \cup B) - C$

---

- (SETx25x10x12x002)
type: HW , topics: DISC SET

  a) Suppose $X = \{2, 3, 5\}$. Find the power set of $X$.

  b) Suppose $Y = \{a \in Z \mid -10 < a < 10\}$. How many elements are in the power set of $Y$? (use an exponential expression to give your answer).

---

### 6.4.1 Proofs with Sets

- (SETx26x03x24x001)
type: LECTURE topics: DISC SET PROOF
Prove that $A \cup \emptyset = A$.

---

- (SETx25x08x21x003)
topics: DISC SET PROOF
Prove that $A - B = A \cap \overline{B}$, for sets $A$ and $B$.

---

- (SETx25x08x21x004)
topics: DISC SET PROOF
Prove that $(A \cap B) \cup (A \cap \overline{B}) = A$ for sets $A$ and $B$.

---

- (SETx25x08x21x001)
topics: DISC SET PROOF
Prove that $(A^c)^c = A$ for any set $A$.

---

- (SETx25x08x21x002)
topics: DISC SET PROOF
Prove that $A \cap \overline{A} = \emptyset$ for any set $A$.

---

- (SETx26x03x14x004)
type: HW topics: DISC SET PROOF
Prove that $A \cup B \subseteq B$

---

- (SETx26x03x14x002)
type: HW topics: DISC SET PROOF
Prove that $A \cup (B - A) \subseteq A \cup B$

---

- (SETx26x03x14x003)
type: HW topics: DISC SET PROOF
Prove that $A \cap (B - A) \subset \emptyset$

---

- (SETx25x09x17x004)
type: TEST topics: DISC SET PROOF
Prove that $(A \cap B) \cup B \subseteq B$.

---

## 6.5 Number Bases

- (MODx25x11x14x005)
type: TEST , topics: DISC NUMBASE
Convert each number to decimal.

a) The binary number 11010.

b) The HEX number $10E$.

---

- (MODx25x11x14x006)
type: TEST , topics: DISC NUMBASE
Convert the decimal number 45 to binary.

---

- (MODx25x12x10x001)
type: TEST , topics: DISC NUMBASE

  a) Convert the binary number 11100 to decimal and to HEX.

  b) Convert the decimal number 25 to binary.

---

- (MODx25x11x14x007)
type: TEST , topics: DISC NUMBASE
Find the sums of the following.

  a) The binary numbers 10110 and 11.

  b) The HEX numbers $2A9$ and $5E$.

---

## 6.6 Basic Number Theory

- (MODx25x11x14x001)
type: TEST , topics: DISC NUMTHY
What are the quotient and remainder when

  a) 17 is divided by 3.

  b) $-5$ is divided by 2.

---

- (MODx25x11x14x008)
type: TEST , topics: DISC NUMTHY
Find the prime factorizations of 450 and 54, then use these factorizations to find the GCD and LCM.

---

- (MODx25x12x10x002)
type: TEST , topics: DISC NUMTHY
Consider the two integers $x = 80$ and $y = 36$.

  a) Find the prime factorizations of $x$ and $y$.

  b) Find the GCD and LCM of $x$ and $y$.

  c) Are $x$ and $y$ relatively prime or not?

  d) Find any positive integer that is relatively prime with $x$.

---

- (MODx25x11x18x001)
type: TEST , topics: DISC NUMTHY
For each pair of numbers, are they relatively prime? If not, find an integer that is relatively prime with both integers in the pair.

  a) 20 and 27

  b) 10 and 9

---

## 6.7 Modular Arithmetic

- (MODx25x08x30x001)
topics: DISC MODARITH
What is the multiplicative inverse of 3 (mod 10)?

---

- (MODx25x11x14x002)
type: TEST , topics: DISC MODARITH

  a) What is the additive inverse of 4 (mod 7)?

  b) What is the multiplicative inverse of 4 (mod 7)?

---

- (MODx25x11x14x003)
type: TEST , topics: DISC MODARITH
Suppose $x \equiv 3 \pmod{10}$

  a) Calculate $4x \pmod{10}$

  b) Calculate $x^{16} \pmod{10}$

---

- (MODx25x11x14x004)
type: TEST , topics: DISC MODARITH

  a) Find the smallest positive integer congruent to 20 modulo 15.

  b) Find the largest negative integer congruent to 20 modulo 15.

---

## 6.8 Cryptography

- (CRYx25x11x18x001)
type: TEST , topics: DISC CRYPTO
Use a shift cipher with key 4 to encode the message "WIN". In your work, 1) Show the translations between letters and numbers (using the standard translation of 0 for 'A', 1 for 'B', and so on), 2) Show the encoding calculations with numbers, 3) Give your final encoded answer as a 3 letter word.

---

- (CRYx25x12x10x001)
type: TEST , topics: DISC CRYPTO
Suppose Alice sends encrypted messages to Bob using a shift cipher with key = +10. To keep it simple, instead of sending a word, Alice is just going to send a single integer that is between 0 and 25 (inclusive), calculating mod 26.

  a) If Alice wants to send the message consisting of the single number 20, what is the encrypted message (i.e. number between 0 and 25) she actually sends?

  b) If Bob receives the encrypted message consisting of the single number 7, what is the decrypted message? (i.e. between 0 and 25).

  c) Alice tries being clever and uses the following affine cipher instead of the above shift cipher: $f(x) = 26^{212} \cdot x + 3 \pmod{26}$; that is to encrypt a number $x$, she finds $f(x)$ and sends that. If Alice wants to send 24, what is the encrypted message she sends?

---

- (CRYx25x08x30x001)
topics: DISC CRYPTO
Consider RSA encryption. Suppose your private key is $(65, 3)$. Someone sends you the encrypted message "6". Decode it.

---

- (CRYx25x08x30x003)
topics: DISC CRYPTO
Consider RSA encryption. Now you are an eavesdropper. Bob's public key is $(51, 13)$. Someone sent Bob the encrypted message "12". Decode the message. You can do this without even knowing Bob's private key.

---

- (CRYx25x08x30x004)
topics: DISC CRYPTO
Consider RSA encryption. Use the primes 3 and 11, and $e = 13$.

  a) Find your public key.

  b) Find your private key.

  c) If someone wants to send you the message "2", what is the encrypted message they would actually send.

  d) If you receive the encrypted message "10", what is the person's message?

---

- (CRYx25x11x17x001)
type: GROUP , topics: DISC CRYPTO
We will partition the class into 4 groups, named for the elements of $Z_4$. Each group will get space on the blackboard.

  - Part 1: Each group creates a public and private key (keep it small!). Write your public key in your group's board space.
  - Part 2: If you are group $x$, send a single secret letter to group $x + 1 \pmod 4$, using their public key. Write your encoded message in their board space.
  - Part 3: Decode the secret message passed to you, using your private key.

---

- (CRYx25x08x30x002)
type: GROUP , topics: DISC CRYPTO
We will partition the class into 3 groups: "Alice", "Bob", and "Eve".

  Part 1: Alice chooses a Shift Cipher key (integer between 1 and 25). Alice secretly tells Bob this key. Alice chooses a Plaintext Message (2 standard English words), then encodes this message to create the Coded Message. The Coded Message is shared with both Bob and Eve. Bob uses their key to decipher. Eve uses their cleverness to try to break the code and decipher.

  Part 2: Repeat the process with an affine cipher (so the key is a pair of integers).

---

## 6.9 Basic Counting

- (DISCx25x08x10x001)
NOTAGS
Scrabble experts memorize 3 letter words. How many are there? (assume we include all possibile listings of 3 letters from the English alphabet)

---

- (DISCx25x08x10x002)
type: TEST
Scrabble experts memorize 3 letter words (from the usual English alphabet with 26 letters). For this problem we include all sequences of 3 letters even if they are not really English words - for example, we include the actual word HOG and the nonsense word HHG.

  a) How many such 3-letter words are there?

  b) How many such 3-letter words are there that *start* with a vowel (such as ACT)? (the vowels are: A, E, I, O, U)

  c) How many such 3-letter words contain *no* vowels (such as BBX)?

  d) How many such 3-letter words contain *at least one* vowel (such as: ALL, UFO)

---

- (DISCx25x12x10x009)
type: TEST
Suppose a class of students has 25 Mets fans and 40 Yankee fans, and amazingly exactly 10 of these students are fans of both the Mets and the Yankees.

  a) How many students are there?

  b) If you select a random student, what is the probability that they are a Yankee fan?

---

- (DISCx25x08x10x003)
NOTAGS
Scrabble experts memorize 3 letter words. Are there more that start with a vowel or more that start with a consonant? Justify your answer without a calculator. (assume we include all possible listings of 3 letters from the English alphabet)

---

## 6.10 Combinations and Permutations

- (DISCx25x12x10x004)
type: TEST
Suppose a subway train with 9 cars pulls into the station.

  a) How many ways can you choose 4 cars from this train?

  b) Suppose 5 of the 9 cars have graffiti and the other 4 have no graffiti. How many ways can you choose 4 cars from the 9 if you need to have an equal number of cars with graffiti as ones without graffiti?

---

- (DISCx25x12x10x006)
type: TEST
Recall that a byte can store a binary string of length exactly 8.

  a) How many binary strings can a byte store?

  b) How many binary strings can a byte store if its first 3 positions must contain 1s (and the other positions can be anything)?

  c) How many binary strings can the byte store if exactly 3 of its positions must contain a 1 (and the rest are 0)?

---

- (DISCx25x08x10x007)
NOTAGS
Suppose a subway has 7 cars and two people get on. How many ways can this be done? ... Answer this question for each of the following two interpretations.

  a) The two people are named Vic and Mary, so it matters which one of them gets into which train car.

  b) The two people appear as indistinguishable icons from a surveillance camera.

---

## 6.11 Basic Discrete Probability

- (DISCx25x08x10x004)
NOTAGS
Suppose someone flips a coin a number of times in a row in order to get a sequence like $HHT$ - meaning first heads, second heads, and third tails. Which is more likely: HHHHH or THHTH?

---

- (DISCx25x08x10x005)
type: TEST
Consider a bridge that connects The Bronx to Manhattan. On The Bronx side there are 4 people and on the Manhattan side there are 6 people.

  a) If one of the people is chosen at random, what is the probability they are on The Bronx side? On the Manhattan side?

  b) If you chose two people at random, what is the probability they are both on The Bronx side?

  c) If you chose two people at random, what is the probability they are both on same side?

---

- (DISCx25x08x10x006)
diff:CREATIVE
Consider a bridge that connects The Bronx to Manhattan. There is one person on each side. Consider a process: Pick one of the people at random and send them to the other side. Keep repeating this process, one step at a time.

  a) What is the probability that they are on the same side after one step of this process?

  b) What is the probability that they are on the same side after two steps of this process?

  c) Give a full answer for what is happening in this process.

---

- (DISCx25x08x10x008)
type:GROUP
(Requires Expectation) Decide what you think is a reasonable probability of nuclear war on any given day. How many days have to pass before the expected number of nuclear wars is 1? (oh, and why do I not ask about the expected number of days will the number of nuclear wars is 2??)

---

## 6.12 Recursive Definitions

- (RECx25x08x20x004)
topics: DISC RECURSIVE
We recursively define $f$ as follows: $f(1) = 5$ and $f(n+1) = 10 \cdot f(n) + 5$. Evaluate the following

  a) $f(1)$

  b) $f(3)$

  c) $f(4)$

  d) Describe $f(n)$ in words without recursion

---

- (RECx25x08x20x005)
topics: DISC RECURSIVE
We recursively define $g$ as follows: $g(1) = 1$ and $g(n+1) = (g(n))^2 + g(n)$. Evaluate the following

  a) $g(2)$

  b) $g(3)$

  c) $g(4)$

---

- (RECx25x08x20x003)
topics: DISC RECURSIVE
We recursively define $f$ as follows: $f(0) = 9$ and $f(n+1) = -f(n)$. Evaluate the following

  a) $f(3)$

  b) $f(4)$

  c) $f(1000)$

---

- (RECx25x09x29x001)
type: HW topics: DISC RECURSIVE
We recursively define $f$ as follows: $f(0) = 1$, $f(1) = 1$ and $f(n) = f(n-1) - f(n-2)$. Evaluate the following.

  a) $f(1)$

  b) $f(3)$

  c) $f(5)$

  Can you describe $f$ without recursion?

---

- (RECx25x12x11x001)
type: TEST topics: DISC RECURSIVE
Recursively define $f$ as follows: $f(1) = 0$, $f(2) = 2$ and $f(n) = f(n-1) + 2 \cdot f(n-2)$.

  a) Evaluate $f(3)$ and $f(5)$.

  b) Write a recursive Python function that computes $f$ (you may assume the function input is a positive integer).

---

- (RECx25x09x29x002)
type: TEST topics: DISC RECURSIVE
We recursively define $f$ as follows: $f(0) = 1$, $f(1) = 3$ and $f(n) = 2f(n-1) - f(n-2)$. Evaluate the following.

  a) $f(1)$

  b) $f(3)$

  c) $f(5)$

---

- (RECx25x09x18x001)
type: TEST topics: DISC RECURSIVE
We recursively define $f$ as follows: $f(1) = 2$ and $f(n) = 5 \cdot f(n-1) + 10$.
Evaluate the following.

  a) $f(1)$

  b) $f(2)$

  c) $f(3)$

---

- (RECx25x08x20x006)
topics: DISC RECURSIVE
Let $G(n) = 7^n$, defined on integers $n \geq 0$. Find a recursive definition for $G$.

---

- (RECx25x08x20x007)
topics: DISC RECURSIVE
Give a recursive definition of the sequence $a_n$, $n \geq 1$, if $a_n = 8n$.

---

- (RECx25x09x29x003)
type: HW , topics: DISC RECURSIVE
Let $g(n) = (-3)^n$, defined on integers $n \geq 1$. Find a recursive definition for $g$.

---

- (RECx25x09x19x001)
type: TEST , topics: DISC RECURSIVE
Let $F(n) = -3n$, defined on integers $n \geq 1$. Find a recursive definition for $F$.

---

- (RECx25x08x20x002)
topics: DISC RECURSIVE
Let $F(n)$ be the number of strings of length $n$ using just the letter $b$. Find a recursive definition for $F(n)$ (Warning: This question is so simple, it might be tricky! ...).

---

- (RECx25x08x20x001)
topics: DISC RECURSIVE
Let $F(n, k)$ be the number of strings of length $n$ using letters in $\{a, b\}$ with exactly $k$ $b$'s. Find a recursive definition for $F(n, k)$.

---

- (RECx25x08x19x001)
topics: DISC RECURSIVE
Let $F(n, k)$ be the number of strings of length $n$ using letters in $\{a, b, c\}$ with exactly $k$ $b$'s. Find a recursive definition for $F(n, k)$.

---

- (RECx25x08x20x009)
topics: DISC RECURSIVE
Give a recursive definition of the set of positive odd integers.

---

- (RECx25x08x20x010)
topics: DISC RECURSIVE
Give a recursive definition of the set of even integers (including zero and the negatives).

---

- (RECx25x09x19x010)
type: TEST , topics: DISC RECURSIVE
Consider the set $S$ which is defined recursively as follows:

$2 \in S$ and $5 \in S$

If $x \in S$ and $y \in S$, then $x + y \in S$.

a) Is 20 in $S$? Is 13 in $S$?

b) Find all the positive integers that are **not** in $S$.

---

- (RECx25x12x11x002)
type: TEST , topics: DISC RECURSIVE
Consider the set $S$ which is defined recursively as follows:

$2 \in S$ and $5 \in S$

If $x \in S$ and $y \in S$, then $x \cdot y \in S$

a) Show that $8 \in S$. Show that $100 \in S$.

b) Prove that every power of 10 (that is 10 or larger), is in $S$ (i.e. show that the following numbers are all in $S$: $10, 100, 1000, \cdots$)

---

- (RECx25x12x18x002)
type: TEST , topics: DISC RECURSIVE
Consider the set $S$ which is defined recursively as follows:

$2 \in S$ and $5 \in S$

If $x \in S$ and $y \in S$, then $x + y \in S$

a) Show that $17 \in S$. Show that $25 \in S$.

b) Use induction to prove that every multiple of 10 is in $S$ (i.e. show that the following numbers are all in $S$: $10, 20, 30, \cdots$)

---

## 6.13 Recursive Programs

- (RECx25x09x29x102)
type: HW , topics: DISC RECURSIVE PROG
Let $g$ be defined by recursion as follows: $g(2) = 3$ and $g(n) = 7g(n-1) - 2$. Write the definition of a Python function that takes one integer as input and returns the output of $g$ on that input.

---

- (RECx25x09x29x103)
type: TEST , topics: DISC RECURSIVE PROG
Let $h$ be defined by recursion as follows: $h(3) = 2$ and $h(n+1) = 2h(n) - 3$. Write the definition of a Python function that takes one integer as input and returns the output of $h$ on that input.

---

## 6.14 Relations

- (RELx25x09x29x003)
type: HW , topics: DISC RELATION
List the ordered pairs in the relation from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$ defined as follows.

  a) $(a, b) \in R \Leftrightarrow gcd(a, b) = 1$ b) $(a, b) \in S \Leftrightarrow lcm(a, b) = 4$

---

- (RELx25x09x29x001)
type: HW , topics: DISC RELATION
For each of these relations on the set $\{A, B, C, D, E\}$ decide whether it is reflexive, whether it is irreflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

  a) $\{(B, B), (B, C), (B, D), (C, B), (C, C), (C, D)\}$

  b) $\{(A, E), (E, A)\}$

---

- (RELx25x10x29x001)
type: TEST , topics: DISC RELATION
Consider the following binary relation on the set $\{A, B, C, D\}$. Determine whether or not it is 1) reflexive, 2) irreflexive, 3) symmetric, 4) antisymmetric, 5) transitive.

$$\{(A, A), (B, B), (C, C), (B, C), (B, D), (C, B), (D, B)\}$$

---

- (RELx25x09x29x002)
type: HW , topics: DISC RELATION
For each of these relations on the integers, decide whether it is reflexive, whether it is irreflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

  a) $(x, y) \in R \Leftrightarrow xy \geq 2$ b) $(x, y) \in T \Leftrightarrow x \neq y$

---

### 6.14.1 Equivalence Relations

- (RELx25x10x13x001)
type: HW , topics: DISC RELATION
Consider the relation on the positive integers defined by $(x, y) \in R$ if: $x$ and $y$ "have the same parity" (which means: both $x$ and $y$ are even, or both $x$ and $y$ are odd). Prove that $R$ is an equivalence relation.

---

- (RELx25x10x29x002)
type: TEST , topics: DISC RELATION
The set of binary strings is the set of all finite sequences of 0's and 1's (example: 011, 10100, etc). Define the following relation $R$ on the binary strings.

$$(x, y) \in R \text{ if strings } x \text{ and } y \text{ have the same number of 0s}$$

For example: $(1001, 01011) \in R$, but $(010, 011) \notin R$. Prove that $R$ is an equivalence relation.

---

- (RELx25x10x13x002)
type: HW , topics: DISC RELATION
Consider the relation on the positive integers defined by $xRy$ if: $x \leq y$. Show that $R$ is NOT an equivalence relation.

---

- (RELx25x10x13x003)
type: HW , topics: DISC RELATION
The set of binary strings is the set of all finite sequences of 0's and 1's (example: 011, 10100, etc). Define the relation $S$ on binary strings of length at least 4. $(x, y) \in S$ if: strings $x$ and $y$ are the same on their two rightmost bits. For example: $(10011, 011) \in S$, but $(0100, 0101) \notin S$. Is $S$ an equivalence relation?

---

### 6.14.2 Posets

- (RELx25x10x13x004)
type: HW , topics: DISC RELATION
Which of these relations on $\{A, B, C, D\}$ are partial orderings? If a relation is not a partial ordering, determine which relevant properies it lacks.

  a) $\{(A, B), (B, C), (C, D), (A, C), (A, D), (B, D), (A, A), (B, B), (C, C)\}$

  b) $\{(A, A), (B, B), (B, A), (C, C), (A, B)\}$

---

- (RELx25x10x13x005)
type: HW , topics: DISC RELATION
Draw the Hasse diagram for the divisibility poset for each set of integers (i.e. $(a, b)$ is in the poset when $a|b$). Indicate any least or greatest elements.

  a) $\{1, 2, 3, 4, 6, 8, 24\}$ b) $\{3, 6, 9, 12, 15, 18, 36\}$

---

- (RELx25x10x29x003)
type: TEST , topics: DISC RELATION
Draw the Hasse diagram for the divisibility poset for $\{2, 3, 6, 9, 18\}$ (i.e. $(a, b)$ is in the poset when $a|b$). Indicate any least or greatest elements. What is the least upper bound of 2 and 3? What is the greatest lower bound of 6 and 18?

---

- (RELx25x10x13x006)
type: HW , topics: DISC RELATION
Draw the Hasse diagram for the in inclusion relation on the power set of $\{s, t, v, w\}$

---

### 6.14.3 More Relations

- (RELx26x03x16x001)
type: GROUP , topics: DISC RELATION
Create a sqlite database for some topic (use class example, but do something different!). Other people from class will make queries of your database. Write CREATE and INSERT commands below.

---

- (RELx25x12x11x001)
type: TEST , topics: DISC RELATION
Consider the binary relation $R$ on the set of positive integers defined by $(x, y) \in R$ if $x + y$ is even. For example: $(3, 5) \in R$ because $3 + 5 = 8$, and 8 is even.

  a) Determine whether or not $R$ is 1) reflexive, 2) irreflexive, 3) symmetric, 4) antisymmetric, 5) transitive.

  b) Is $R$ an equivalence relation? Is $R$ a partial order?

---

- (RELx25x12x18x001)
type: TEST , topics: DISC RELATION
Consider the binary relation $R$ on the set of positive integers defined by $(x, y) \in R$ if $x \cdot y$ is odd. For example: $(3, 5) \in R$ because $3 \cdot 5 = 15$, and 15 is odd.

  a) Determine whether or not $R$ is 1) reflexive, 2) irreflexive, 3) symmetric, 4) antisymmetric, 5) transitive.

  b) Is $R$ an equivalence relation? Is $R$ a partial order?

---

## 6.15 Graph Theory

### 6.15.1 Basics

- (GRAPHx25x11x03x001)
type: HW , topics: DISC GRAPHTHEORY
Consider the the graph on vertices $\{1, 2, 3, 4, 5, 6\}$ with edges:

$$\{1, 2\}, \{2, 3\}, \{1, 3\}, \{4, 5\}\{5, 6\}$$

Draw it, and write down its adjacency matrix.

---

- (GRAPHx25x11x04x001)
type: HW , topics: DISC GRAPHTHEORY
Consider the following adjacency matrix for a graph on vertex set $\{1, 2, 3, 4, 5\}$.

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Draw the graph, write down its edge set, and find its adjacency list representation. What is the common name for this graph? (i.e. from among the classes $P_n$, $C_n$, $K_n$, $K_{a,b}$, etc)

---

- (GRAPHx25x11x04x002)
type: HW , topics: DISC GRAPHTHEORY
Draw $K_4$. Find a connected subgraph which is not an induced subgraph. Find a different graph from $K_4$ such that every connected subgraph is in fact an induced subgraph (choose a small example! and verify by checking all connected subgraphs).

---

- (GRAPHx25x11x04x003)
type: HW , topics: DISC GRAPHTHEORY
How many edges do each of the following graphs have? ($n, a, b$ are all positive integers)

$$P_n \quad C_n \quad K_{a,b} \quad K_n$$

---

- (GRAPHx25x11x04x004)
type: HW , topics: DISC GRAPHTHEORY
Consider $C_4$ expressed as follows: Its vertex set is $\{1, 2, 3, 4\}$; its edge set is $\{\ \{1, 2\},\ \{2, 3\},\ \{3, 4\},\ \{4, 1\}\ \}$.
Find all the graphs (on the same vertex set) which are not the same, but are isomorphic.

---

- (GRAPHx25x11x04x005)
type: HW , topics: DISC GRAPHTHEORY
Consider $P_4$ expressed as follows: Its vertex set is $\{1, 2, 3, 4\}$; its edge set is $\{\ \{1, 2\},\ \{2, 3\},\ \{3, 4\},\ \}$.
Find all the graphs (on the same vertex set) which are not the same, but are isomorphic.

---

- (GRAPHx25x11x04x006)
type: HW , topics: DISC GRAPHTHEORY
Find two graphs with the same degree sequence which are not isomorphic (don't look it up!)

---

- (GRAPHx25x11x04x007)
type: HW , topics: DISC GRAPHTHEORY
Draw a strongly connected directed graph with 4 arcs and 4 vertices. Write down its adjacency matrix.

---

- (GRAPHx25x11x04x008)
type: HW , topics: DISC GRAPHTHEORY
Suppose the degree sequence of a graph is $1, 2, 2, 3, 3, 3$. How many edges are in the graph?

---

- (GRAPHx25x11x04x009)
type: HW , topics: DISC GRAPHTHEORY
Suppose someone tells you they have a graph with degree sequence $1, 2, 2, 2, 4, 5, 5$. How do you know they are lying? (or at least mistaken ...)

---

- (GRAPHx25x11x04x010)
type: HW , topics: DISC GRAPHTHEORY
Consider the graph on vertex set $\{A, B, C, D\}$, with edges $\{A, B\}, \{A, C\}, \{A, D\}$. Draw all its subgraphs which have at least one edge.

---

- (GRAPHx25x11x04x011)
type: HW , topics: DISC GRAPHTHEORY
Consider the graph on vertex set $\{A, B, C, D\}$, with edges

$$\{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}.$$

Draw all its *induced* subgraphs which have at least one edge.

---

- (GRAPHx25x11x04x012)
type: HW , topics: DISC GRAPHTHEORY
Consider the graph given by the following adjacency list representation.

| | |
|---|---|
| $A$ | $B, C$ |
| $B$ | $A, C, D$ |
| $C$ | $A, B, D$ |
| $D$ | $B, C$ |

Draw the graph. Find its adjacency matrix. What is its connectivity?

---

- (GRAPHx25x12x13x001)
type: TEST , topics: DISC GRAPHTHEORY
The adjacency list representation of a graph is given below.

a) Draw the graph.

b) Find its adjacency matrix.

c) Is there a matching from $\{A, B, C\}$ to $\{D, E, F\}$?

d) Is there a matching from $\{A, B, E\}$ to $\{C, D, F\}$?

| | |
|---|---|
| $A$ | $C, D$ |
| $B$ | $C, F$ |
| $C$ | $A, B$ |
| $D$ | $A, E$ |
| $E$ | $D$ |
| $F$ | $B$ |

---

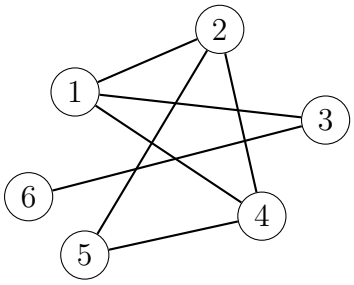
- (GRAPHx25x11x19x009)
type: TEST , topics: DISC GRAPHTHEORY
Consider the graph below.

  a) How many vertices does it have? How many edges does it have?

  b) On the graph, label the degree of each vertex.

  c) Draw the induced subgraph that has vertices 1, 2, 4, and 5.

  d) Draw any subgraph whose vertices are 1, 2, 4, and 5; but which is **not** an induced subgraph.




---

- (GRAPHx25x12x11x001)
type: TEST , topics: DISC GRAPHTHEORY
Answer the following questions for the graph below.

  a) On the graph, label the degree of each vertex.

  b) What is its degree sequence?

  c) How many vertices does it have?

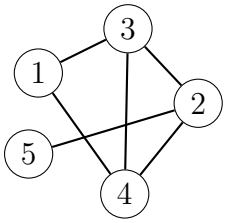
  d) How many edges does it have?

  e) What is its diameter?

  f) What are the neighbors of vertex 2?

  g) What is the distance between vertex 1 and vertex 2?

  h) Draw the induced subgraph that has vertices 1, 3, 4. What are two common names for this subgraph?

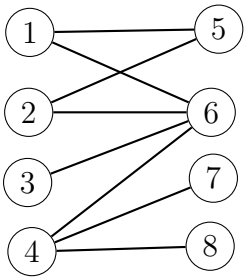



---

- (GRAPHx25x11x19x001)
type: TEST , topics: DISC GRAPHTHEORY
Draw the graph $K_{3,4}$. What is its chromatic number?
How many edges does $K_{a,b}$ have?

---

- (GRAPHx25x11x19x002)
type: TEST , topics: DISC GRAPHTHEORY
Consider the following graph, where $L = \{1, 2, 3, 4\}$ and $R = \{5, 6, 7, 8\}$.




Show that there is **not** a matching from $L$ to $R$ by finding a violation of Hall's Condition.

---

- (GRAPHx25x11x19x003)
type: TEST , topics: DISC GRAPHTHEORY
Consider the graph given by the following adjacency list representation. Draw the graph. What is its connectivity?

| | |
|---|---|
| $A$ | $B, C$ |
| $B$ | $A, D, E$ |
| $C$ | $A, D, E$ |
| $D$ | $B, C, E$ |
| $E$ | $B, C, D$ |

---

- (GRAPHx25x11x19x005)
type: TEST , topics: DISC GRAPHTHEORY
Consider the degree sequence $1, 2, 2, 2, 3$. Draw two **non-isomorphic** graphs which both have this degree sequence.

---

- (GRAPHx25x11x19x004)
type: TEST , topics: DISC GRAPHTHEORY
Consider the two proposed degree sequences. For each one state whether or not it is actually the degree sequence of a graph. If you say that it is the degree sequence of a graph, then draw that graph. If you say that it is **not** the degree sequence of any graph, prove this.

a) $2, 2, 3, 3$

b) $2, 2, 3, 3, 3$

---

- (GRAPHx25x12x13x003)
type: TEST , topics: DISC GRAPHTHEORY
For each graph: 1) draw it, 2) find its connectivity, 3) find its chromatic number.

  a) $C_5$ b) $C_6$ c) $K_4$ d) $K_{3,4}$

---

- (GRAPHx25x12x18x003)
type: TEST , topics: DISC GRAPHTHEORY
For each graph: 1) draw it, 2) find its connectivity, 3) find its chromatic number.

  a) $P_5$ b) $C_5$ c) $K_5$ d) $K_{3,4}$

---

- (GRAPHx25x11x19x008)
type: TEST , topics: DISC GRAPHTHEORY
Consider any particular graph whose diameter is 3. Define the following relation $R$ on the vertices of the graph: $(x, y) \in R$ if the distance between $x$ and $y$ is at most 2.

  a) Determine whether or not $R$ is 1) reflexive, 2) irreflexive, 3) symmetric, 4) antisymmetric, 5) transitive.

  b) Is $R$ an equivalence relation?

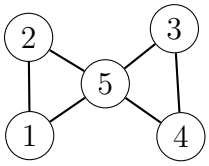
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### 6.15.2 Paths, Trails, Walks

- (GRAPHx25x12x13x002)
type: TEST , topics: DISC GRAPHTHEORY
Consider the graph below.

  a) Does it have an Eulerian circuit? Does it have an Eulerian path which is not a circuit?

  b) Does it have a Hamiltonian circuit? Does it have a Hamiltonian path which is not a circuit?

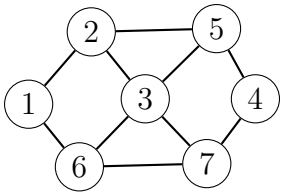



---

- (GRAPHx25x11x23x001)
type: TEST , topics: DISC GRAPHTHEORY
Consider the graph below.

  a) Prove that it does **not** have an Eulerian Circuit.

  b) Prove that it **does** have a Hamiltonian Circuit by finding one.




---

- (GRAPHx25x11x12x001)
type: HW , topics: DISC GRAPHTHEORY
Write down the adjacency matrix $A$ for $P_3$. Find $A^2$ and $A^3$.
What is the meaning of the number at entry $(2,2)$ in $A^2$? and in $A^3$? (justify the meaning of these numbers and write down all the corresponding walks)

---

- (GRAPHx25x11x19x007)
type: TEST , topics: DISC GRAPHTHEORY

Suppose a graph has vertex set $\{1,2,3,4\}$ and adjacency matrix $A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

You can trust that by a calculation $A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 3 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

  a) How many length two walks are there from vertex 1 to vertex 2?

  b) How many length three walks are there from vertex 1 to vertex 2?

---

### 6.15.3 Trees

- (TREEx25x12x13x001)
type: TEST , topics: DISC GRAPHTHEORY TREE
Consider the rooted tree below.

  a) Which vertices are leaves?

  b) What are the children of vertex 3?

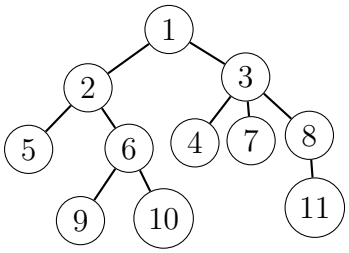
  c) What is the parent of vertex 6? Which vertex has no parent?

  d) What is the subtree at 2?

  e) Suppose 1000 vertices are added to the tree, along with edges, so that the new graph is still a tree.

  How many edges does the new graph have?

  What is the result of summing all the degrees of all the vertices in this new tree?




---

- (TREEx25x12x12x001)
type: TEST , topics: DISC GRAPHTHEORY TREE
Build the binary search tree you get by inserting the following integers *in the given order*: $5, 3, 2, 8, 7, 6, 1$.
Construct a different binary search tree with the same numbers which has as few levels as possible.

---

- (TREEx25x12x12x002)
type: TEST , topics: DISC GRAPHTHEORY TREE
Consider the following postfix expression: 2 5 + 8 2 ÷ 1 − ∗

  a) Evaluate it:

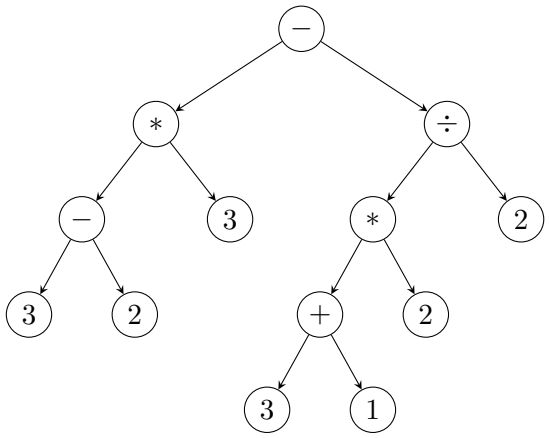
  b) Draw the expression tree for the postfix expression.

  c) Do a preorder traversal of that expression tree in order to find the prefix version of this expression.

---

- (TREEx25x12x19x001)
type: TEST , topics: DISC GRAPHTHEORY TREE

Consider the following tree:




a) Find the numerical value of the expression tree.

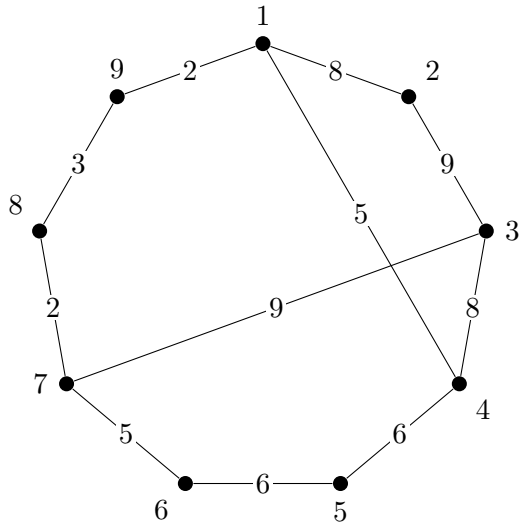
b) Write the expression using prefix notation.

c) Write the expression using infix notation.

d) Write the expression using postfix notation.

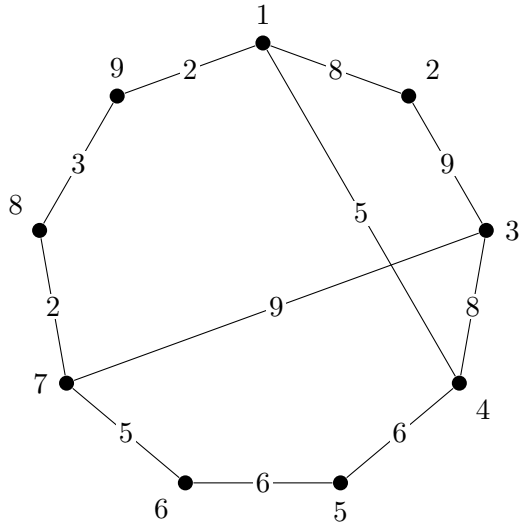
e) Evaluate the expression written in postfix notation: $5\ 3\ -\ 8\ *\ 2\ 3\ *\ +\ 2\ /$

- (TREEx25x12x13x002)
  type: TEST , topics: DISC GRAPHTHEORY TREE
  Find TWO different minimum cost spanning trees of the following graph.




---

- (TREEx25x12x18x002)
  type: TEST , topics: DISC GRAPHTHEORY TREE
  For the following graph, find 1) a minimum cost spanning tree, and 2) a maximum cost spanning tree.




---

## 6.16 Proofs

### 6.16.1 Propositional Proofs

- (INFx25x08x18x002)
topics: DISC PROOF
Suppose you are given the following premises:

  a) "It is sunny"

  b) "If it is sunny then it is daytime"

  c) "If it is daytime then you should study"

  Conclude from these premises that "You should study". Do this by translating the phrases into formal logic. Then give a structured proof where each line is justified by a named inference rule.

---

- (INFx25x08x18x003)
topics: DISC PROOF
Suppose you are given the following premises:

  a) "NYC is not the best city in the world"

  b) "If there were no homeless in NYC, then NYC would be the best city in the world"

  c) "If there are homeless in NYC, then taxes in NYC should increase"

  Conclude from these premises that "Taxes in NYC should increase". Do this by translating the phrases into formal logic. Then give a structured proof where each line is justified by a named inference rule.

---

- (INFx25x08x18x001)
topics: DISC PROOF
Suppose you are given the following premises:

  a) "BCC is a university"

  b) "Every university is on earth"

  Conclude from these premises that "BCC is on earth". Do this by translating the phrases into formal logic (using quantifiers), using the following predicates (whose domain are all buildings): $U(x)$ says "$x$ is a university" and $E(x)$ says "$x$ is on earth". Give a structured proof where each line is justified by a named inference rule.

---

- (INFx26x01x30x001)
topics: DISC PROOF
Suppose you are given the following premises:

$$P \to Q, \quad A \to P, \quad A$$

  Conclude $Q$, by giving a structured proof where each line is justified by a named inference rule.

---

- (INFx26x01x30x002)
topics: DISC PROOF
Suppose you are given the following premises:

$$P \to Q, \quad A \to P, \quad \neg Q$$

Conclude $\neg A$, by giving a structured proof where each line is justified by a named inference rule.

---

- (INFx26x01x30x003)
topics: DISC PROOF
Suppose you are given the following premises:

$$\neg X, \quad P \wedge Q, \quad P \to (X \vee Y)$$

Conclude $Y$, by giving a structured proof where each line is justified by a named inference rule.

---

- (INFx25x08x18x004)
topics: DISC PROOF
Suppose you are given the following premises:

a) Every person is either politically engaged or rich (or both)

b) Bob is not rich.

Conclude from these premises that Bob is politically engaged. Do this by translating the phrases into formal logic (using quantifiers), letting $P(x)$ mean "$x$ is politically engaged" and $R(x)$ mean "$x$ is rich".

---

- (INFx26x01x30x004)
topics: DISC PROOF
Suppose you are given the following premises:

$$P \to \forall x \ A(x), \quad \neg A(c),$$

Conclude $\neg P$, by giving a structured proof where each line is justified by a named inference rule.

---

- (INFx26x01x30x005)
topics: DISC PROOF
Suppose you are given the following premises:

$$(\exists x A(x)) \to (\forall y B(y)), \quad A(c)$$

Conclude $B(c)$, by giving a structured proof where each line is justified by a named inference rule.

---

- (INFx26x02x23x001)
type: TEST , topics: DISC PROOF
Consider the following two inferences. For each, which inference rule from the "Inference Rules" sheet justifies it? In each case, you must state the name of the inference rule, and how the variables on sheet (some combination of $p, q, r$) match the inferences here.

  a) From $(\neg X) \to (\neg Y)$ and $\neg X$, conclude $\neg Y$.

  b) From $(\neg Y) \to (A \wedge B)$ and $X \to (\neg Y)$, conclude $X \to (A \wedge B)$.

---

### 6.16.2 Basic Proofs

- (DISCx25x08x17x001)
topics: DISC PROOF
Consider summing two odd integers. What is your conjecture about the result? (even or odd or something else?) Prove it.

---

- (DISCx25x08x17x002)
topics: DISC PROOF
Prove that the product of any two consecutive integers is even.

---

- (DISCx25x08x18x001)
topics: DISC PROOF
Prove or disprove: If $n$ is an odd integer then $\dfrac{n-1}{2}$ is odd.

---

- (DISCx25x08x18x005)
topics: DISC PROOF
Prove that for all positive integers $n$, $n^3$ is odd if and only if $n$ is odd.

  You may use the following two facts (but reference every usage of them!).
  *Fact A*: The product of two odd integers is odd.
  *Fact B*: The product of two even integers is even.

---

- (DISCx25x08x18x002)
topics: DISC PROOF

  Prove that the product of any 3 consecutive positive integers is divisible by 3.

---

- (DISCx25x08x18x109)
topics: DISC PROOF

  Prove that for all positive integers $n$, $n^2$ is odd if and only if $n$ is odd.

---

- (DISCx25x08x17x104)
topics: DISC PROOF
Prove that if $A|B$ and $B|C$, then $A|C$.

---

- (DISCx25x08x18x003)
topics: DISC PROOF
One of the following is true, and the other false (notice they are converses of eachother). Find a counter-example to the false claim, and prove the true one.

  a) If $ab|c$, then both $a|c$ and $b|c$.

  b) If both $a|c$ and $b|c$, then $ab|c$.

---

- (DISCx25x08x18x105)
topics: DISC PROOF
Prove or disprove: If $a|b$ then $a^2|b^2$.

---

- (DISCx25x08x18x008)
topics: DISC PROOF
Prove that 20 is *not* a perfect square (use cases).

---

- (DISCx25x08x18x007)
topics: DISC PROOF
Prove that if $x$ and $y$ are integers and $xy$ is even, then $x$ is even or $y$ is even (Hint: Use Proof-by-Contradiction)

---

- (DISCx25x08x18x009)
topics: DISC PROOF
Consider the following program:

  if $x > 5$ :
  $\quad z = x + 1$
  else:
  $\quad z = x - 1$

  Suppose the initial assertion is that $x = 3$. Determine what the final value of $z$ will be, and prove it.

---

- (DISCx25x09x19x004)
type: TEST , topics: DISC PROOF
Consider the following program:

    if $x < 0$ :
        $x = x + 7$
    else:
        $x = 6x$

    Suppose you are given that $x$ is initialized to some positive integer. Show that after running the program, the value of $x$ is even.

---

- (DISCx25x08x18x011)
topics: DISC PROOF
Consider the following program:

    for $x$ in range(1, 1000):
        $y = y + 3$

    Given that the initial value of $y$ is even, prove that the final value is odd.

---

- (DISCx25x08x18x012)
topics: DISC PROOF
Consider the following program:

    $a = 1$
    while $a > 0$ :
        $a = a + 1$

    Prove that the program does *not* terminate.

---

- (DISCx25x08x25x001)
type: HW topics: DISC PROOF
Prove that the sum of any three odd integers is odd.

---

- (DISCx25x08x25x002)
topics: DISC PROOF
Consider the following program:

if $x > 9$ :
    $x = 9$
elif $x < -9$ :
    $x = -9$
else:
    $x = 0$

a) Suppose the initial assertion is that $x = -70$. Determine what the final value of $x$ will be, and prove it.

b) What are all the possible final values of $x$? Prove it (Hint: Use cases)

---

- (DISCx26x01x30x006)
type: HW topics: DISC PROOF
Prove that the sum of two integers which are both multiples of 5, results in a multiple of 5.

---

- (DISCx25x08x18x004)
type: HW topics: DISC PROOF
Prove or disprove: If $a|bc$ then $a|b$ or $a|c$.

---

- (DISCx25x08x18x205)
type: HW topics: DISC PROOF
Suppose $n$ is an integer. Prove $n$ is odd if and only if $n + 101$ is even.

---

- (DISCx25x12x10x008)
type: TEST topics: DISC PROOF
Suppose $n$ is an odd integer. Prove that $n^2$ is odd.

---

- (DISCx25x08x18x103)
type: HW , topics: DISC PROOF

We know that the product of 2 consecutive positive integers is divisible by 2, and that the product of any 3 consecutive positive integers is divisible by 3.
Make a conjecture about $k$ integers. Prove it!

---

- (DISCx25x08x18x006)
type: HW , topics: DISC PROOF
Prove that a right triangle *cannot* have all its side lengths equal to prime numbers (Hint: use Pythagorean Theorem and proof by contradiction).

---

- (DISCx25x08x17x003)
type: HW , topics: DISC PROOF
A **perfect square** is an integer that is equal to the square of another integer; for example 9 is a perfect square because $9 = 3^2$, but 18 is *not* a perfect square.

    a) Prove that if $m$ and $n$ are both perfect squares, then $nm$ is also a perfect square.

    b) Prove or disprove: The sum of two perfect squares is a perfect square.

---

### 6.16.3 Inductive Proofs

- (INDx25x09x01x003)
type: HW , topics: DISC PROOF INDUCTION
Prove that $5 + \cdots + 5 = 5n$ for positive integers $n$ (i.e. 5 added together $n$ times is $5n$). You must use induction! (not logically necessary, but just for practice).

---

- (INDx25x09x01x001)
type: HW , topics: DISC PROOF INDUCTION
Prove that $1 \cdot 1 \cdots 1 < 2$ (i.e. 1 times itself $n$ times is less than 2). You must use induction! (not logically necessary, but just for practice).

---

- (INDx25x09x01x002)
type: HW , topics: DISC PROOF INDUCTION
Prove that $2^n < 3^n$ for positive integers $n$. You must use induction! (not logically necessary, but just for practice).

---

- (INDx26x01x20x002)
type: CLASS , topics: DISC PROOF INDUCTION
Suppose $P(n)$ is the statement: "$3|(n^3 - n)$''. Suppose you want to prove $P(n)$ holds for all $n \geq b$, for some integer $b$.

    a) What is the smallest $b$ you can choose?

    b) Verify $P(n)$ for three distinct integers $n$.

    c) Write down the statment $P(k + 1)$.

---

- (INDx26x01x20x001)
type: CLASS , topics: DISC PROOF INDUCTION
Consider the proposition: $n^2 + n$ is even, for $n \geq 1$.
Do just the start of the inductive proof. For the full inductive proof you should 1) state what statement $P(n)$ you are doing induction on, 2) show $P(n)$ is true for the base case, and 3) do the inductive step.
Just do (1) and (2). That is, state exactly what $P(n)$ is, then state and prove the base case. Do **not** do the inductive step.

---

- (INDx26x03x23x001)
type: CLASS , topics: DISC PROOF INDUCTION
Use induction to show that $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$, for $n \geq 1$.

---

- (INDx25x09x19x003)
type: TEST , topics: DISC PROOF INDUCTION
Consider the proposition: 5 divides $n^5 - n$ for $n \geq 2$.
In order to prove this by induction you should 1) state what statement $P(n)$ you are doing induction on, 2) show $P(n)$ is true for the base case, and 3) do the inductive step.
Just do (1) and (2). That is, state exactly what $P(n)$ is, then state and prove the base case. Do **not** do the inductive step.

---

- (INDx25x10x13x001)
type: HW , topics: DISC PROOF INDUCTION
Prove that $2^n > n^2$ if $n$ is an integer greater than 4 (use induction).

---

- (INDx25x09x19x002)
type: TEST , topics: DISC PROOF INDUCTION
Prove that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$
(i.e. the sum of the first $n$ positive even integers is $n(n + 1)$. You must use induction to prove this.

---

- (INDx25x12x12x001)
type: TEST , topics: DISC PROOF INDUCTION
Use induction to prove that $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$

---

- (INDx25x12x18x001)
type: TEST , topics: DISC PROOF INDUCTION
Use induction to prove that $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$

---

- (INDx25x09x19x001)
type: TEST , topics: DISC PROOF INDUCTION
Prove that every amount of postage of 6 cents or more can be formed using just 3-cent and 4-cent stamps. Use strong induction with the proof starting as follows:

  - Let $P(n)$ be "postage of $n$ cents can be formed using just 3-cent and 4-cent stamps."
  - Base cases: $P(6), P(7)$, and $P(8)$. [You need to state and prove each base case]

  Now finish the proof by doing the inductive step.

---

### 6.16.4 Recursive Proofs

- (RECPFx25x08x20x001)
type: HW , topics: DISC PROOF RECURSIVE
Consider the set $S$ defined by $14 \in S$ and $s + t \in S$ whenever $s \in S$ and $t \in S$. Show that every element of $S$ is even.

---

- (RECPFx25x08x20x002)
topics: DISC PROOF RECURSIVE
Let $g$ be defined recursively by $g(0) = g(1) = 7$ and $g(n+1) = 2 \cdot g(n) + g(n-1)$. Prove that $g(n)$ is always odd.

---

- (RECPFx25x09x29x001)
type: HW topics: DISC PROOF RECURSIVE
Let $r$ be defined recursively by $r(1) = 3$, $r(2) = 5$ and $r(n+1) = r(n) + 4 \cdot r(n-1)$. Prove that $r(n)$, for $n \geq 1$ is always odd.

---

- (RECPFx25x10x29x001)
type: HW topics: DISC PROOF RECURSIVE
Let $f$ be defined recursively by $f(1) = 2$, $f(2) = 6$, and $f(n) = f(n-1) + f(n-2)$.

  a) Evaluate $f(3)$ and $f(4)$.

  b) Prove that for $n \geq 1$, we have: $f(n)$ is even.

---

- (RECPFx25x08x20x003)
topics: DISC PROOF RECURSIVE
Define a subset of binary strings $S$ as follows (implicit operation is concatenation): $101 \in S$ and if $x \in S$ then $x01 \in S$ and $x10 \in S$.

  a) List all the length 7 binary strings in $S$

  b) Prove that the strings in S all have one more 1 than 0.

---

# 7 Statistics *(STAT)*

- (26x09x15x001)
topics: STAT
Consider the following data:

15, 41, 12, 12, 32, 60, 9, 41, 42, 59, 40

a) Find the second quartile (i.e. median).

b) Find the first quartile.

c) Find the third quartile.

d) Find the range, and the interquartile range.

e) Draw a Box plot.

---

- (26x09x15x002)
topics: STAT

The table below shows the favorite color of each of 12 students.

| Student | Favorite Color | Student | Favorite Color |
|---|---|---|---|
| 1 | red | 7 | green |
| 2 | blue | 8 | red |
| 3 | green | 9 | blue |
| 4 | red | 10 | red |
| 5 | blue | 11 | green |
| 6 | red | 12 | blue |

a) Complete the frequency table and relative frequency table below.

| Favorite Color | Frequency | Relative Frequency |
|---|---|---|
| red | | |
| blue | | |
| green | | |

b) Draw a bar graph showing the frequency of each favorite color.

c) Draw another bar graph showing the *relative* frequency of each favorite color.

d) Make a pie chart showing the percentage breakdown of the favorite colors.

---

- (26x09x15x003)
topics: STAT
Consider the CUNY community colleges. Find the student population of each college. Then calculate the percentage of the total CUNY community-college population enrolled at each college, and use these percentages to create a pie chart.
(while you may work together to collect his data! ... Make your own pie chart!)

---

- (26x09x14x001)
topics: STAT
A group of 25 people were surveyed about the number of books they read during the past year and were found to have read the following numbers of books:

| | | | | |
|---|---|---|---|---|
| 2 | 3 | 5 | 6 | 6 |
| 7 | 8 | 8 | 9 | 10 |
| 11 | 11 | 12 | 13 | 13 |
| 14 | 15 | 15 | 16 | 17 |
| 18 | 19 | 20 | 21 | 23 |

In order to display the data in clearer form,

a) determine the class width for five (5) classes,

b) construct a frequency distribution showing the class limits for the five classes,

c) in the table, show the class boundaries,

d) construct a histogram, labeling the class boundaries.

---

# 8 Linear Algebra

## 8.1 Matrix Terminology

- (MATx25x10x12x001)
type: HW , topics: MATRIX
Consider the matrix $M = \begin{bmatrix} 11 & 2 & 8 & 9 \\ 6 & 4 & 13 & -1 \\ -7 & 5 & -3 & 0 \end{bmatrix}$

a) What are the dimensions of $M$?

b) What is the value at entry $M_{2,3}$

c) Using the "$M_{i,j}$" notation, name the entries which are negative.

---

## 8.2 Matrix Addition

- (MATx25x10x12x002)
type: HW , topics: MATRIX
Let $M = \begin{bmatrix} 1 & 12 & 0 & 9 \\ -6 & 0 & 3 & -1 \end{bmatrix}$ and $N = \begin{bmatrix} 11 & 2 & 8 & -9 \\ 0 & 4 & -2 & -1 \end{bmatrix}$
a) Find $M + N$ b) Find $M - N$

---

- (MATx25x10x27x002)
type: TEST , topics: MATRIX
Let $M = \begin{bmatrix} 1 & 1 & 0 \\ -3 & 0 & 3 \end{bmatrix}$ and $N = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$
a) Find $M + N$
b) What is the value of $M_{2,1} + N_{1,1}$?
c) Use the subscript notation to name the entries of $M$ and $N$ which are zero.

---

- (MATx25x11x05x005)
type: TEST , topics: MATRIX
Suppose
$$\begin{bmatrix} 1 & 1 & 0 \\ -3 & 0 & 3 \end{bmatrix} + \begin{bmatrix} a & b & c \\ x & y & z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$
Find $a, b, c, x, y$, and $z$.

---

## 8.3 Matrix Multiplication

- (MATx25x10x12x003)
type: HW , topics: MATRIX
Let $M = \begin{bmatrix} 1 & 2 \\ -6 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} 3 & 2 \\ 0 & 4 \end{bmatrix}$
a) Find $M \cdot N$ b) Find $N \cdot M$

---

- (MATx25x10x12x004)
type: HW , topics: MATRIX
Find two 2 by 2 matrices $M$ and $N$ (which are not the same) such that $M \cdot N = N \cdot M$.

---

- (MATx25x10x27x001)
type: TEST , topics: MATRIX
Let $A = \begin{bmatrix} 1 & 2 \\ -2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$
For each, evaluate or state why it is undefined.

  a) $A \cdot B$ b) $B \cdot A$

---

## 8.4 Neural Net Applications of Matrices

- (MATx26x03x23x001)
type: CLASS , topics: MATRIX
Draw the neural net that has 2 inputs and 4 outputs given by the following weight matrix. $\begin{bmatrix} 3 & 0 & -2 & 0 \\ 0 & 2 & 0 & -1 \end{bmatrix}$

---

- (MATx26x03x23x002)
type: CLASS , topics: MATRIX
Suppose a 3-layer neural net uses matrix $A$ to go from layer 1 to layer 2, and matrix $B$ to go from layer 2 to layer 3.
$A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 2 & 0 \end{bmatrix}, B = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 1 & -2 \end{bmatrix}$

  a) How many neurons does each layer have?

  b) Find one matrix which can be used to go directly from layer 1 to layer 3.

---

- (MATx26x03x23x008)
type: CLASS , topics: MATRIX
Design a neurel net with input layer of size 2, and one output layer of size 1 (which uses the Heaviside function), which computes the AND-function; that is: you just care about the 0, 1 inputs, and it must give AND on those inputs. Draw your neural net and describe it using matrices.

---

- (MATx26x03x23x004)
type: CLASS , topics: MATRIX
Design a neurel net (show the matrices) with input layer of size 4, and one output layer of size 4, which does not change the input, that is it is the identity function, assigning input vector $x$ to output vector $x$.

---

- (MATx26x03x23x003)
type: CLASS , topics: MATRIX
Design a neurel net (show the matrices) with input layer of size 2, output layer of size 1, with one hidden layer (which uses a ReLU), so that if the inputs are both positive, then the final output is $> 0$, and if the inputs are both negative, then the final is $\leq 0$.

---

# 9 UNCATAGORIZED *(XXX)*

- (XXXx25x01x22x001)
NOTAGS
For this question only, just give your answer (no work required).

  a) Which of the following are integers? $7,\ 9.5,\ 0,\ 7/4,\ \sqrt{10},\ -220$

  b) Which of the following are polynomials? $3x^5-4,\ \sqrt{2x}+3,\ \frac{20x}{x^3+7},\ -8x$

---

- (XXXx25x01x22x101)
NOTAGS

  a) Find any 4 numbers between $-23$ and $-22$ (not including them).

  b) How many integers are between $-2$ and 108, including both?

  c) How many rational numbers are between $-2$ and 108, including both?

---

- (XXXx25x01x22x002)
NOTAGS
For each item, say whether it is a *definition* or a *theorem*.

  a) For any real $x, y$, $2x + 2y = x + x + y + y$.

  b) A prime number is an integer $> 1$ with only 1 and itself as positive divisors.

  c) There is a real number whose square equals itself. (Also write this in symbols.)

---

- (XXXx25x01x22x003)
NOTAGS

a) Simplify $3u + 3u^2 - u - 2u^2$.

b) Evaluate the original expression and the simplified one at $u = 2$.

c) Write down the identity you conclude from the simplification.

---

- (XXXx25x01x22x004)
NOTAGS
Perform the operation and simplify (express results without negative exponents; positive rational exponents are ok):

a) $6^7 \cdot 6^{-10}$

b) $\dfrac{22^{20}}{22^{12}}$

---

- (XXXx25x01x22x005)
NOTAGS
(continued):

a) $x^{\frac{7}{2}} \cdot x^{\frac{2}{3}}$

b) $(3x^{-3}y^5)^4$

---

- (XXXx25x01x22x006)
NOTAGS
Evaluate each expression:

a) $\log_2(8)$

b) $\log_2\left(\frac{1}{8}\right)$

c) $\log_2(1)$

d) $\log_3\left(3^{75}\right)$

e) $5^{\log_5(20x)}$

---

- (XXXx25x01x22x008)
NOTAGS
Simplify (express results without exponents):

a) $8^{2/3}$

b) $8^{-2/3}$

---

- (XXXx25x01x22x009)
NOTAGS
Simplify $3\sqrt{8}\left(2\sqrt{5} - \sqrt{6}\right)$.

---

- (XXXx25x01x22x010)
NOTAGS
Simplify $\dfrac{2}{3+\sqrt{5}}$.

---

- (XXXx25x01x22x012)
NOTAGS
Perform and simplify: $\dfrac{m^3p^5}{3m^2p} \cdot \dfrac{9mp^2}{m^3p}$.

---

- (XXXx25x01x22x016)
NOTAGS
Let $h(y) = 10y^2 + \sqrt{y+3}$. Find $h(-2)$ and evaluate $h(1)$.

---

- (XXXx25x01x22x013)
NOTAGS
Consider $g(x) = \dfrac{x-3}{(x+2)(x-9)}$. Where is $g$ undefined? Where is it defined? What is the domain of $g$? Compute $g(0)$.

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- (XXXx25x01x22x015)
NOTAGS
Graph $f(x) = 2x + 1$. From the graph, guess its $x$ and $y$ intercepts.

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- (XXXx25x01x22x116)
NOTAGS
Consider the expression $\dfrac{y-20}{y+10}$.

  a) Evaluate it when $y = 0$ and when $y = 10$.

  b) When is the expression undefined?

  c) When does $\dfrac{y-20}{y+10} = 0$?

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- (XXXx25x01x22x017)
NOTAGS
Evaluate $3x^{3/4}$ at $x = 16$ and at $x = 0$.

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- (XXXx25x01x22x018)
NOTAGS
Consider $h(y) = y^{1/3}$.

  a) Evaluate $h(8)$ and $h(-8)$.

  b) Evaluate $h(0)$.

  c) What is the domain of $h$?

---

- (XXXx25x01x22x019)
NOTAGS

  a) Write $x^{7/4}$ using radicals.

  b) Write $\sqrt[5]{y^7}$ using exponents.

  c) Write $\left(\sqrt[12]{u}\right)^{13}$ using exponents.

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- (XXXx25x01x22x020)
NOTAGS
Evaluate: $\log_3(9)$, $\log_3(\frac{1}{9})$, $\log_3(3)$, $\log_3(1)$.

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- (XXXx25x01x22x021)
NOTAGS
Evaluate (the short way):

  a) $\log_5(5^{19})$

  b) $7^{\log_7(81)}$

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- (XXXx25x01x22x022)
NOTAGS
Factor $16v^3w^2 - w^4v$.

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- (XXXx25x01x22x023)
NOTAGS
Factor $x^2 - x - 6$.

---

- (XXXx25x01x22x024)
NOTAGS
Simplify $\left(\frac{1000}{27}\right)^{-\frac{1}{3}}$.

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- (XXXx25x01x22x025)
NOTAGS
Simplify $\dfrac{x^{-2}y^7}{x^4u^{-2}}$.

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- (XXXx25x01x22x026)
NOTAGS
Simplify $\left(4x^2y^{-3}z^4\right)^{\frac{1}{2}}$.

---

- (XXXx25x01x22x027)
NOTAGS
Simplify $\sqrt{72}$.

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- (XXXx25x01x22x028)
NOTAGS
Simplify $\dfrac{5}{\sqrt{5}}$.

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- (XXXx25x01x22x030)
NOTAGS
Simplify $3\sqrt{2}+\sqrt{50}$.

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- (XXXx25x01x22x031)
NOTAGS
Simplify $(\sqrt{12}+\sqrt{5})(\sqrt{12}-\sqrt{5})$.

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- (XXXx25x01x22x032)
NOTAGS
Simplify $\dfrac{2}{4-\sqrt{3}}$.

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- (XXXx25x01x22x033)
NOTAGS
Simplify $\sqrt{30x^9w}$.

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- (XXXx25x01x22x034)
NOTAGS
Simplify $\sqrt{\dfrac{a^5b^6c^7}{50a^2b^9}}$.

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- (XXXx25x01x22x035)
NOTAGS
Simplify $\dfrac{15xy^9}{12z^3w}\times\dfrac{8z^8w^8}{3xy}$.

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- (XXXx25x01x22x036)
NOTAGS
Simplify $\dfrac{9x(x-3)(x-7)}{10(2x+1)(x-12)} \div \dfrac{3(x-7)}{10x(x-12)}$.

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- (XXXx25x01x22x037)
NOTAGS
Simplify $\dfrac{8x^2}{x^3+7x^2-30x} \div \dfrac{12x+36}{x^2-3x}$.

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- (XXXx25x01x22x038)
NOTAGS
Simplify $\dfrac{x^3y^2}{3zw^9} - \dfrac{15y}{3z^5w^3}$.

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- (XXXx25x01x22x039)
NOTAGS
Simplify $\dfrac{7(x+7)^4}{4(a-2)^3(y-9)^{12}} + \dfrac{(x+7)^5e}{3(a-2)^{10}(y-9)}$.

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- (XXXx25x01x22x040)
NOTAGS
Simplify $\dfrac{1+\frac{1}{8}}{1-\frac{5}{12}}$.

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- (XXXx25x01x22x041)
NOTAGS
Simplify $\dfrac{\frac{3}{x}-\frac{9}{y}}{\frac{3}{xy}}$.

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- (XXXx25x01x22x042)
NOTAGS
Solve the equation (justify each step):
$-7u+1=8$

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- (XXXx25x01x22x043)
NOTAGS
Solve the equation (justify each step):
$9+4x=3(1+2x)$

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- (XXXx25x01x22x044)
NOTAGS
Solve the equation (complex solutions allowed), justifying every step.
$\sqrt{2x}-1=9$

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- (XXXx25x01x22x045)
NOTAGS
Solve the equation (complex solutions allowed), justifying every step.
$3\sqrt{x+2}+15=6$

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- (XXXx25x01x22x046)
NOTAGS
Solve the equation (complex solutions allowed), justifying every step.
$\sqrt[3]{x+1}=-2$

---

- (XXXx25x01x22x047)
NOTAGS
Simplify:

a) $\sqrt{1000}$

b) $\dfrac{10+\sqrt{1000}}{4}$

---

- (XXXx25x01x22x048)
NOTAGS
Solve each equation (complex solutions allowed):

a) $(3x-1)(x)=0$

b) $2x^2-x=0$

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- (XXXx25x01x22x052)
NOTAGS
Solve, justifying every step:
$\dfrac{5}{x-1}=5.$

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- (XXXx25x01x22x053)
NOTAGS
Solve, justifying every step:
$\dfrac{4x-1}{5}+\dfrac{2-x}{15}=\dfrac{2x+1}{3}.$

---

- (XXXx25x01x22x054)
NOTAGS
Solve, justifying every step:
$\dfrac{x}{x+5}+\dfrac{1}{x+1}=\dfrac{20}{x^2+6x+5}.$

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- (XXXx25x01x22x055)
NOTAGS

a) Solve $10^x=1$

b) Solve $x+2=\log_3(81)$

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- (XXXx25x01x22x056)
NOTAGS

a) Solve $5^{3y+1} = 125$

b) Solve $3 = \log_{10}(x + 50)$

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- (XXXx25x01x22x057)
NOTAGS
Solve $4 = \log_2(3x - 2)$

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- (XXXx25x01x22x058)
NOTAGS
The length of a rectangle is 2 meters more than its width. If the area is 80 $m^2$, find the rectangle's dimensions.

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- (XXXx25x01x22x059)
NOTAGS
A 21-foot board is cut into two pieces with ratio 3:4. Find their lengths.

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